

High Speed Amplifier Techniques

A Designer's Companion for Wideband Circuitry

Jim Williams

PREFACE

This publication represents the largest LTC commitment to an application note to date. No other application note absorbed as much effort, took so long or cost so much. This level of activity is justified by our belief that high speed monolithic amplifiers greatly interest users.

Historically, monolithic amplifiers have represented packets of inexpensive, precise and controllable gain. They have partially freed engineers from the constraints and frustrations of device level design. Monolithic operational amplifiers have been the key to practical implementation of high level analog functions. As good as they are, one missing element in these devices has been speed.

Devices presently coming to market are addressing monolithic amplifiers' lack of speed. They bring with them the ease of use and inherent flexibility of op amps. When

Philbrick Researches introduced the first mass produced op amp in the 1950's (K2-W) they knew it would be used. What they couldn't possibly know was just how widely, and how many different types of applications there were. As good a deal as the K2-W was (I paid \$24.00 for mine - or rather, my father did), monolithic devices are far better. The combination of ease of use, economy, precision and versatility makes modern op amps just too good to be believed.

Considering all this, adding speed to op amps' attractions seems almost certain to open up new application areas. We intend to supply useful high speed products and the level of support necessary for their successful application (such high minded community spirit is, of course, capitalism's deputy). We hope you are pleased with our initial efforts and look forward to working together.

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INTRODUCTION

Most monolithic amplifiers have been relatively slow devices. Wideband operation has been the province of discrete and hybrid technologies. Some fast monolithic amplifiers have been available, but the exotic and expensive processing required has inflated costs, precluding widespread acceptance. Additionally, many of the previous monolithic designs were incapable of high precision and prone to oscillation or untoward dynamics, making them unattractive.

Recent processing and design advances have made inexpensive, precision wideband amplifiers practical. Figure 1 lists some amplifiers, along with a summary of their characteristics. Reviewing this information reveals extraordinarily wideband devices, with surprisingly good DC characteristics. All of these amplifiers utilize standard op amp architecture, except the LT1223 and LT1228, which are so-called current mode feedback types (see Appendix H, "About Current Mode Feedback"). It is clear that the raw speed capabilities of these devices, combined with their inherent flexibility as op amps, permit a wide range of applications. What is required of the user is a familiarity with the devices and respect for the requirements of high speed circuitry.

This effort's initial sections are devoted to familiarizing the reader with the realities and difficulties of high speed circuit work. The mechanics and subtleties of achieving precision circuit operation at DC and low frequency have been well documented. Relatively little has appeared which discusses, in practical terms, how to get fast circuitry to work. In developing such circuits, even veteran designers sometimes feel that nature is conspiring against them. In some measure this is true. Like all engineering endeavors, high speed circuits can only work if negotiated compromises with nature are arranged. Ignorance of, or contempt for, physical law is a direct route to frustration. Mother Nature laughs at dilettantism and crushes arrogance without even knowing she did it. Even without Einstein's revelations, the world of high speed is full of surprises. Working with events measured in nanoseconds requires the greatest caution, prudence and respect for Mother Nature. Absolutely nothing should be taken for granted, because nothing is. Circuit design is very much the art of compromise with parasitic effects. The "hidden

schematic" (this descriptive was originated by Charly Gullett of Intel Corporation) usually dominates the circuit's form, particularly at high speed.

In this regard, much of the text and appendices are directed at developing awareness of, and respect for, circuit parasitics and fundamental limitations. This approach is maintained in the applications section, where the notion of negotiated compromises is expressed in terms of resistor values and compensation techniques. Many of the application circuits use the amplifier's speed to improve on a standard circuit. Some utilize the speed to implement a traditional function in a non-traditional way, with attendant advantages. A (very) few operate at or near the state-of-the-art for a given circuit type, regardless of approach. Substantial effort has been expended in developing these examples and documenting their operation. The resultant level of detail is justified in the hope that it will be catalytic. The circuits should stimulate new ideas to suit particular needs, while demonstrating fast amplifiers' capabilities in an instructive manner.

PERSPECTIVES ON HIGH SPEED DESIGN

A substantial amount of design effort has made Figure 1's amplifiers relatively easy to use. They are less prone to oscillation and other vagaries than some much slower amplifiers. Unfortunately, laws of physics dictate that the circuit's *environment* must be properly prepared. The performance limits of high speed circuitry are often determined by parasitics such as stray capacitance, ground impedance and layout. Some of these considerations are present in digital systems where designers are comfortable describing bit patterns, delays and memory access times in terms of nanoseconds. Figure 2's test circuit provides valuable perspective on just how fast these amplifiers are. Here, the pulse generator (Trace A, Figure 3) drives a 74S04 Schottky TTL inverter (Trace B), an LT1223 op amp connected as an inverter (Trace C), and a 74HC04 high speed CMOS inverter (Trace D). The LT1223 doesn't fare too badly. Its delay and fall times are about 2ns slower than the 74S04, but significantly faster than the 74HC04. In fact, the LT1223 has completely finished its transition before the 74HC04 even begins to move! Linear circuits operating with this kind of speed make many engineers justifiably wary. Nanosecond domain linear circuits are widely associated with oscillations, mysterious shifts in

PARAMETER	LT1122	LT1190	LT1191	LT1192	LT1193	LT1194	LT1220	LT1221	LT1222	LT1223	LT1224
Slew Rate	60V/ μ s	450V/ μ s	450V/ μ s	450V/ μ s	450V/ μ s	450V/ μ s	250V/ μ s	250V/ μ s	200V/ μ s	1000V/ μ s	300V/ μ s
Bandwidth	14MHz	50MHz	90MHz	400MHz	70MHz	70MHz	45MHz	150MHz	350MHz	100MHz	45MHz
Delay-Rise Time	15ns-65ns	4ns-7ns	3.5ns-1.6ns	5ns-7ns	4ns-7ns	4ns-7ns	4ns-4ns	5ns-5ns	5ns-5ns	3.5ns-3.5ns	4ns-4ns
Settling Time	340ns-0.01%	100ns-0.1%	100ns-0.1%	80ns-0.1%	100ns-0.1%	100ns-0.1%	90ns-0.1%	90ns-0.1%	90ns-0.1%	75ns-0.1%	90ns-0.1%
Output Current	6mA	50mA	50mA	50mA	50mA	50mA	24mA	24mA	24mA	50mA	40mA
Offset	600 μ V	4mV	2mV	2mV	3mV	3mV	2mV	1mV	1mV	3mV	1mV
Drift	6 μ V/ $^{\circ}$ C						20 μ V/ $^{\circ}$ C	15 μ V/ $^{\circ}$ C	10 μ V/ $^{\circ}$ C		20 μ V/ $^{\circ}$ C
Bias Current	75pA	500nA	500nA	500nA	500nA	500nA	300nA	300nA	300nA	3 μ A	6 μ A
Gain	500,000	22,000	45,000	200,000	Adjustable	10	20,000	50,000	100,000	90dB	80dB
Gain Error (Minimum Gain)				$A_{V\text{MIN}} = 10$	0.1%	0.1%		$A_{V\text{MIN}} = 4$	$A_{V\text{MIN}} = 10$		$A_{V\text{MIN}} = 1$
Gain Drift											
Power Supply	40V	18V _{MAX}	18V _{MAX}	18V _{MAX}	18V _{MAX}	18V _{MAX}	36V	36V	36V	36V _{MAX}	36V

Figure 1. Characteristics of Some Different Fast IC Amplifiers

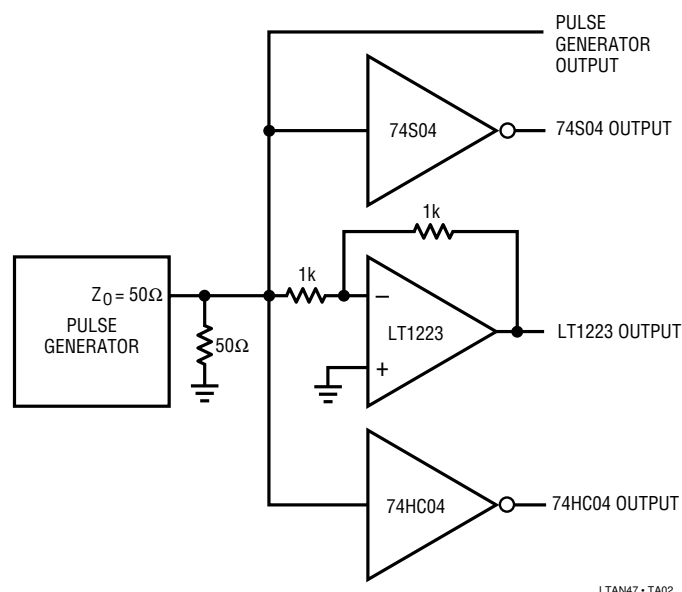


Figure 2. A Race Between the LT1223 Amplifier and Some Fast Logic Inverters

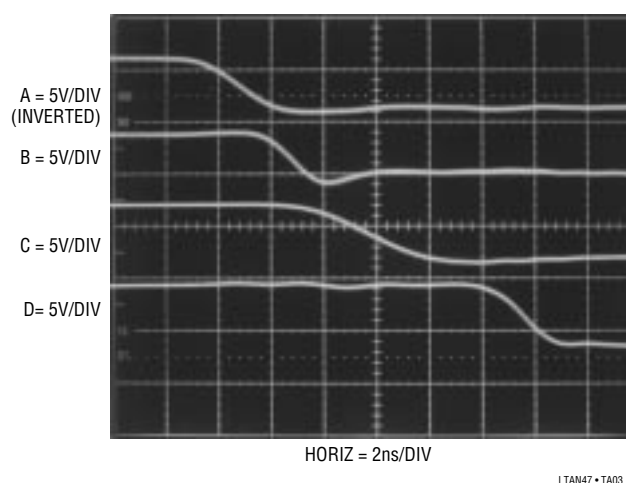


Figure 3. The Amplifier (Trace C) is 3ns Slower than 74S Logic (Trace B), but 5ns Faster than High Speed HCMOS (Trace D)!

circuit characteristics, unintended modes of operation and outright failure to function.

Other common problems include different measurement results using various pieces of test equipment, inability to make measurement connections to the circuit without inducing spurious responses, and dissimilar operation between two identical circuits. If the components used in the circuit are good and the design is sound, all of the above problems can usually be traced to failure to provide a proper circuit environment. To learn how to do this

requires studying the causes of the aforementioned difficulties.

The following segments, “Mr. Murphy’s Gallery of High Speed Amplifier Problems” and the “Tutorial Section”, address this. The “Problems” section alerts the reader to trouble areas, while the “Tutorial” highlights theory and techniques which may be applied towards solving the problems shown. The tutorials are arranged in roughly the same order as the problems are presented.

MR. MURPHY’S GALLERY OF HIGH SPEED AMPLIFIER PROBLEMS

It sometimes seems that Murphy’s Law dominates all physical law. For a complete treatise on Murphy’s Law, see Appendix J, “The Contributions of Edsel Murphy to the Understanding of the Behavior of Inanimate Objects”, by D.L. Klipstein. The law’s consequences weigh heavily in high speed design. As such, a number of examples are given in the following discussion. The average number of phone calls we receive per month due to each “Murphy” example appears at the end of each figure caption.

Problems can start even before power is applied to the amplifier. Figure 4 shows severe ringing on the pulse edges at the output of an unterminated pulse generator cable. This is due to reflections and may be eliminated by terminating the cable. *Always terminate the source in its characteristic impedance when looking into cable or long PC traces. Any path over 1 inch long is suspect.*

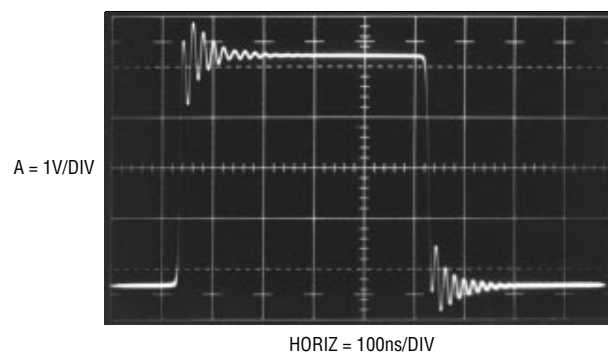


Figure 4. An Unterminated Pulse Generator Cable Produces Ringing Due to Reflections – 3 ☎

In Figure 5 the cable is terminated, but ripple and aberration are still noticeable following the high speed edge transitions. In this instance the terminating resistor’s leads are lengthy ($\approx 3/4"$), preventing a high integrity wideband termination.

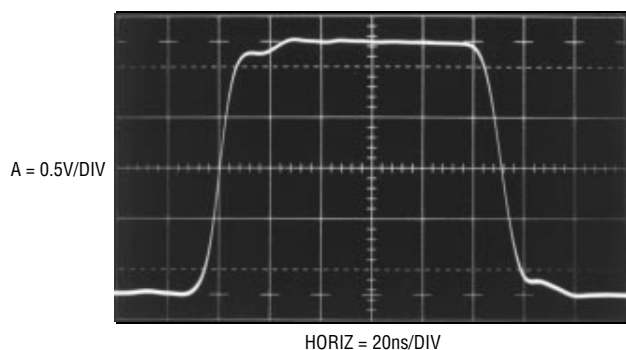


Figure 5. Poor Quality Termination Results in Pulse Corner Aberrations – 1 📞

The best termination for 50Ω cable is the BNC coaxial type. These devices should not simply be resistors in an enclosure. Good grade 50Ω terminators maintain true coaxial form. They use a carefully designed 50Ω resistor with significant effort devoted to connections to the actual resistive element. In particular, the largest possible connection surface area is utilized to minimize high speed losses. While these type terminators are practical on the test bench, they are rarely used as board level components. In general, the best termination resistors for PC board use are carbon or metal film types with the shortest possible lead lengths. These resistor's end-cap connections provide better high speed characteristics than the rod-connected composition types. Wirewound resistors, because of their inherent and pronounced inductive characteristics, are completely unsuitable for high speed work. This includes so-called non-inductive types.

Another termination consideration is disposal of the current flowing through the terminator. The terminating resistor's grounded end should be placed so that the high speed currents flowing from it do not disrupt circuit operation. For example, it would be unwise to return terminator current to ground near the grounded positive input of an inverting op amp. The high speed, high density (5V pulses through a 50Ω termination generates 100mA current spikes) current flow could cause serious corruption of the desired zero volt op amp reference. This is another reason why, for bench testing, the coaxial BNC terminators are far preferable to discrete, breadboard mounted resistors. With BNC types in use the termination current returns directly to the source generator and never flows in the breadboard. (For more information see the Tutorial section.) *Select terminations carefully and evaluate the effects of their placement in the test set-up.*

Figure 6 shows an amplifier output which rings and distorts badly after rapid movement. In this case, the probe ground lead is too long. For general purpose work, most probes come with ground leads about 6 inches long. At low frequencies this is fine. At high speed, the long ground lead looks inductive, causing the ringing shown. High quality probes are always supplied with some short ground straps to deal with this problem. Some come with very short spring clips which fix directly to the probe tip to facilitate a low impedance ground connection. For fast work, the ground connection to the probe should not exceed 1 inch in length. (Probes are covered in the Tutorial section; also see Appendix A, "ABC's of Probes", guest written by the engineering staff of Tektronix, Inc.). *Keep the probe ground connection as short as possible. The ideal probe ground connection is purely coaxial. This is why probes mated directly to board mounted coaxial connectors give the best results.*

In Figure 7 the probe is properly grounded, but a new problem pops up. This photo shows an amplifier output

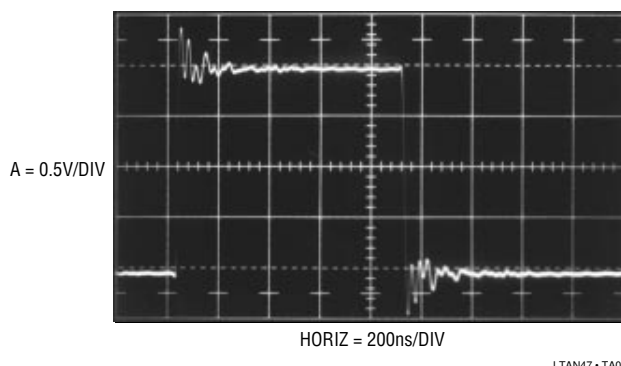


Figure 6. Poor Probe Grounding Badly Corrupts the Observed Waveform – 53 📞

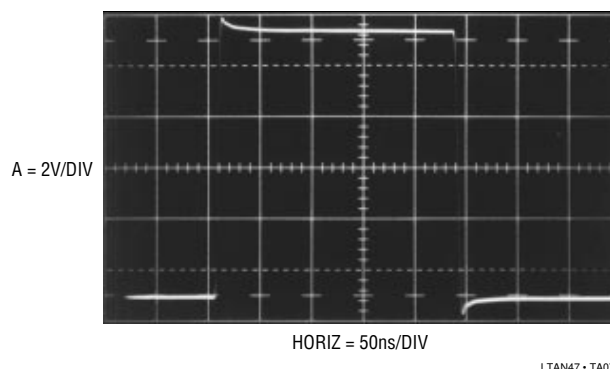
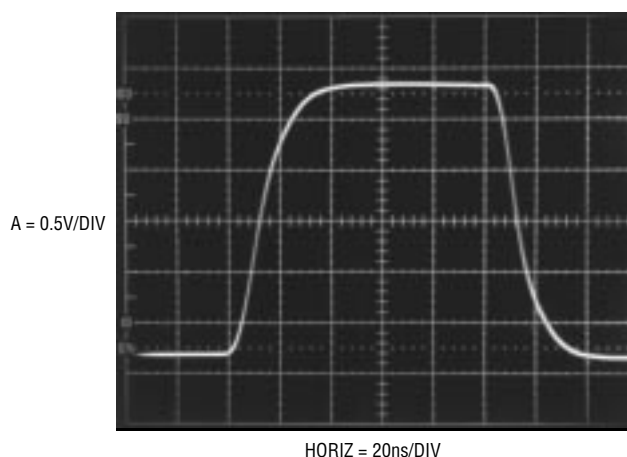


Figure 7. Improper Probe Compensation Causes Seemingly Unexplainable Amplitude Error – 12 📞

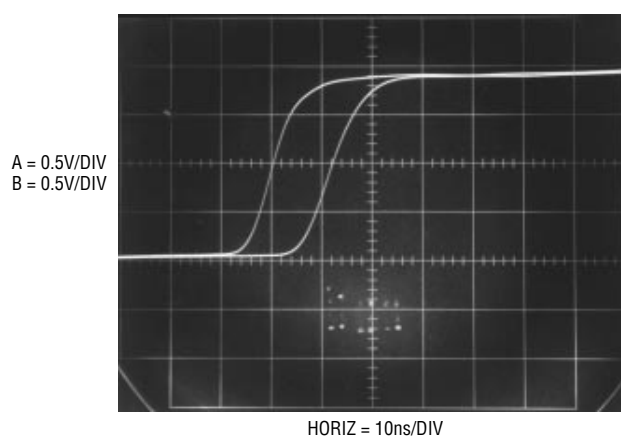
excursion of 11V — quite a trick from an amplifier running from $\pm 5V$ rails. This is a commonly reported problem in high speed circuits and can be quite confusing. It is not due to suspension of natural law, but is traceable to a grossly miscompensated or improperly selected oscilloscope probe. *Use probes which match your oscilloscope's input characteristics and compensate them properly.* (For discussions on probes, see Appendix A, “ABC’s of Probes”, guest written by the engineering staff of Tektronix, Inc. and the Tutorial section.) Figure 8 shows another probe-induced problem. Here the amplitude seems correct but the amplifier appears slow with pronounced edge rounding. In this case, the probe used is too heavily compensated or slow for the oscilloscope. Never use 1X or straight probes. Their bandwidth is 20MHz or less and capacitive loading is high. *Check probe bandwidth to ensure it is adequate for the measurement. Similarly, use an oscilloscope with adequate bandwidth.*



LTAN47 • TA08

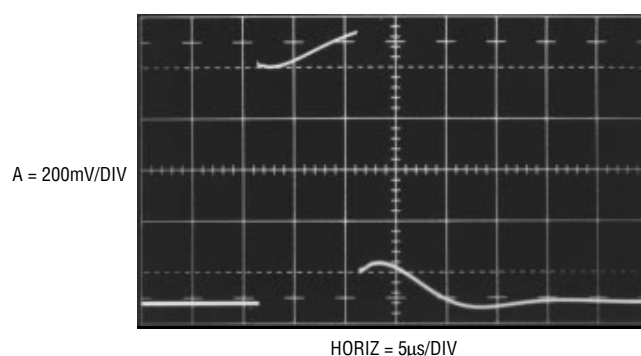
Figure 8. Overcompensated or Slow Probes Make Edges Look Too Slow – 2 📞

Mismatched probes account for the apparent excessive amplifier delay in Figure 9. Delay of almost 12ns (Trace A is the input, Trace B the output) is displayed for an amplifier specified at 6ns. Always keep in mind that various types of probes have different signal transit delay times. At high sweep speeds, this effect shows up in multi-trace displays as time skewing between individual channels. Using similar probes will eliminate this problem, but measurement requirements often dictate dissimilar probes. In such cases the differential delay should be measured and then mentally factored in to reduce error when interpreting the display. It is worth noting that active probes,



LTAN47 • TA09

Figure 9. Probes with Mismatched Delays Produce Apparent Time Skewing in the Display – 4 📞



LTAN47 • TA10

Figure 10. Overdriven FET Probe Produces Excessive Waveform Distortion and Tailing. Saturation Effects can Also Cause Delayed Response – 1 📞

such as FET and current probes, have signal transit times as long as 25ns. A fast 10X or 50 Ω probe delay can be inside 3ns. *Account for probe delays in interpreting oscilloscope displays.*

The difficulty shown in Figure 10 is a wildly distorted amplifier output. The output slews quickly, but the pulse top and bottom recoveries have lengthy, tailing responses. Additionally, the amplifier output seems to clip well below its nominal rated output swing. A common oversight is responsible for these conditions. A FET probe monitors the amplifier output in this example. The probe's common-mode input range has been exceeded, causing it to overload, clip and distort badly. When the pulse rises, the probe is driven deeply into saturation, forcing internal circuitry away from normal operating points. Under these conditions the displayed pulse top is illegitimate. When the output falls, the probe's overload recovery is lengthy and uneven, causing the tailing. More subtle forms of FET

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probe overdrive may show up as extended delays with no obvious signal distortion. *Know your FET probe. Account for the delay of its active circuitry. Avoid saturation effects due to common-mode input limitations (typically $\pm 1V$). Use 10X and 100X attenuator heads when required.*

Figure 11's probe-caused problem results in amplifier output peaking and ringing. In other respects the display is acceptable. This output peaking characteristic is caused by a second 10X probe connected to the amplifier's summing junction. Because the summing point is so central to analyzing op amp operation, it is often monitored. At high speed the 10pF probe input capacitance causes a significant lag in feedback action, forcing the amplifier to overshoot and hunt as it seeks the null point. Minimizing this effect calls for the lowest possible probe input capacitance, mandating FET types or special passive probes. (Probes are covered in the Tutorial section; also see Appendix A, "ABC's of Probes", guest written by the engineering staff of Tektronix, Inc.). *Account for the effects of probe capacitance, which often dominates its impedance characteristics at high speeds. A standard 10pF 10X probe forms a 10ns lag with a 1K Ω source resistance.*

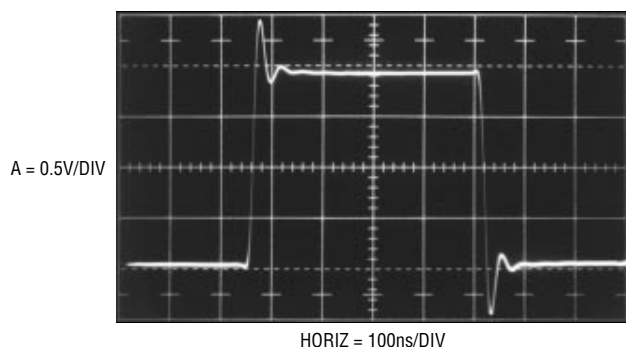


Figure 11. Effect of a 10X, 10pF 'Scope Probe at the Summing Point – 2 📞

A peaked, tailing response is Figure 12's characteristic. The photo shows the final 40mV of a 2.5V amplifier excursion. Instead of a sharp corner which settles cleanly, peaking occurs, followed by a lengthy tailing decay. This waveform was recorded with an inexpensive off-brand 10X probe. Such probes are often poorly designed, and constructed from materials inappropriate for high speed work. The selection and integration of materials for wideband probes is a specialized and difficult art. Sub-

stantial design effort is required to get good fidelity at high speeds. *Never use probes unless they are fully specified for wideband operation. Obtain probes from a vendor you trust.*

Figure 13 shows the final movements of an amplifier output excursion. At only 1mV per division the objective is to view the settling residue to high resolution. This response is characterized by multiple time constants, non-linear slew recovery and tailing. Note also the high speed event just before the waveform begins its negative going transition. What is actually being seen is the oscilloscope recovering from excessive overdrive. Any observation that requires off-screen positioning of parts of the waveform should be approached with the greatest caution. Oscilloscopes vary widely in their response to overdrive, bringing displayed results into question. Complete treatment of high resolution settling time measurements and oscilloscope overload characteristics is given in the Tutorial section, "About Oscilloscopes" and Appendix B "Measuring Amplifier Settling Time". *Approach all oscilloscope*

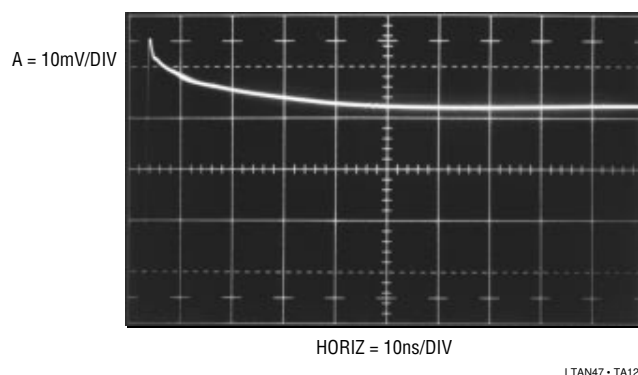


Figure 12. Poor Quality 10X Probe Introduces Tailing – 2 📞

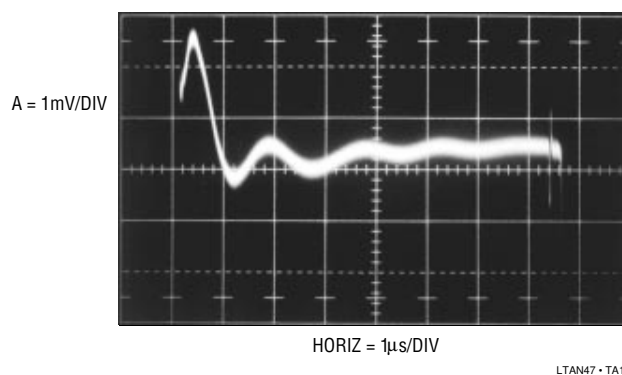


Figure 13. Overdriven Oscilloscope Display Says More About the Oscilloscope than the Circuit it's Connected to – 6 📞

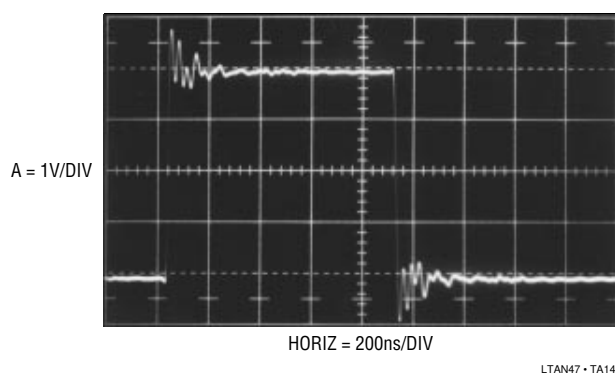


Figure 14. Instabilities Due to No Ground Plane Produce a Display Similar to a Poorly Grounded Probe – 62 ☎

measurements which require off-screen activity with caution. Know your instrument's capabilities and limitations.

Sharp eyed readers will observe that Figure 14 is a duplicate of Figure 6. Such lazy authorship is excusable because almost precisely the same waveform results when no ground plane is in use. A ground plane is formed by using a continuous conductive plane over the surface of the circuit board. (The theory behind ground planes is discussed in the Tutorial section). The only breaks in this plane are for the circuit's necessary current paths. The ground plane serves two functions. Because it is flat (AC currents travel along the surface of a conductor) and covers the entire area of the board, it provides a way to access a low inductance ground from anywhere on the board. Also, it minimizes the effects of stray capacitance in the circuit by referring them to ground. This breaks up potential unintended and harmful feedback paths. *Always use a ground plane with high speed circuitry.*

By far the most common error involves power supply bypassing. Bypassing is necessary to maintain low supply impedance. DC resistance and inductance in supply wires and PC traces can quickly build up to unacceptable levels. This allows the supply line to move as internal current levels of the devices connected to it change. This will almost always cause unruly operation. In addition, several devices connected to an unbypassed supply can “communicate” through the finite supply impedances, causing erratic modes. Bypass capacitors furnish a simple way to eliminate this problem by providing a local reservoir of energy at the device. The bypass capacitor acts as an electrical flywheel to keep supply impedance low at high

frequencies. The choice of what type of capacitors to use for bypassing is a critical issue and should be approached carefully (see Tutorial, “About Bypass Capacitors”). An unbypassed amplifier with a 100Ω load is shown in Figure 15. The power supply the amplifier sees at its terminals has high impedance at high frequency. This impedance forms a voltage divider with the amplifier and its load, allowing the supply to move as internal conditions in the comparator change. This causes local feedback and oscillation occurs. *Always use bypass capacitors.*

In Figure 16 the 100Ω load is removed, and a pulse output is displayed. The unbypassed amplifier responds surprisingly well, but overshoot and ringing dominate. *Always use bypass capacitors.*

Figure 17's settling is noticeably better, but some ringing remains. This response is typical of lossy bypass capacitors, or good ones placed too far away from the amplifier. *Use good quality, low loss bypass capacitors, and place them as close to the amplifier as possible.*

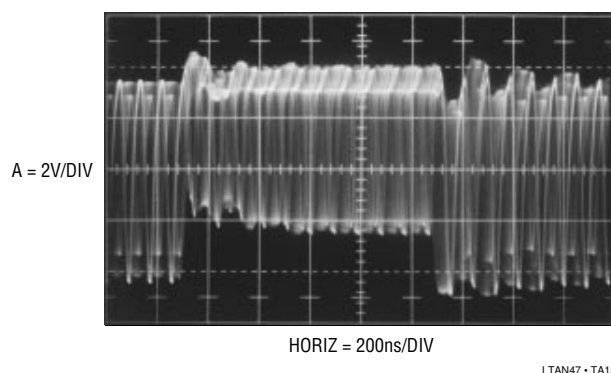


Figure 15. Output of an Unbypassed Amplifier Driving a 100Ω Load Without Bypass Capacitors – 58 ☎

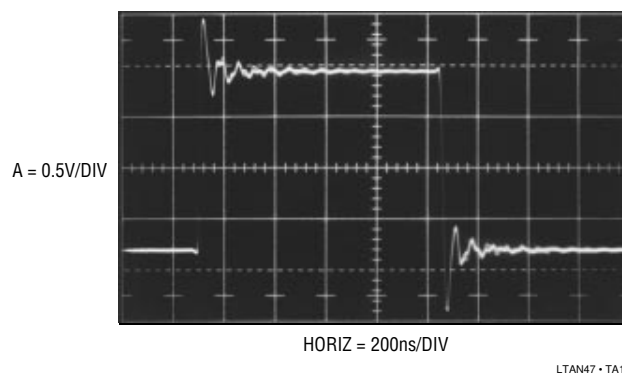


Figure 16. An Unbypassed Amplifier Driving No Load is Surprisingly Stable...at the Moment – 49 ☎

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The multiple time constant ringing in Figure 18 often indicates poor grade paralleled bypassing capacitors or excessive trace length between the capacitors. While paralleling capacitors of different characteristics is a good way to get wideband bypassing, it should be carefully considered. Resonant interaction between the capacitors can cause a waveform like this after a step.

This type response is often aggravated by heavy amplifier loading. *When paralleling bypass capacitors, plan the layout and breadboard with the units you plan to use in production.*

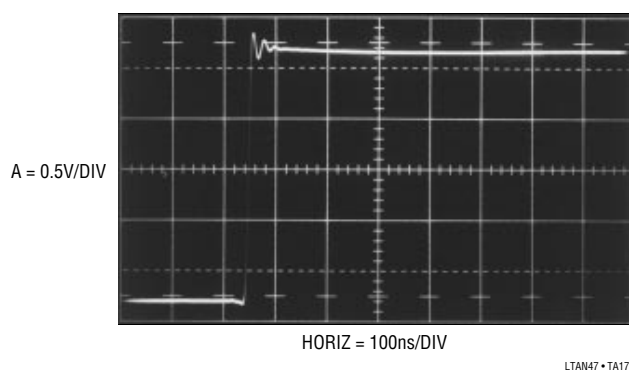


Figure 17. Poor Quality Bypass Capacitor Allows Some Ringing – 28 📞

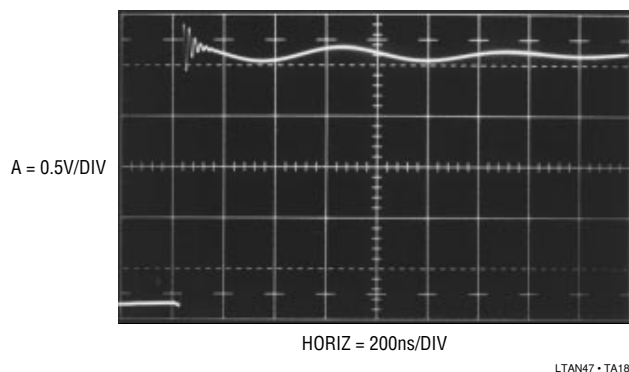


Figure 18. Paralleled Bypass Capacitors Form a Resonant Network and Ring – 2 📞

Figure 19 addresses a more subtle bypassing problem. The trace shows the last 40mV excursion of a 5V step almost settling cleanly in 300ns. The slight overshoot is due to a loaded (500Ω) amplifier without quite enough bypassing. Increasing the total supply bypassing from 0.1μF to 1μF cured this problem. *Use large value paralleled bypass capacitors when very fast settling is required, particularly if the amplifier is heavily loaded or sees fast load steps.*

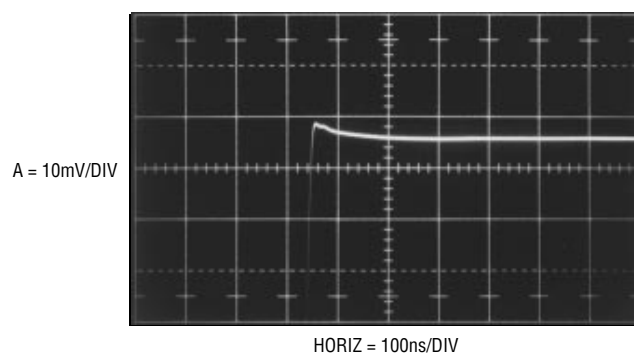


Figure 19. A More Subtle Bypassing Problem. Not-Quite-Good-Enough Bypassing Causes a Few Millivolts of Peaking – 1 📞

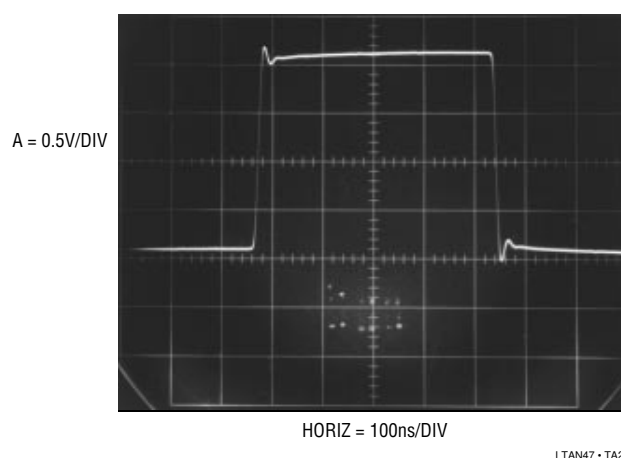


Figure 20. 2pF Stray Capacitance at the Summing Point Introduces Peaking – 4 📞

The problem shown in Figure 20, peaking on the leading and trailing corners, is typical of poor layout practice (see Tutorial section on “Breadboarding Techniques”). This unity gain inverter suffers from excessive trace area at the summing point. Only 2pF of stray capacitance caused the peaking and ring shown. *Minimize trace area and stray capacitance at critical nodes. Consider layout as an integral part of the circuit and plan it accordingly.*

Figure 21’s low level square wave output appears to suffer from some form of parasitic oscillation. In actuality, the disturbance is typical of that caused by fast digital clocking or switching regulator originated noise getting into critical circuit nodes. *Plan for parasitic radiative or conductive paths and eliminate them with appropriate layout and shielding.*

Figure 22 underscores the previous statement. This output was taken from a gain-of-ten inverter with 1kΩ input

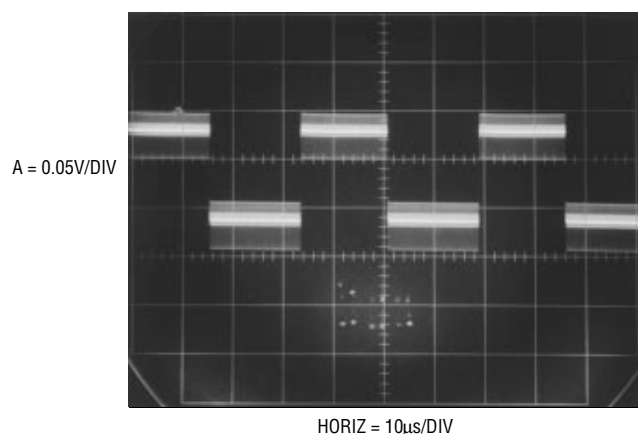


Figure 21. Clock or Switching Regulator Noise Corrupts Output Due to Poor Layout – 3 📞

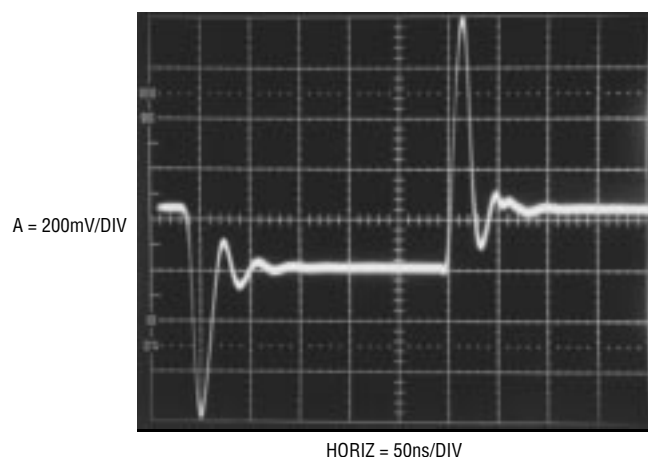


Figure 22. Output of an X10 Amplifier with 1pF Coupling from the Summing Point to the Input. Careful Shielding of the Input Resistor Will Eliminate the Peaked Edges and Ringing – 2 📞

resistance. It shows severe peaking induced by *only 1pF* of parasitic capacitance across the 1k resistor. The 50Ω terminated input source provides only 20mV of drive via a divider, but that's more than enough to cause problems, even with only 1pF stray coupling. In this case the solution was a ground referred shield at a right angle to, and encircling, the 1kΩ resistor. *Plan for parasitic radiative paths and eliminate them with appropriate shielding.*

A decompensated amplifier running at too low a gain produced Figure 23's trace. The price for decompensated amplifiers' increased speed is restrictions on minimum allowable gain. Decompensated amplifiers are simply not stable below some (specified) minimum gain, and no amount of ignorance or wishing will change this. This is a

common applications oversight with these devices, although the amplifier never fails to remind the user. *Observe gain restrictions when using decompensated amplifiers.*

Oscillation is also the problem in Figure 24, and it is due to excessive capacitive loading (see Tutorial section on "Oscillation"). Capacitive loading to ground introduces lag in the feedback signal's return to the input. If enough lag is introduced (e.g., a large capacitive load) the amplifier may oscillate. Even if a capacitively loaded amplifier doesn't oscillate, it's always a good idea to check its response with step testing. It's amazing how close to the edge of the cliff you can get without falling off, except when you build 10,000 production units. *Avoid capacitive loading. If such loading is necessary, check performance margins and isolate or buffer the load if necessary.*

Figure 25 appears to contain one cycle of oscillation. The output waveform initially responds, but abruptly reverses direction, overshoots and then heads positive again. Some

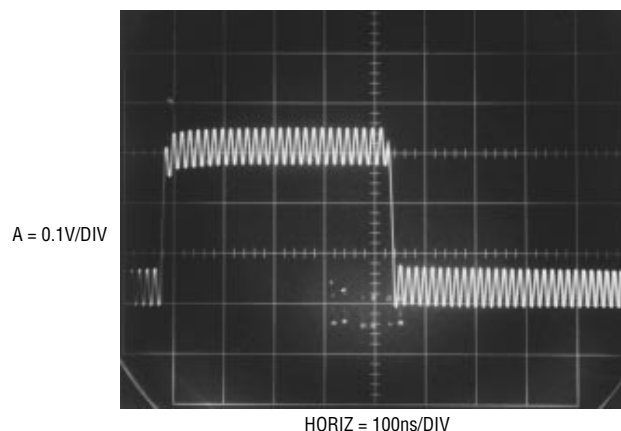


Figure 23. Decompensated Amplifier Running at Too Low a Gain – 22 📞

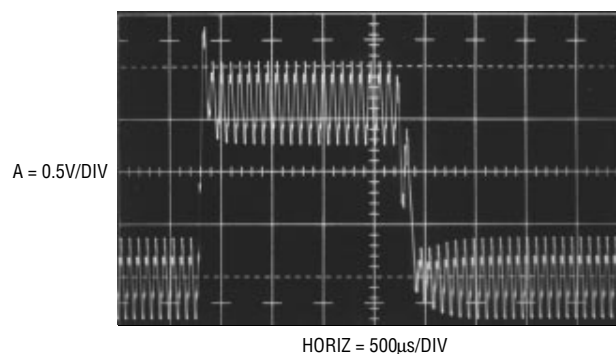


Figure 24. Excessive Capacitive Load Upsets the Amplifier – 165 📞

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overshoot again occurs, with a long tail and a small dip well before a non-linear slew returns the waveform to zero. Ugly overshoot and tailing completes the cycle. This is certainly strange behavior. What is going on here? The input pulse is responsible for all these anomalies. Its amplitude takes the amplifier outside its common-mode limits, inducing the bizarre effects shown. *Keep inputs inside specified common-mode limits at all times.*

Figure 26 shows an oscillation laden output (Trace B) trying to unity gain invert the input (Trace A). The input's form is distinguishable in the output, but corrupted with very high frequency oscillation and overshoot. In this case the amplifier includes a booster within its loop to provide increased output current. The disturbances noted are traceable to local instabilities within the booster circuit. (See Appendix C, "The Oscillation Problem — Frequency Compensation Without Tears"). *When using output booster stages, insure they are inherently stable before placing them inside an amplifier's feedback loop. Wideband booster stages are particularly prone to device level parasitic high frequency oscillation.*

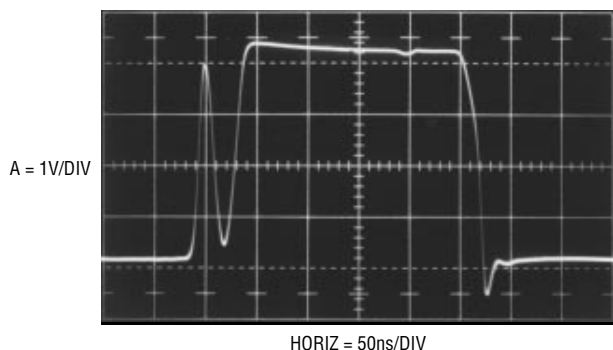


Figure 25. Input Common Mode Overdrive Generates Odd Outputs – 3 📞

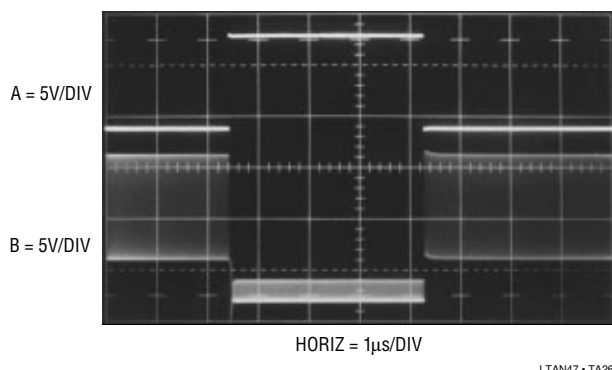


Figure 26. Local Oscillations in a Booster Stage. Frequency is Typically High – 12 📞

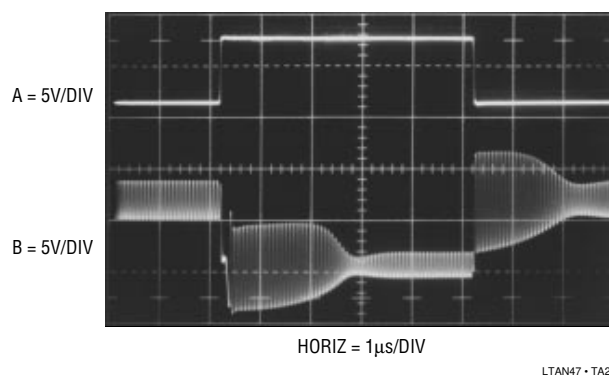


Figure 27. Loop Oscillations in a Booster Stage. Note Lower Frequency than Local Oscillations in Previous Example – 28 📞

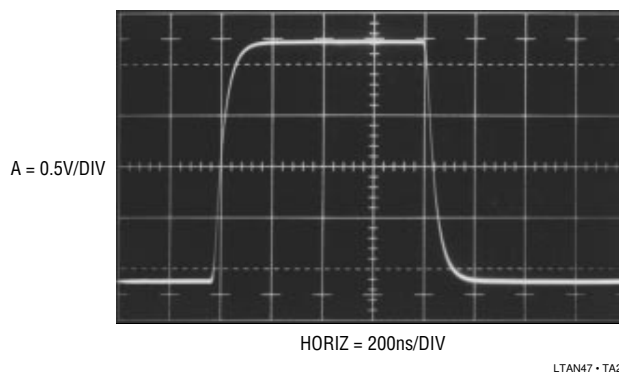


Figure 28. Excessive Source Impedance Gives Serene But Undesired Response – 6 📞

Figure 27's booster augmented unity gain inverting op amp also oscillates, but at a much lower frequency. Additionally, overshoot and non-linear recovery dominate the waveform's envelope. Unlike the previous example, this behavior is not due to local oscillations within the booster stage. Instead, the booster is simply too slow to be included in the op amp's feedback loop. It introduces enough lag to force oscillation, even as it hopelessly tries to maintain loop closure. *Insure booster stages are fast enough to maintain stability when placed in the amplifier's feedback loop.*

The serene rise and fall of Figure 28's pulse is a welcome relief from the oscillatory screaming of the previous photos. Unfortunately, such tranquilized behavior is simply too slow. This waveform, reminiscent of Figure 8's bandlimited response, is due to excessive source impedance. The high source impedance combines with amplifier input capacitance to band limit the input and the output reflects this action. *Minimize source impedance to levels*

which maintain desired bandwidth. Keep stray capacitance at inputs down.

TUTORIAL SECTION

An implied responsibility in raising the aforementioned issues is their solution or elimination. What good is all the rabble-raising without suggestions for fixes? It is in this spirit that this tutorial section is presented. Theory, techniques, prejudice and just plain gossip are offered as tools which may help avoid or deal with difficulties. As previously mentioned, the tutorials appear in roughly the same order as the problems were presented.

About Cables, Connectors and Terminations

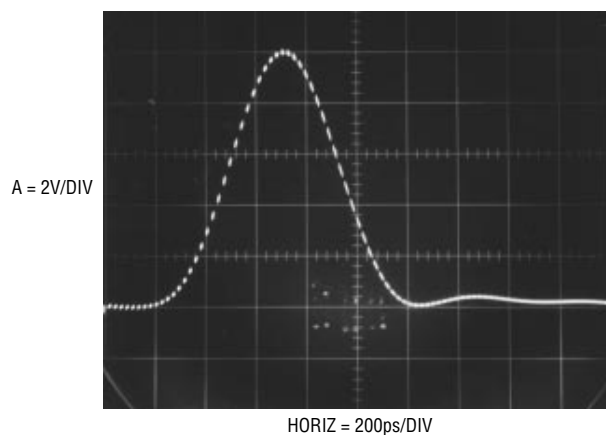
Routing of high speed signals to and from the circuit board should always be done with good quality coaxial cable. The cable should be driven and terminated in the system's characteristic impedance at the drive and load points. The driven end is usually an instrument (e.g., pulse or signal generator), presumably endowed with proper characteristics by its manufacturer. It is the cable and its termination, selected by the experimenter, that often cause problems.

All coaxial cable is not the same. Use cable appropriate to the system's characteristic impedance and of good quality. Poorly chosen cable materials or construction methods can introduce odd effects at very high speeds, resulting in observed waveform distortion. A poor cable choice can adversely effect 0.01% settling in the 100ns-200ns region. Similarly, poor cable can preclude maintenance of even the cleanest pulse generator's 1ns rise time or purity. Typically, inappropriate cable can introduce tailing, rise time degradation, aberrations following transitions, non-linear impedance and other undesirable characteristics.

Termination choice is equally important. Good quality BNC coaxial type terminators are usually the best choice for breadboarding. Their impedance vs frequency is flat into the GHz range. Additionally, their construction insures that the (often substantial) drive current returns directly to the source, instead of being dumped into the breadboard's

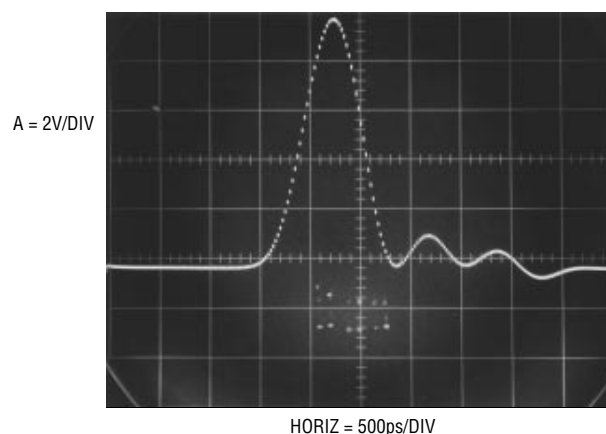
Note 1: The ability to generate such a pulse proves useful for a variety of tasks, including testing terminators, cables, probes and oscilloscopes for response. The requirements for this pulse generator are surprisingly convenient and inexpensive. For a discussion and construction details see Appendix D "Measuring Probe - Oscilloscope Response".

ground system. As previously discussed, BNC coaxial terminators are not simply resistors in a can. Special construction techniques insure optimum wideband response. Figures 29 and 30 demonstrate this nicely. In Figure 29 a 1ns pulse with 350ps rise and fall times¹ is monitored on a 1GHz sampling 'scope (Tektronix 556 with 1S1 sampling plug-in and P6032 probe). The waveform is clean, with only a slight hint of ring after the falling edge. This photo was taken with a high grade BNC coaxial type terminator in use. Figure 30 does not share these attributes. Here, the generator is terminated with a 50Ω carbon composition resistor with lead lengths of about 1/8 inch. The waveform rings and tails badly on turn-off before finally settling. Note that the sweep speed required a 2.5X reduction to capture these unwanted events.



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Figure 29. 350ps Rise and Fall Times are Preserved by a Good Quality Termination



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Figure 30. Poor Grade Termination Produces Pronounced Ringing and Tailing in the GHz Range

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Connectors, such as BNC barrel extensions and tee-type adaptors, are convenient and frequently employed. Remember that these devices represent a discontinuity in the cable, and can introduce small but undesirable effects. In general it is best to employ them as close as possible to a terminated point in the system. Use in the middle of a cable run provides minimal absorption of their mismatch and reflections. The worst offenders among connectors are adapters. This is unfortunate, as these devices are necessitated by the lack of connection standardization in wideband instrumentation. The mismatch caused by a BNC-to-GR874 adaptor transition at the input of a wideband sampling 'scope is small, but clearly discernible in the display. Similarly, mismatches in almost all adaptors, and even in "identical" adaptors of different manufacture, are readily measured on a high-frequency network analyzer such as the Hewlett-Packard 4195A² (for additional wisdom and terror along these lines see Reference 1).

BNC connections are easily the most common, but not necessarily the most desirable, wideband connection mechanism. The ingenious GR874 connector has notably superior high frequency characteristics, as does the type N. Unfortunately, it's a BNC world out there.

About Probes and Probing Techniques

The choice of which oscilloscope probe to use in a measurement is absolutely crucial. The probe must be considered as an inherent part of the circuit under test. Rise time, bandwidth, resistive and capacitive loading, delay and other limitations must be kept in mind.

Sometimes, the best probe is no probe at all. In some circumstances it is possible and preferable to connect critical breadboard points *directly* to the oscilloscope (see Figure 31). This arrangement provides the highest possible grounding integrity, eliminates probe attenuation, and maintains bandwidth. In most cases this is mechanically inconvenient, and often the oscilloscope's electrical characteristics (particularly input capacitance) will not permit it. This is why oscilloscope probes were developed, and why so much effort has been put into their development (Reference 42 is excellent). In addition to the mate-

Note 2: Almost no one believes any of this until they see it for themselves. I didn't. Photos of the network analyzer's display aren't included in the text because no one would believe them. I wouldn't.

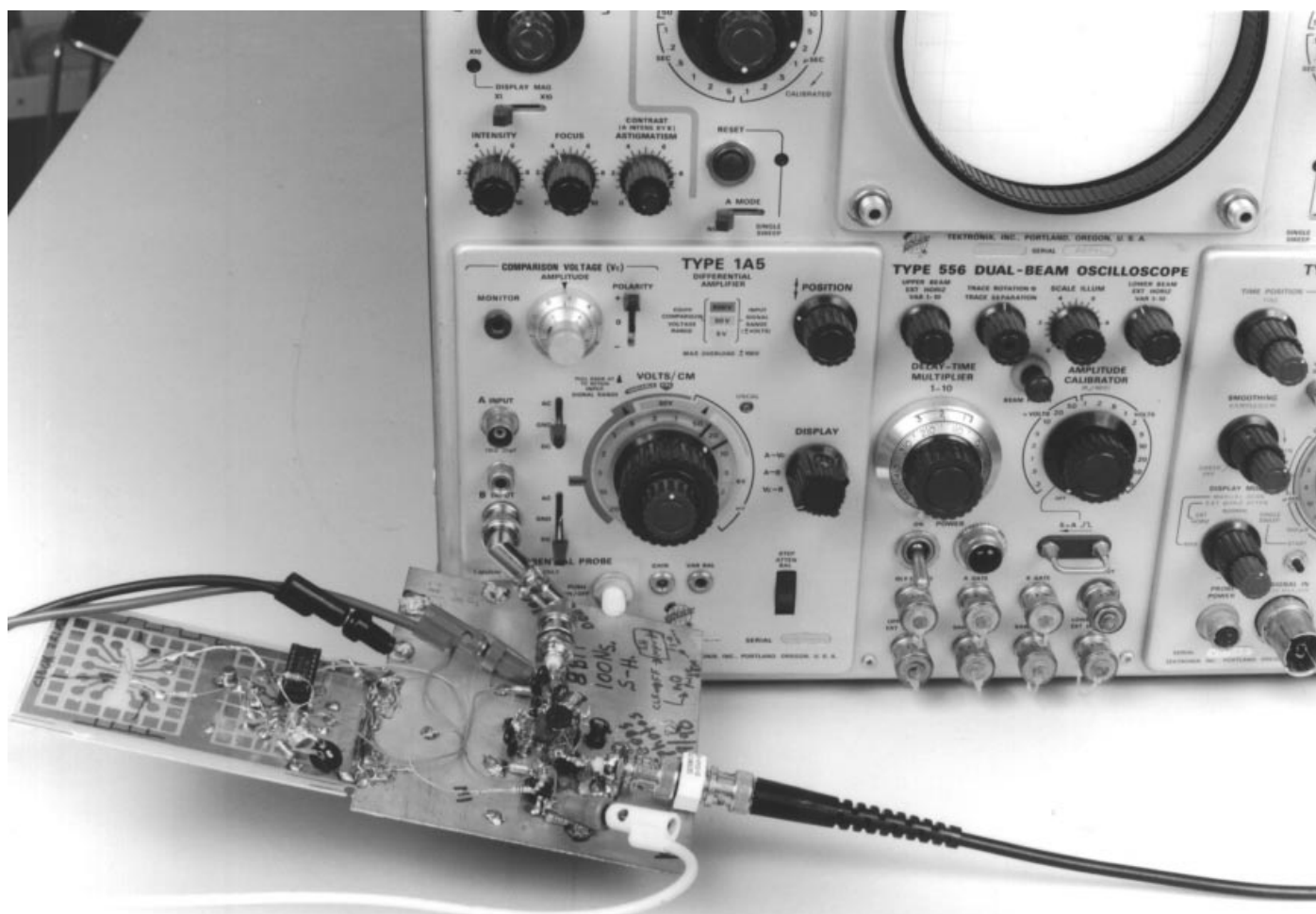
rial presented here, an in-depth treatment of probes appears in Appendix A, "ABC's of Probes", guest written by the engineering staff of Tektronix, Inc.

Probes are the most overlooked cause of oscilloscope mismeasurement. All probes have some effect on the point they are measuring. The most obvious is input resistance, but input capacitance usually dominates in a high speed measurement. Much time can be lost chasing circuit events which are actually due to improperly selected or applied probes. An 8pF probe looking at a 1k Ω source impedance forms an 8ns lag — substantially longer than a fast amplifier's delay time! Pay particular attention to the probe's input capacitance. Standard 10M Ω , 10X probes typically have 8pF-10pF of input capacitance, with 1X types being much higher. In general, 1X probes are not suitable for fast work because their bandwidth is limited to about 20MHz. Remember that all 10X probes cannot be used with all oscilloscopes; the probe's compensation range must match the oscilloscope's input capacitance. Low impedance probes (with 500 Ω to 1k Ω resistance), designed for 50 Ω inputs, usually have input capacitance of 1pF or 2pF. They are a very good choice if you can stand the low resistance. FET probes maintain high input resistance and keep capacitance at the 1pF level but have substantially more delay than passive probes. FET probes also have limitations on input common-mode range which must be adhered to or serious measurement errors will result. Contrary to popular belief, FET probes *do not* have extremely high input resistance — some types are as low as 100k Ω . It is possible to construct a wideband FET probe with very high input impedance, although input capacitance is somewhat higher than standard FET probes. For measurements requiring these characteristics, such a probe is useful. See Appendix E, "An Ultra Fast High Impedance Probe".

Regardless of which type probe is selected remember that they all have bandwidth and rise time restrictions. The displayed rise time on the oscilloscope is the vector sum of source, probe and 'scope rise times.

$$t_{\text{RISE}} = \sqrt{(t_{\text{RISE Source}})^2 + (t_{\text{RISE Probe}})^2 + (t_{\text{RISE Oscilloscope}})^2}$$

This equation warns that some rise time degradation must occur in a cascaded system. In particular, if probe and oscilloscope are rated at the same rise time, the system response will be slower than either.



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Figure 31. Sometimes the Best Probe is No Probe. Direct Connection to the Oscilloscope Eliminates a 10X Probe's Attenuation and Possible Grounding Problems in a Sample-Hold (Figure 124) Settling Time Measurement

Current probes are useful and convenient.³ The passive transformer-based types are fast and have less delay than the Hall effect-based versions. The Hall types, however, respond at DC and low frequency and the transformer types typically roll off around 100Hz to 1kHz. Both types have saturation limitations which, when exceeded, cause odd results on the CRT which will confuse the unwary. The Tektronix type CT-1 current probe, although not nearly as versatile as the clip-on probes, bears mention. Although this is not a clip-on device, it may be the least electrically intrusive way of extracting wideband signal information. Rated at 1GHz bandwidth, it produces 5mV/mA output with only 0.6pF loading. Decay time constant of this AC current probe is $\approx 1\%/50\text{ns}$, resulting in a low frequency limit of 35kHz.

Note 3: A more thorough discussion of current probes is given in LTC Application Note 35, "Step Down Switching Regulators". See Reference 2.

A very special probe is the differential probe. A differential probe may be thought of as two matched FET probes contained within a common probe housing. This probe literally brings the advantage of a differential input oscilloscope to the circuit board. The probes matched, active circuitry provides greatly improved high frequency common mode rejection over single ended probing or even matched passive probes used with a differential amplifier. The resultant ability to reject common-mode signals and ground noise at high frequency allows this probe to deliver exceptionally clean results when monitoring small, fast signals. Figure 32 shows a differential probe being used to verify the waveshape of a 2.5mV input to a wideband, high gain amplifier (Figure 76 of the Applications section).

When using different probes, remember that they all have different delay times, meaning that apparent timing errors

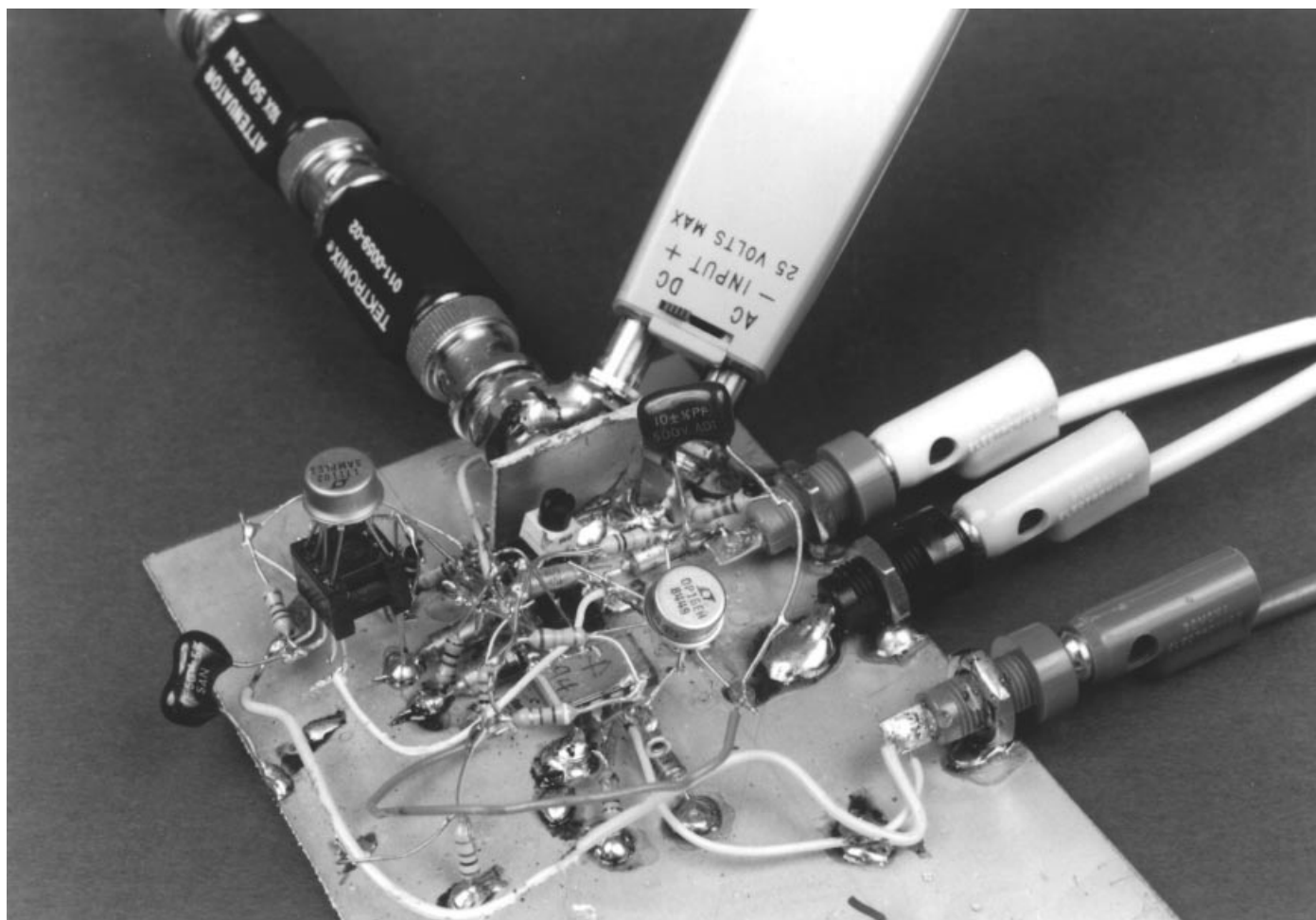
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will occur on the CRT. Know what the individual probe delays are and account for them in interpreting the CRT display.

By far the greatest source of error in probe use is grounding. Poor probe grounding can cause ripples and discontinuities in the waveform observed. In some cases the choice and placement of a probe's ground strap will affect waveforms on another channel. In the worst case, connecting the probe's ground wire will virtually disable the circuit being measured. The cause of these problems is due to parasitic inductance in the probe's ground connection. In most oscilloscope measurements this is not a problem, but at nanosecond speeds it becomes critical. Fast probes are always supplied with a variety of spring clips and accessories designed to aid in making the lowest possible inductive connection to ground. Most of these attachments

assume a ground plane is in use, which it should be. Always try to make the shortest possible connection to ground – anything longer than 1 inch may cause trouble. Sometimes it's difficult to determine if probe grounding is the cause of observed waveform aberrations. One good test is to disturb the grounding set-up and see if changes occur. Nominally, touching the ground plane or jiggling probe ground connectors or wires should have no effect. If a ground strap wire is in use try changing its orientation or simply squeezing it together to change and minimize its loop area. *If any waveform change occurs while doing this the probe grounding is unacceptable, rendering the oscilloscope display unreliable.*

The simple network of Figure 33 shows just how easy it is for poorly chosen or used probes to cause bad results. A 9pF input capacitance probe with a 4 inch long ground



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Figure 32. Using a Differential Probe to Verify the Integrity of a 2.5mV High Speed Input Pulse (Figure 76's X1000 Amplifier)

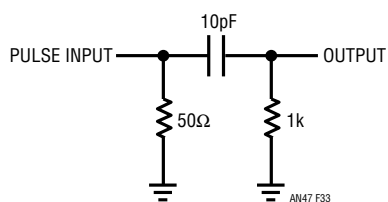


Figure 33. Probe Test Circuit

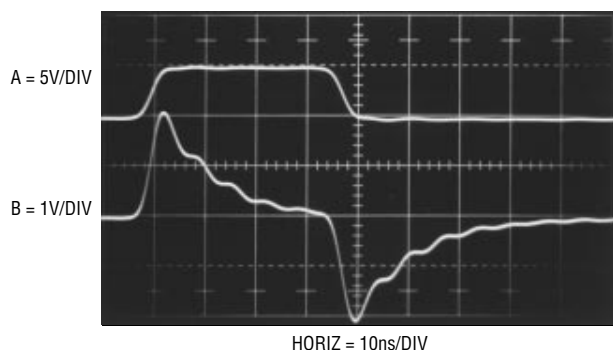


Figure 34. Test Circuit Output with 9pF Probe and 4 Inch Ground Strap

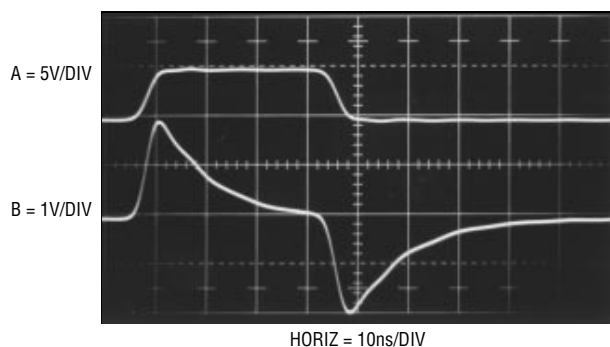


Figure 35. Test Circuit Output with 9pF Probe and 0.25 Inch Ground Strap

strap monitors the output (Trace B, Figure 34). Although the input (Trace A) is clean, the output contains ringing. Using the same probe with a 1/4 inch spring tip ground connection accessory seemingly cleans up everything (Figure 35). However, substituting a 1pF FET probe (Figure 36) reveals a 50% output amplitude error in Figure 35! The FET probe's low input capacitance allows a more accurate version of circuit action. The FET probe does, however, contribute its own form of error. Note that the probe's response is tardy by 5ns due to delay in its active circuitry. Hence, separate measurements with each probe are required to determine the output's amplitude and timing parameters.

A final form of probe is the human finger. Probing the circuit with a finger can accentuate desired or undesired effects, giving clues that may be useful. The finger can be used to introduce stray capacitance to a suspected circuit node while observing results on the CRT. Two fingers, lightly moistened, can be used to provide an experimental resistance path. Some high speed engineers are particularly adept at these techniques and can estimate the capacitive and resistive effects created with surprising accuracy.

Examples of some of the probes discussed, along with different forms of grounding implements, are shown in Figure 37. Probes A, B, E, and F are standard types equipped with various forms of low impedance grounding attachments. The conventional ground lead used on G is more convenient to work with but will cause ringing and

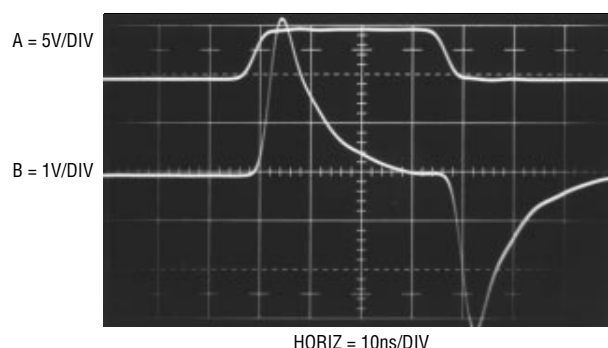


Figure 36. Test Circuit Output with FET Probe

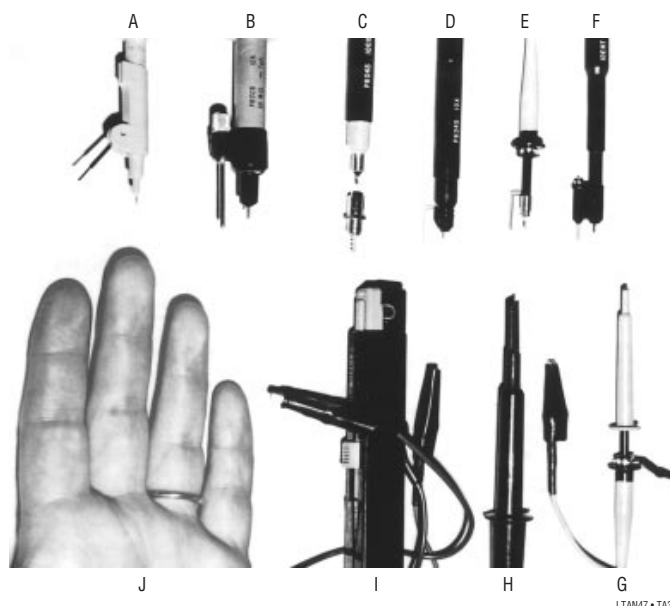


Figure 37. Various Probe-Ground Strap Configurations

other effects at high frequencies, rendering it useless. H has a very short ground lead. This is better, but can still cause trouble at high speeds. D is a FET probe. The active circuitry in the probe and a very short ground connector ensure low parasitic capacitance and inductance. C is a separated FET probe attenuator head. Such heads allow the probe to be used at higher voltage levels (e.g., $\pm 10\text{V}$ or $\pm 100\text{V}$). The miniature coaxial connector shown can be mounted on the circuit board and the probe mated with it. This technique provides the lowest possible parasitic inductance in the ground path and is especially recommended. I is a current probe. A ground connection is not usually required. However, at high speeds the ground connection may result in a cleaner CRT presentation. Because no current flows in the ground lead of these probes, a long strap is usually permissible. J is typical of the finger probes described in the text. Note the ground strap on the third finger.

The low inductance ground connectors shown are available from probe manufacturers and are always supplied with good quality, high frequency probes. Because most oscilloscope measurements do not require them, they invariably become lost. There is no substitute for these devices when they are needed, so it is prudent to take care of them. This is especially applicable to the ground strap on the finger probe.

About Oscilloscopes

The modern oscilloscope is one of the most remarkable instruments ever constructed. The protracted and intense development effort put toward these machines is perhaps equaled only by the fanaticism devoted to timekeeping.⁴ It is a tribute to oscilloscope designers that instruments manufactured over 25 years ago still suffice for over 90% of today's measurements. The oscilloscope-probe combination used in high speed work is the most important equipment decision the designer must make. Ideally, the oscilloscope should have at least 150MHz bandwidth, but

Note 4: In particular, the marine chronometer received ferocious and abundant amounts of attention. See References 4, 5, and 6. For an enjoyable stroll through the history of oscilloscope vertical amplifiers, see Reference 3. See also Reference 41.

Note 5: See Appendix D, "Measuring Probe - Oscilloscope Response", for complete details on this pulse generator.

Note 6: This sequence of photos was shot in my home lab. I'm sorry, but 1GHz is the fastest 'scope in my house.

slower instruments are acceptable if their limitations are well understood. Be certain of the characteristics of the probe-oscilloscope combination. Rise time, bandwidth, resistive and capacitive loading, delay, noise, channel-to-channel feedthrough, overdrive recovery, sweep nonlinearity, triggering, accuracy and other limitations must be kept in mind. High speed linear circuitry demands a great deal from test equipment and countless hours can be saved if the characteristics of the instruments used are well known. Obscene amounts of time have been lost pursuing "circuit problems" which in reality are caused by misunderstood, misapplied or out-of-spec equipment. Intimate familiarity with your oscilloscope is invaluable in getting the best possible results with it. In fact, it is possible to use seemingly inadequate equipment to get good results if the equipment's limitations are well known and respected. All of the circuits in the Applications section involve rise times and delays well above the 100MHz-200MHz region, but 90% of the development work was done with a 50MHz oscilloscope. Familiarity with equipment and thoughtful measurement technique permit useful measurements seemingly beyond instrument specifications. A 50MHz oscilloscope cannot track a 5ns rise time pulse, but it can measure a 2ns delay between two such events. Using such techniques, it is often possible to deduce the desired information. There are situations where no amount of cleverness will work and the right equipment (e.g., a faster oscilloscope) must be used. Sometimes, "sanity-checking" a limited bandwidth instrument with a higher bandwidth oscilloscope is all that is required. For high speed work, brute force bandwidth is indispensable when needed, and no amount of features or computational sophistication will substitute. Most high speed circuitry does not require more than two traces to get where you are going. Versatility and many channels are desirable, but if the budget is limited, spend for bandwidth!

Dramatic differences in displayed results are produced by probe-oscilloscope combinations of varying bandwidths. Figure 38 shows the output of a very fast pulse⁵ monitored with a 1GHz sampling 'scope (Tektronix 556 with 1S1 sampling plug-in). At this bandwidth the 10V amplitude appears clean, with just a small hint of ringing after the falling edge. The rise and fall times of 350ps are suspicious, as the sampling oscilloscope's rise time is also specified at 350ps.⁶

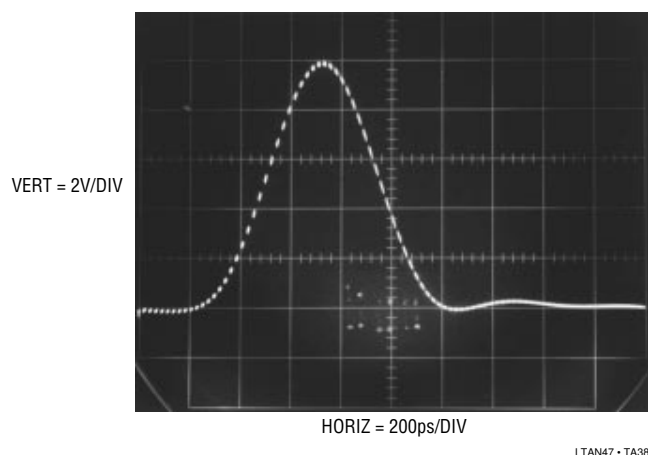


Figure 38. A 350ps Rise/Fall Time 10V Pulse Monitored on 1GHz Sampling Oscilloscope. Direct 50Ω Input Connection is Used

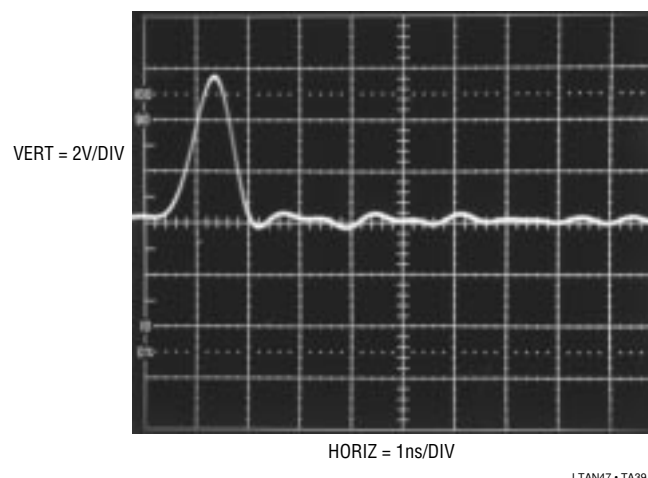


Figure 39. The Test Pulse Appears Smaller and Slower On a 350MHz Instrument ($t_{RISE} = 1ns$). Deliberate Poor Grounding Creates Rippling After the Pulse Falls. Direct 50Ω Connection is Used

Figure 39 shows the same pulse observed on a 350MHz instrument with a direct connection to the input (Tektronix 485/50Ω input). Indicated rise time balloons to 1ns, while displayed amplitude shrinks to 6V, reflecting this instrument's lesser bandwidth. To underscore earlier discussion, poor grounding technique (1 1/2" of ground lead to the ground plane) created the prolonged rippling after the pulse fall.

Figure 40 shows the same 350MHz (50Ω input) oscilloscope with a 3GHz 10X probe (Tektronix P6056). Displayed results are nearly identical, as the probe's high bandwidth contributes no degradation. Again, deliberate poor grounding causes overshoot and rippling on the pulse fall.

Figure 41 equips the same oscilloscope with a 10X probe specified at 290MHz bandwidth (Tektronix P6047). Additionally, the oscilloscope has been switched to its 1MΩ input mode, reducing bandwidth to a specified 250MHz. Amplitude degrades to less than 4V and edge times similarly increase. The deliberate poor grounding contributes the undershoot and underdamped recovery on pulse fall.

In Figure 42 a 100MHz 10X probe (Hewlett-Packard Model 10040A) has been substituted for the 290MHz unit. The oscilloscope and its set-up remain the same. Amplitude shrinks below 2V, with commensurate rise and fall times. Cleaned up grounding eliminates aberrations.

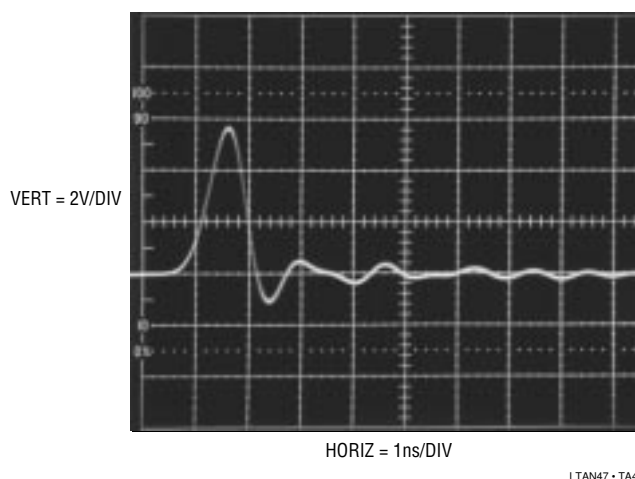


Figure 40. Test Pulse on the Same 350MHz Oscilloscope Using a 3GHz 10X Probe. Deliberate Poor Grounding Maintains Rippling Residue

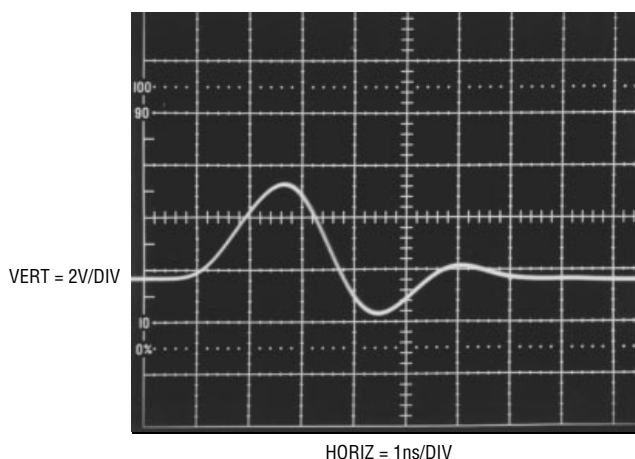


Figure 41. Test Pulse Measures Only 3V High on a 250MHz 'Scope with Significant Waveform Distortion. 250MHz 10X Probe Used

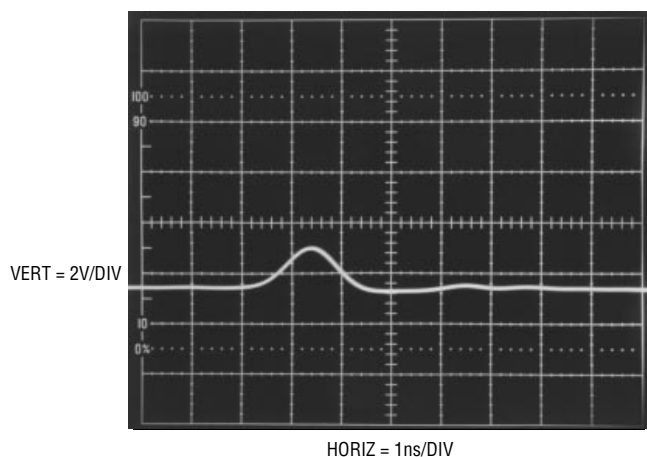


Figure 42. Test Pulse Measures Under 2V High Using 250MHz 'Scope and a 100MHz Probe

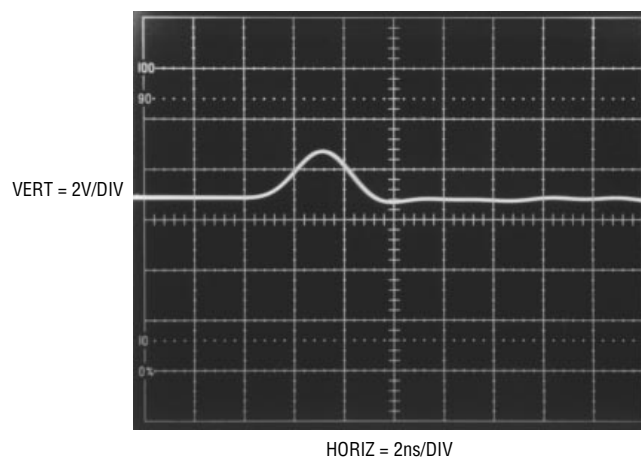


Figure 43. 150MHz Oscilloscope ($t_{RISE} = 2.4ns$) with Direct Connection Responds to the Test Pulse

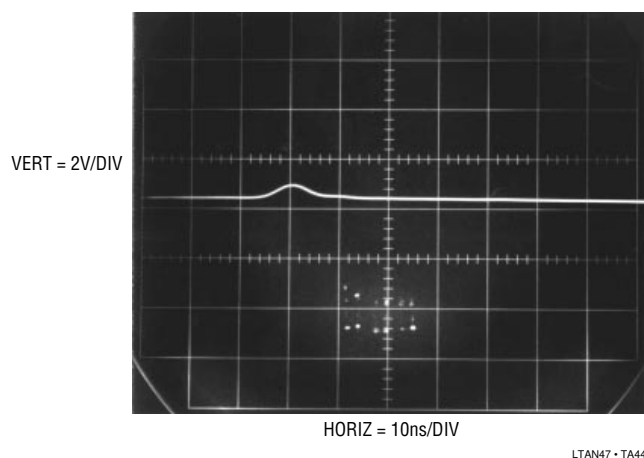


Figure 44. A 50MHz Instrument Barely Grunts. 10V, 350ps Test Pulse Measures Only 0.5V High with 7ns Rise and Fall Times!

A Tektronix 454A (150MHz) produced Figure 43's trace. The pulse generator was directly connected to the input. Displayed amplitude is about 2V, with appropriate 2ns edges. Finally, a 50MHz instrument (Tektronix 556 with 1A4 plug-in) just barely grunts in response to the pulse (Figure 44). Indicated amplitude is 0.5V, with edges reading about 7ns. That's a long way from the 10V and 350ps that's really there!

A final oscilloscope characteristic is overload performance. It is often desirable to view a small amplitude portion of a large waveform. In many cases the oscilloscope is required to supply an accurate waveform after the display has been driven off screen. How long must one wait after an overload before the display can be taken seriously? The answer to this question is quite complex. Factors involved include the degree of overload, its duty cycle, its magnitude in time and amplitude, and other considerations. Oscilloscope response to overload varies widely between types and markedly different behavior can be observed in any individual instrument. For example, the recovery time for a 100X overload at 0.005V/division may be very different than at 0.1V/division. The recovery characteristic may also vary with waveform shape, DC content and repetition rate. With so many variables, it is clear that measurements involving oscilloscope overload must be approached with caution. Nevertheless, a simple test can indicate when the oscilloscope is being deleteriously affected by overdrive.

The waveform to be expanded is placed on the screen at a vertical sensitivity which eliminates all off-screen activity. Figure 45 shows the display. The lower right hand portion is to be expanded. Increasing the vertical sensitivity by a factor of two (Figure 46) drives the waveform off-screen, but the remaining display appears reasonable. Amplitude has doubled and waveshape is consistent with the original display. Looking carefully, it is possible to see small amplitude information presented as a dip in the waveform at about the third vertical division. Some small disturbances are also visible. This observed expansion of the original waveform is believable. In Figure 47, gain has been further increased and all the features of Figure 46 are amplified accordingly. The basic waveshape appears clearer and the dip and small disturbances are also easier to see. No new waveform characteristics are observed. Figure 48 brings some unpleasant surprises. This increase in gain

causes definite distortion. The initial negative-going peak, although larger, has a different shape. Its bottom appears less broad than in Figure 47. Additionally, the peak's positive recovery is shaped slightly differently. A new rippling disturbance is visible in the center of the screen. This kind of change indicates that the oscilloscope is having trouble. A further test can confirm that this waveform is being influenced by overloading. In Figure 49 the gain remains the same, but the vertical position knob has

been used to reposition the display at the screen's bottom. This shifts the oscilloscope's DC operating point which, under normal circumstances, should not affect the displayed waveform. Instead, a marked shift in waveform amplitude and outline occurs. Repositioning the waveform to the screen's top produces a differently distorted waveform (Figure 50). It is obvious that for this particular waveform, accurate results cannot be obtained at this gain.

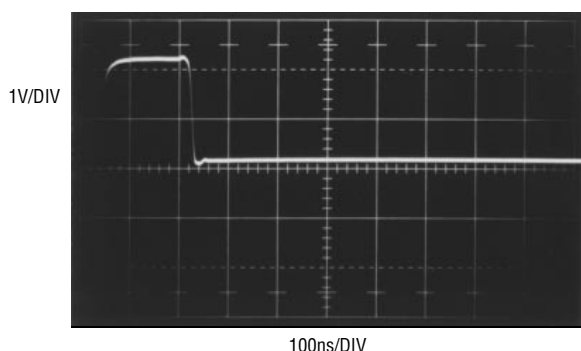


Figure 45

LTAN47 • TA45

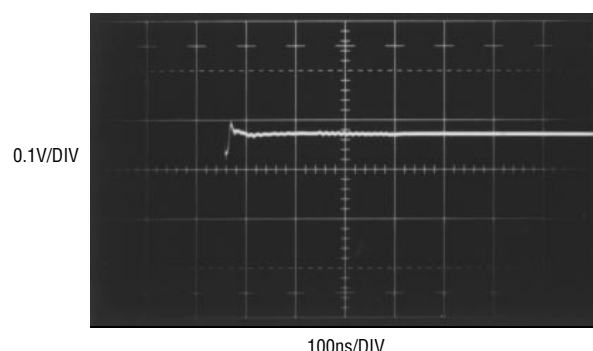


Figure 48

LTAN47 • TA48

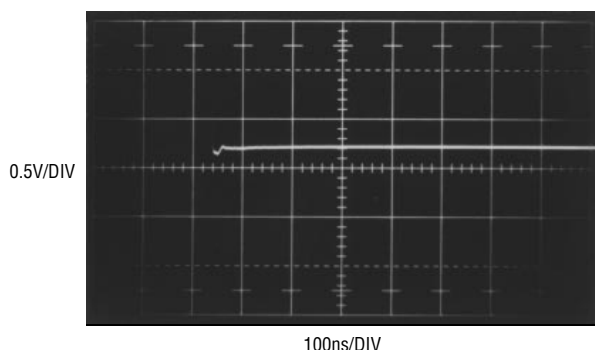


Figure 46

LTAN47 • TA46

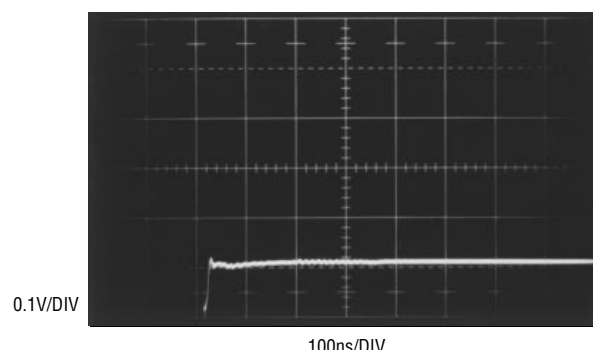


Figure 49

LTAN47 • TA49

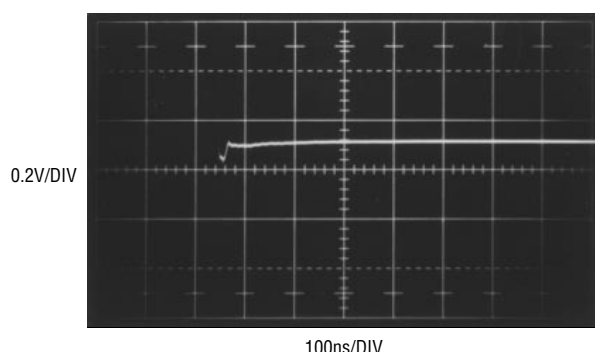


Figure 47

LTAN47 • TA47

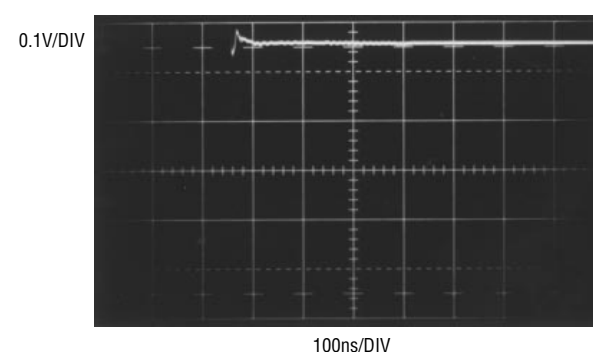


Figure 50

LTAN47 • TA50

Figures 45-50. The Overdrive Limit is Determined by Progressively Increasing Oscilloscope Gain and Watching for Waveform Aberrations

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Differential plug-ins can address some of the issues associated with excessive overdrive, although they cannot solve all problems. Two differential plug-in types merit special mention. At low level, a high sensitivity differential plug-in is indispensable. The Tektronix 1A7, 1A7A and 7A22 feature $10\mu\text{V}$ sensitivity, although bandwidth is limited to 1MHz. The units also have selectable high and low pass filters and good high frequency common-mode rejection. Tektronix type 1A5, W and 7A13 are differential comparators. They have calibrated DC nulling (slideback) sources, allowing observation of small, slowly moving events on top of common-mode DC or fast events riding on a waveform.

A special case is the sampling oscilloscope. By nature of its operation, a sampling 'scope in proper working order is inherently immune to input overload, providing essentially instantaneous recovery between samples. Appendix B, "Measuring Amplifier Settling Time", utilizes this capability. See Reference 8 for additional details.

The best approach to measuring small portions of large waveforms, however, is to eliminate the large signal swing seen by the oscilloscope. Appendix B, "Measuring Amplifier Settling Time" shows ways to do this when measuring DAC-amplifier settling time to very high accuracy at high speed.

In summary, while the oscilloscope provides remarkable capability, its limitations must be well understood when interpreting results.⁷

About Ground Planes

Many times in high frequency circuit layout, the term "ground plane" is used, most often as a mystical and ill-defined cure to spurious circuit operation. In fact, there is little mystery to the usefulness and operation of ground planes, and like many phenomena, their fundamental operating principle is surprisingly simple.

Ground planes are primarily useful for minimizing circuit inductance. They do this by utilizing basic magnetic theory. Current flowing in a wire produces an associated magnetic field. The field's strength is proportional to the current and inversely related to the distance from the conductor. Thus,

we can visualize a wire carrying current (Figure 51) surrounded by radii of magnetic field. The unbounded field becomes smaller with distance. A wire's inductance is defined as the energy stored in the field set up by the wire's current. To compute the wire's inductance requires integrating the field over the wire's length and the total radial area of the field. This implies integrating on the radius from $R = R_W$ to infinity, a very large number. However, consider the case where we have two wires in space carrying the same current in either direction (Figure 52). The fields produced cancel.

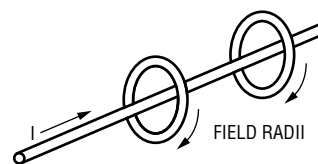


Figure 51. Single Wire Case

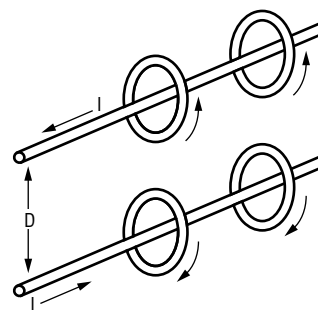


Figure 52. Two Wire Case

In this case, the inductance is much smaller than in the simple wire case and can be made arbitrarily smaller by reducing the distance between the two wires. This reduction of inductance between current carrying conductors is the underlying reason for ground planes. In a normal circuit, the current path from the signal source through its conductor and back to ground includes a large loop area. This produces a large inductance for this conductor which can cause ringing due to LRC effects. It is worth noting that 10nH at 100MHz has an impedance of 6Ω . At 10mA a 60mV drop results.

A ground plane provides a return path directly under the signal carrying conductor through which return current can flow. The conductor's small physical separation means the inductance is low. Return current has a direct path to ground, regardless of the number of branches associated with the conductor. Currents will always flow through the return path of lowest impedance. In a properly designed

Note 7: Additional discourse on oscilloscopes will be found in References 1 and 7 through 11.

ground plane, this path is directly under the signal conductor. In a practical circuit, it is desirable to ground plane one whole side of the PC card (usually the component side for wave solder considerations) and run the signal conductors on the other side. This will give a low inductance path for all the return currents.

Aside from minimizing parasitic inductance, ground planes have additional benefits. Their flat surface minimizes resistive losses due to AC skin effect (AC currents travel along a conductor's surface). Additionally, they aid the circuit's high frequency stability by referring stray capacitances to ground.

Some practical hints for ground planes are:

1. Ground plane as much area as possible on the component side of the board, especially under traces that operate at high frequency.
2. Mount components that conduct substantial fast rise currents (termination resistors, ICs, transistors, decoupling capacitors) as close to the board as possible.
3. Where common ground potential is important (i.e., at comparator inputs), try to single point the critical components into the ground plane to avoid voltage drops.

For example, in Figure 53's common A/D circuit, good practice would dictate that grounds 2, 3, 4 and 6 be as close to single point as possible. Fast, large currents must flow through R1, R2, D1 and D2 during the DAC settle time. Therefore, D1, D2, R1 and R2 should be mounted close to the ground plane to minimize their inductance. R3 and C1 don't carry any current, so their inductance is less important; they could be vertically

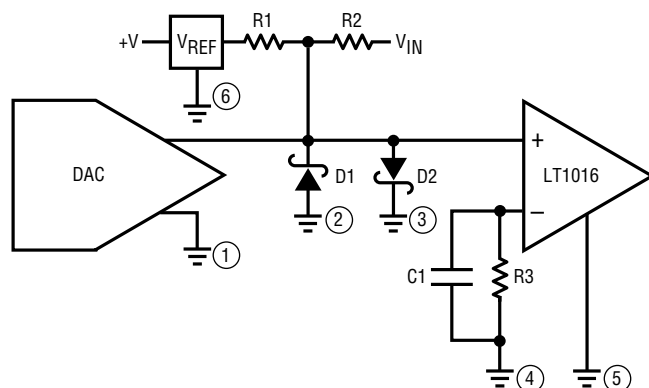


Figure 53. Typical Grounding Scheme

inserted to save space and to allow point 4 to be single point common with 2, 3 and 6. In critical circuits, the designer must often trade off the beneficial effects of lowered inductance versus the loss of single point ground.

4. Keep trace length short. Inductance varies directly with length and no ground plane will achieve perfect cancellation.

About Bypass Capacitors

Bypass capacitors are used to maintain low power supply impedance at the point of load. Parasitic resistance and inductance in supply lines mean that the power supply impedance can be quite high. As frequency goes up, the inductive parasitic becomes particularly troublesome. Even if these parasitic terms did not exist, or if local regulation is used, bypassing is still necessary because no power supply or regulator has zero output impedance at 100MHz. What type of bypass capacitor to use is determined by the application, frequency domain of the circuit, cost, board space and many other considerations. Some useful generalizations can be made.

All capacitors contain parasitic terms, some of which appear in Figure 54. In bypass applications, leakage and dielectric absorption are second order terms but series R and L are not. These latter terms limit the capacitor's ability to damp transients and maintain low supply impedance. Bypass capacitors must often be large values so they can absorb long transients, necessitating electrolytic types which have large series R and L.

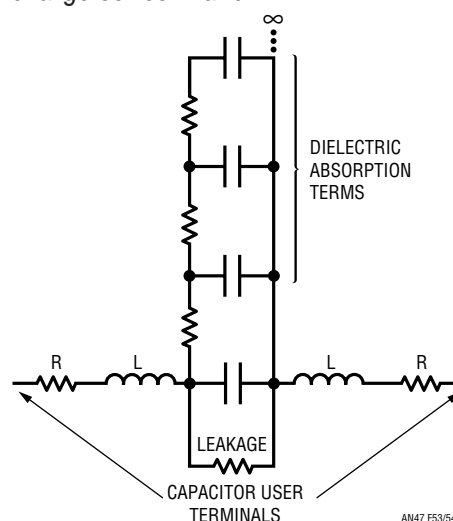


Figure 54. Parasitic Terms of a Capacitor

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Different types of electrolytics and electrolytic-non-polar combinations have markedly different characteristics. Which type(s) to use is a matter of passionate debate in some circles and the test circuit (Figure 55) and accompanying photos are useful. The photos show the response of 5 bypassing methods to the transient generated by the test circuit. Figure 56 shows an unbypassed line which sags and ripples badly at large amplitudes. Figure 57 uses an aluminum $10\mu\text{F}$ electrolytic to considerably cut the disturbance, but there is still plenty of potential trouble. A tantalum $10\mu\text{F}$ unit offers cleaner response in Figure 58 and the $10\mu\text{F}$ aluminum combined with a $0.01\mu\text{F}$ ceramic type is even better in Figure 59. Combining electrolytics with non-polarized capacitors is a popular way to get good response but beware of picking the wrong duo. The right (wrong) combination of supply line parasitics and paralleled dissimilar capacitors can produce a resonant, ringing response, as in Figure 60. Caveat!

Breadboarding Techniques

The breadboard is both the designer's playground and proving ground. It is there that Reality resides, and paper (or computer) designs meet their ruler. More than anything else, breadboarding is an iterative procedure, an odd amalgam of experience guiding an innocent, ignorant,

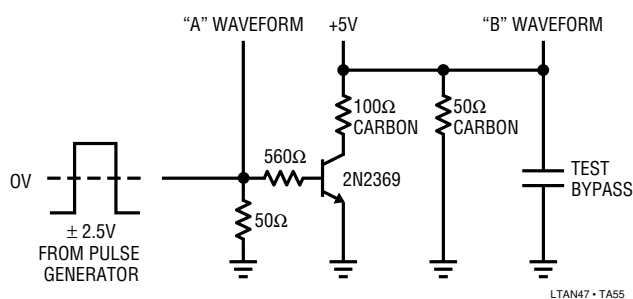


Figure 55. Bypass Capacitor Test Circuit

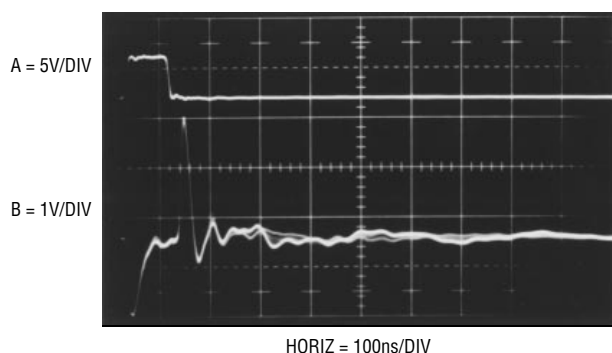


Figure 56. Response of Unbypassed Line

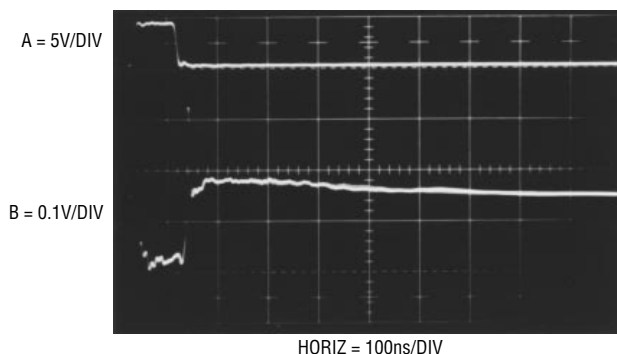


Figure 57. Response of $10\mu\text{F}$ Aluminum Capacitor

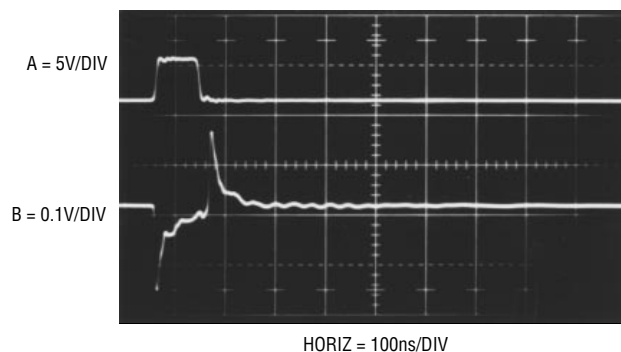


Figure 58. Response of $10\mu\text{F}$ Tantalum Capacitor

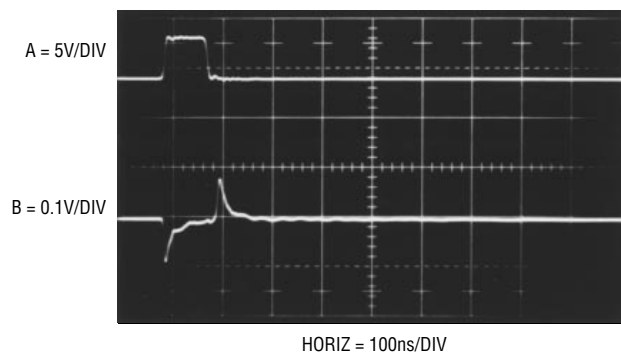


Figure 59. Response of $10\mu\text{F}$ Aluminum Paralleled by $0.01\mu\text{F}$ Ceramic

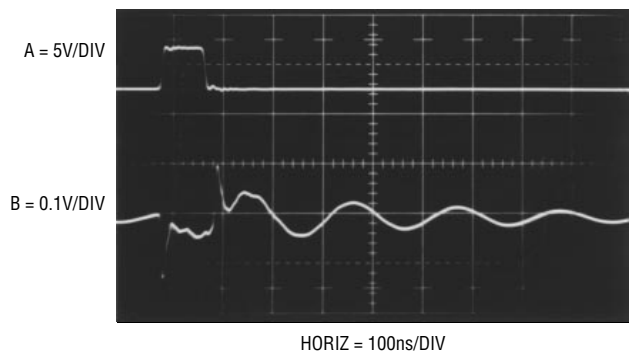


Figure 60. Some Paralleled Combinations can Ring. Try before Specifying!

explorative spirit. A key is to be willing to try things out, sometimes for not very good reasons. Invent problems and solutions, guess carefully and wildly, throw rocks and see what comes loose. Invent and design experiments, and follow them wherever they lead. Reticence to try things is probably the number one cause of breadboards that “don’t work”.⁸ Implementing the above approach to life begins with the physical construction methods used to build the breadboard.

A high speed breadboard must start with a ground plane. Additionally, bypassing, component layout and connections should be consistent with high speed operations. Because of these considerations there is a common misconception that breadboarding high speed circuits is time consuming and difficult. This is simply not true. For high speed circuits of moderate complexity a complete and electrically correct breadboard can be assembled in 10 minutes if all necessary components are on hand. The key to rapid breadboarding is to identify critical circuit nodes and design the layout to suit them. This permits most of the breadboard’s construction to be fairly sloppy, saving time and effort. Additionally, use all degrees of freedom in making connections and mounting components. Don’t be bashful about bending I.C. pins to suit desired low capacitance connections, or air wiring components to achieve rapid or electrically optimum layout. Save time by using components, such as bypass capacitors, as mechanical

supports for other components, such as amplifiers. It is true that eventual printed circuit construction is required, but when initially breadboarding forget about PC and production constraints. Later, when the circuit works, and is well understood, PC adaptations can be taken care of.

Figure 61’s amplifier circuit is a good working example. This circuit, excerpted from the Applications section (where its electrical operation is more fully explained) is a high impedance, wideband amplifier with low input capacitance. Q1 and A1 form the high frequency path, with the 900Ω-100Ω feedback divider setting gain. A2 and Q2 close a DC stabilization loop, minimizing DC offset between the circuit’s input and output. Critical nodes in this circuit include Q1’s gate (because of the desired low input capacitance) and A1’s input related connections (because of their high speed operation). Note that the connections associated with A2 serve at DC and are much less sensitive to layout. These determinations dominate the breadboard’s construction.

Figure 62 shows initial breadboard construction. The copper clad board is equipped with banana type connectors. The connector’s mounting nuts are simply soldered to the clad board, securing the connectors. Figure 63 adds A1 and the bypass capacitors. Observe that A1’s leads have been bent out, permitting the amplifier to sit down on the ground plane, minimizing parasitic capacitance. Also, the bypass capacitors are soldered to the amplifier power pins right at the capacitor’s body. The capacitor’s lead lengths are returned to the banana power jacks. This connection method provides good amplifier bypassing

Note 8: A much more eloquently stated version of this approach is found in Reference 12.

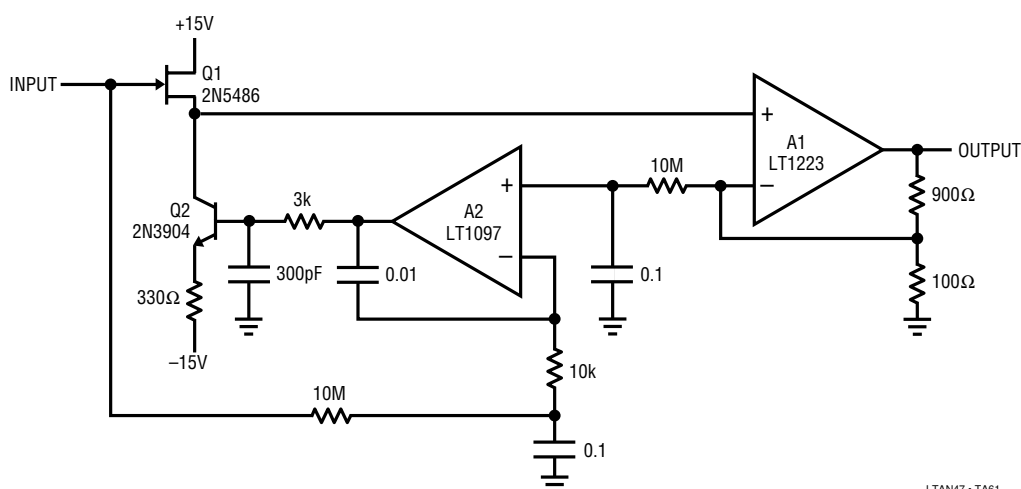


Figure 61. The Stabilized FET Input Amplifier (Applications Figure 73) to be Breadboarded

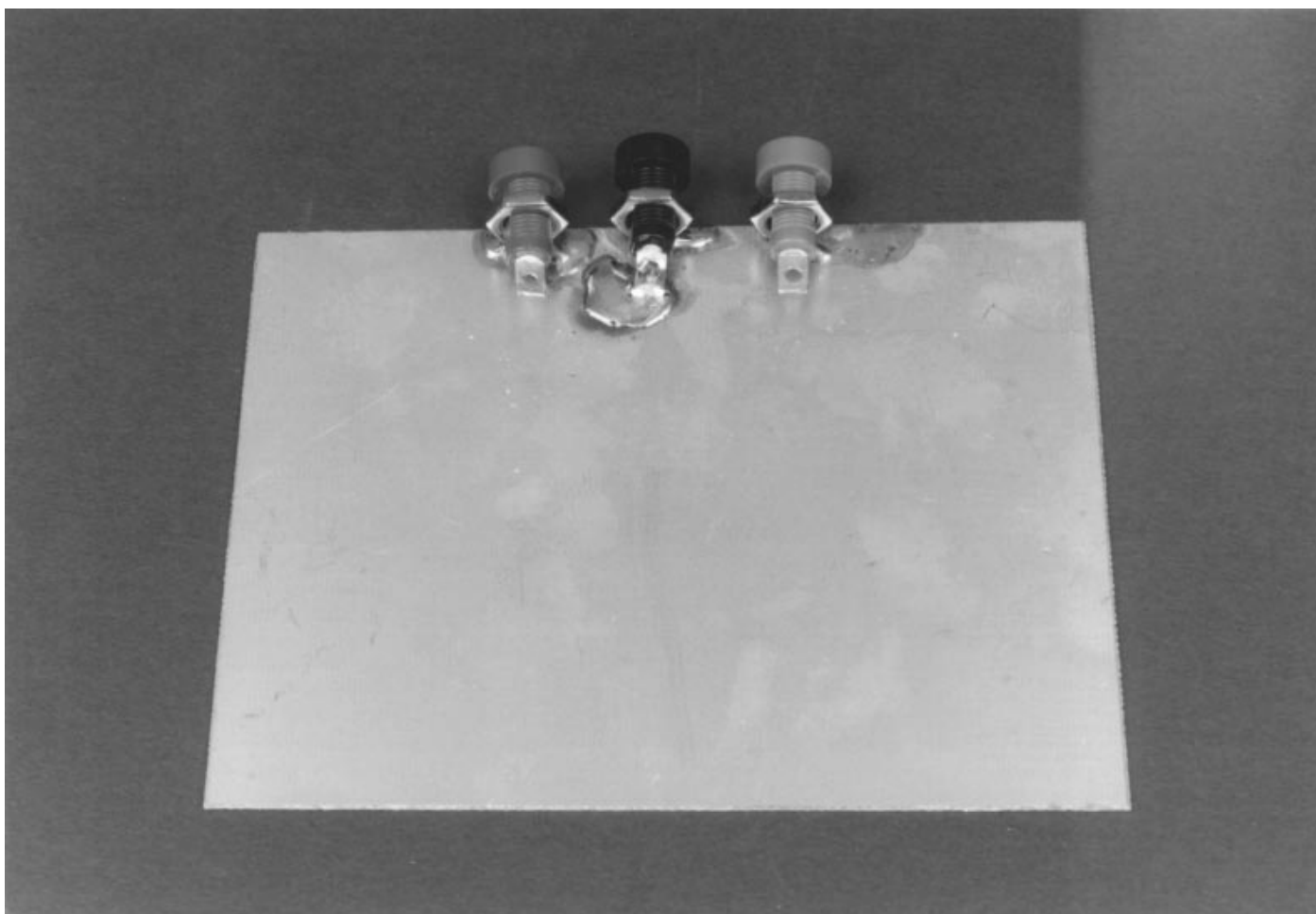


Figure 62. The Banana Jacks are Soldered to the Copper Clad Board

LTAN47 • TA62

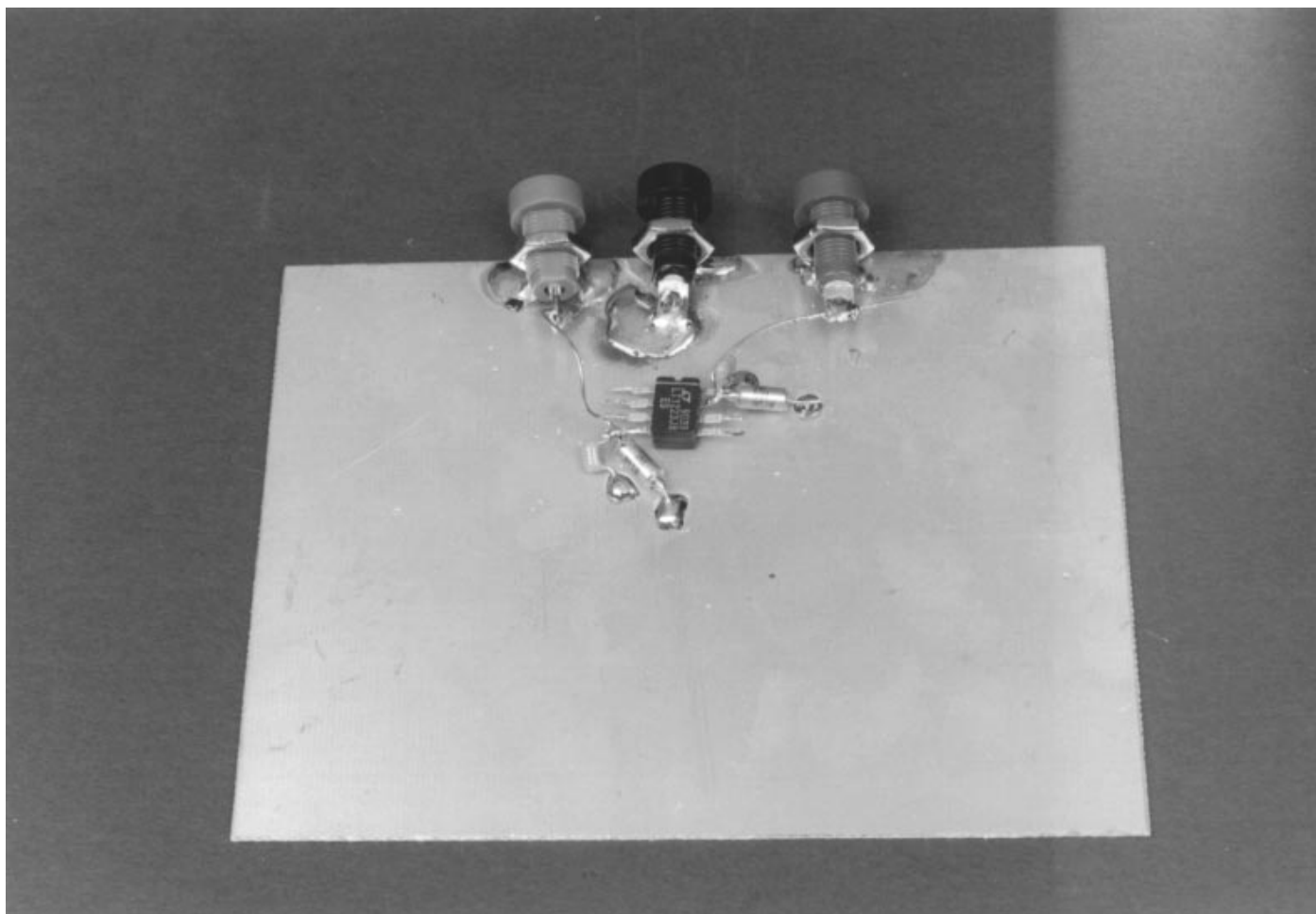
while mechanically supporting the amplifier. It also eliminates separate wire runs to the power pins.

Figure 64 adds the discrete components in the high speed path. Q1's gate is connected directly to the input BNC, as is the $10\text{M}\Omega$ resistor associated with A2's negative input. Note that the end of this resistor that sees high frequency is cut very short, while the other end is left uncut. The 900Ω - 100Ω divider is installed at A1, with very short connections to A1's negative input. A1's $10\text{M}\Omega$ resistor receives similar treatment to the BNC connected $10\text{M}\Omega$ unit; the high frequency end is cut short, while the end destined for connection to A2 remains uncut. Q2's collector and Q1's source, high speed points, are tied closely together with A1's positive input.

Finally, DC amplifier A2 and its associated components are air wired into the breadboard (Figure 65). Their DC opera-

tion permits this, while the construction technique makes connections to the previously wired nodes easy. The previously uncommitted ends of the $10\text{M}\Omega$ resistors may be bent in any way necessary to make connections. All other components associated with A2 receive similar treatment and the circuit is ready for experimentation.

Despite the breadboard's seemingly haphazard construction, the circuit worked well. Input capacitance measured a few pF (including BNC connector) with bias current of about 100pA . Slew rate was $1000\text{V}/\mu\text{s}$, with bandwidth approaching 100MHz . Output, even with 50mA loading, was clean, with no sign of oscillation or other instabilities. Full details on this circuit appear in the Applications section. Additional examples of breadboard construction techniques appear in Appendix F, "Additional Comments on Breadboarding".



LTAN47 • TA63

Figure 63. High Speed Amplifier A1 is Connected to Power. Bypass Capacitors Provide Support. Bending Amplifier Pins Eases Connections and Minimizes Distance to the Ground Plane

Once the breadboard seems to work, it's useful to begin thinking about PC layout and component choice for production. Experiment with the existing layout to determine just how sensitive nominally critical points are. Add controlled parasitic terms (e.g., resistors, capacitors and physical layout changes) to test for sensitivity. Gentle touching of suspect points with a finger can yield preliminary indication of sensitivity, giving clues that can be quite valuable.

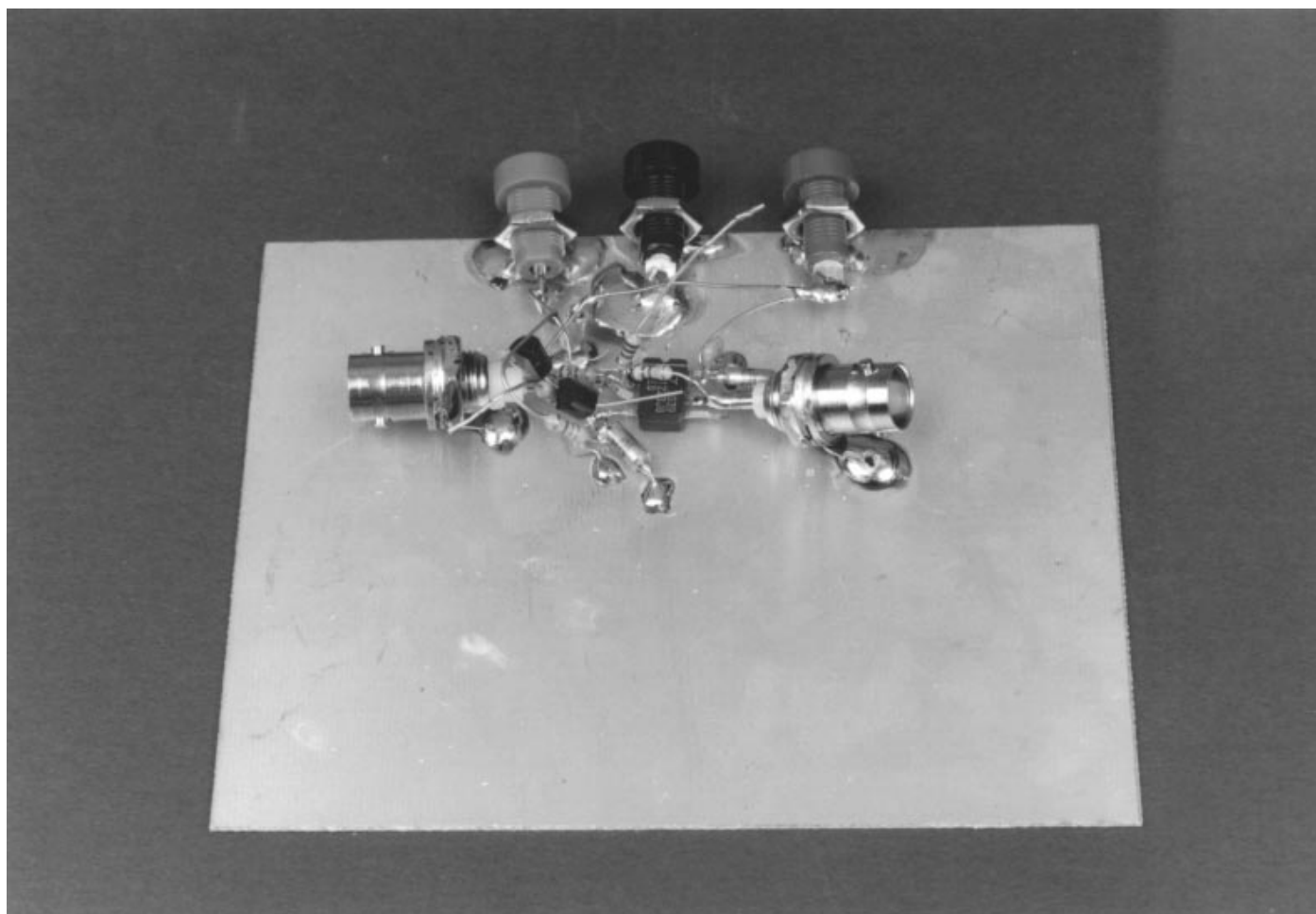
In conclusion, when breadboarding, design the breadboard to be quick and easy to build, work with and modify. Observe the circuit and listen to what it is telling you before trying to get it to some desired state. Finally, don't hesitate to try just about anything; that's what the breadboard is for. Almost anything you do will cause some result —

whether it's good or bad is almost irrelevant. Anything you do that enhances your ability to correlate events occurring on the breadboard can only be beneficial.

Oscillation

The forte of the operational amplifier is negative feedback. It is feedback which stabilizes the operating point and fixes the gain. However, positive feedback or delayed negative feedback can cause oscillation. Thus, a properly functioning amplifier constantly lives in the shadow of oscillation.

When oscillation occurs, several major candidates for blame are present. Power supply impedance must be low. If the supply is unbypassed, the impedance the amplifier sees at its power terminals is high, particularly at high



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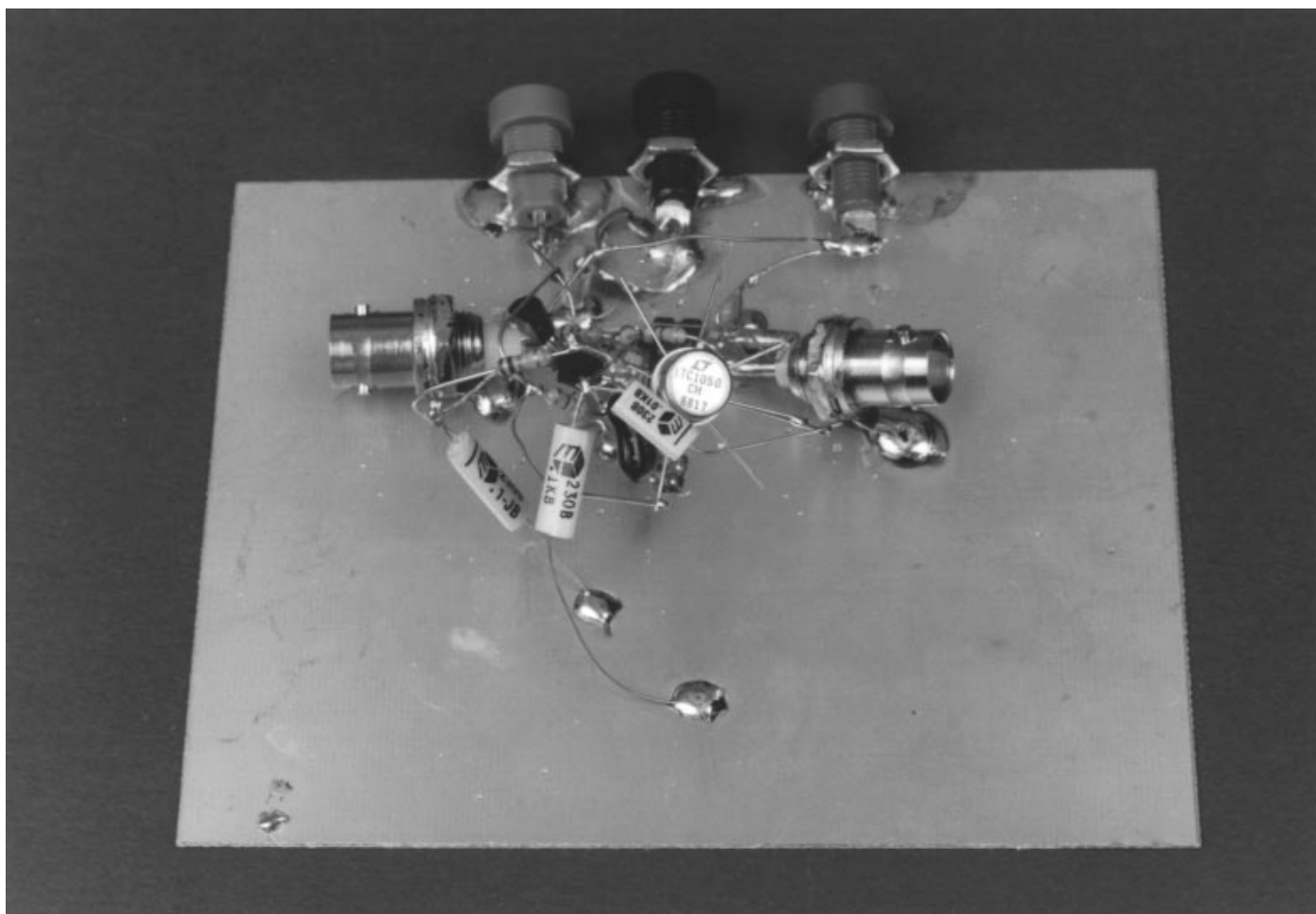
Figure 64. Additional High Speed Discrete Components and Connectors are Added. Note Short Connections at Amplifier Input Pins (Left Side of Package). 10M Resistors Uncommitted Ends are Just Visible

frequency. This impedance forms a voltage divider with the amplifier, allowing the supply to move as internal conditions in the amplifier change. This can cause local feedback and oscillation occurs. The obvious cure is to bypass the amplifier.

A second common cause of oscillation is positive feedback. In most amplifier circuits feedback is negative, although controlled amounts of positive feedback may be used. In a circuit that nominally has only negative feedback unintended positive feedback may occur with poor layout. Check for possible parasitic feedback paths and unwanted or overlooked feedback action. Always minimize (to the extent possible) impedances seen by amplifier inputs. This helps attenuate the effects of parasitic feedback paths to the inputs. Similarly, minimize exposed input trace area. Route amplifier outputs and other signals well away from

sensitive nodes. Sometimes no amount of layout finesse will work and shielding is required. Use shielding only when required — extensive shielding is a sloppy substitute for good layout practice.

A final cause of oscillation is negative feedback arriving well delayed in time. Under these conditions the amplifier hopelessly tries to servo a feedback signal which consistently arrives too late. The servo action takes the form of an electronic tail chase, with oscillation centered around the ideal servo point. The most common causes of this problem are reactive loading of the amplifier (most notably capacitive loads such as cable) and circuitry, such as power amplifiers, placed within the amplifier's feedback path. Reactive loads should be isolated from the amplifier's output (and feedback path) with a resistor or power amplifier. Sometimes rolling off the amplifier's frequency



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Figure 65. DC Servo Amplifier is Wired In and Connections to 10M Resistors Completed. This Part of the Circuit is Not Layout Sensitive

response will fix the problem, but in high speed circuits this may not be an option.

Placing power gain or other type stages within the amplifier's feedback path adds time delay to the stabilizing feedback. If the delay is significant, oscillation commences. Stages operating within the amplifier's loop must contribute minimum time lag compared to the amplifier's speed capability. At lower speeds this is not too difficult, but something destined for operation within a 100MHz amplifier's loop must be *fast*. As mentioned before, rolling off the amplifier's frequency response eases the job, but is usually undesirable in a wideband circuit. Every effort should be expended to maximize the added stages bandwidth before resorting to roll-off of the amplifier. In this way the fastest overall bandwidth is achieved while maintaining stability. Appendix C, "The Oscillation Problem –

Frequency Compensation Without Tears", discusses considerations surrounding operating power gain and other type stages within amplifier loops.

This completes the tutorial section. Hopefully, several notions have been imparted. First, in any measurement situation, test equipment characteristics are an integral part of the circuit. At high speed and high precision this is particularly the case. As such, it is imperative to know your equipment and how it works. There is no substitute for intimate familiarity with your tool's capabilities and limitations.⁹

In general, use equipment you trust and measurement techniques you understand. Keep asking questions and

Note 9: Further exposition and *kvetching* on this point is given in Reference 13.

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don't be satisfied until everything you see on the oscilloscope is accounted for and makes sense.

Fast monolithic amplifiers, combined with the precautionary notes listed above, permit fast linear circuit functions which are difficult or impractical using other approaches. Some of the applications presented represent the state-of-the-art for a particular circuit function. Others show simplified and/or improved ways to implement standard functions by utilizing the amplifier's easily accessed speed. All have been carefully (and painfully) worked out and should serve as good idea sources for potential users of the device. Have fun. I did.

APPLICATIONS SECTION I - AMPLIFIERS

Fast 12-Bit Digital-to-Analog Converter (DAC) Amplifier

One of the most common applications for a high speed amplifier, transforming a 12-bit DAC's current output into a voltage, is also one of the most difficult. Although an op amp can easily do this, care is required to obtain good dynamic performance. A fast DAC can settle to 0.01% in 200ns or less, but its output also includes a parasitic capacitance term, making the amplifier's job more difficult. Normally, the DAC's current output is unloaded directly into the amplifier's summing junction, placing the parasitic capacitance from ground to the amplifier's input. The capacitance introduces feedback phase shift at high frequencies, forcing the amplifier to hunt and ring about the final value before settling. Different DACs have different values of output capacitance. CMOS DACs have the highest output capacitance, in the 100pF-150pF range, and it varies with code. Bipolar DACs typically have 20pF-30pF of capacitance, stable over all codes. As such, bipolar DACs are almost always used where high speed is required. Figure 66 shows the popular AD565A 12-bit DAC with an LT1220 output op amp. Figure 67 shows clean 0.01% settling in 280ns (Trace B) to an all-bits-on input step (Trace A). The requirements for obtaining Trace B's display are not trivial, and are fully detailed in Appendix B, "Measuring Amplifier Settling Time".

2-Channel Video Amplifier

Figure 68 shows a simple way to multiplex two video amplifiers onto a single 75Ω cable. The appropriate ampli-

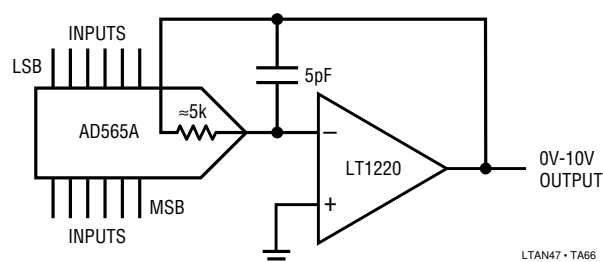


Figure 66. Typical Output Amplifier Configuration for a 12-Bit D-to-A Converter

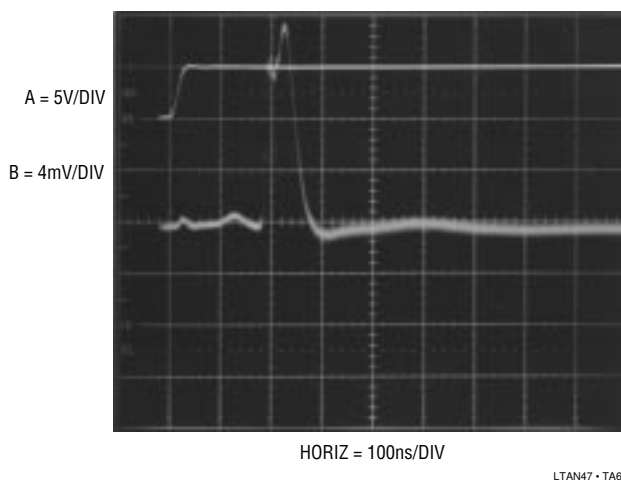


Figure 67. Settling Residue (Trace B) for All Bits Switched On (Trace A). Output is Fully Settled in 280ns

fier is activated in accordance with the truth table in the figure¹⁰. Amplifier performance includes 0.02% differential gain error and 0.1° differential phase error. The 75Ω back termination looking into the cable means the amplifiers must swing 2Vp-p to produce 1Vp-p at the cable output, but this is easily handled.

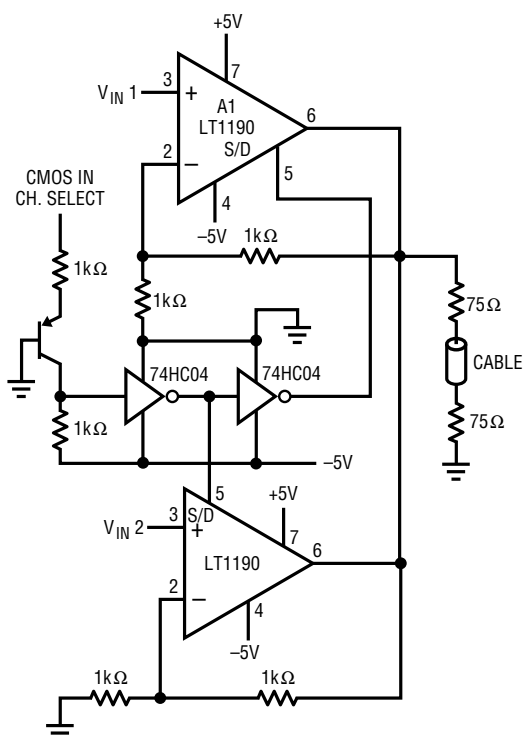
Simple Video Amplifier

Figure 69 is a simpler version of Figure 68. This is a single channel video amplifier, arranged (in this case) for a gain of ten. The double cable termination is retained and the circuit delivers a bandwidth of 55MHz.

Loop Through Cable Receivers

Figure 70 is another cable related circuit. Here, the LT1193 differential amplifier simply hangs across a distribution cable, extracting the signal. The amplifier's true differential inputs reject common-mode signals. As in the previous

Note 10: A truth table in an op amp circuit! Et tu, LTC!!



TRUTH TABLE

INPUT SELECT	A1 OUTPUT	A2 OUTPUT
5V	ACTIVE	INACTIVE
0V	INACTIVE	ACTIVE

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Figure 68. 2-Channel Multiplexed Video Amplifier

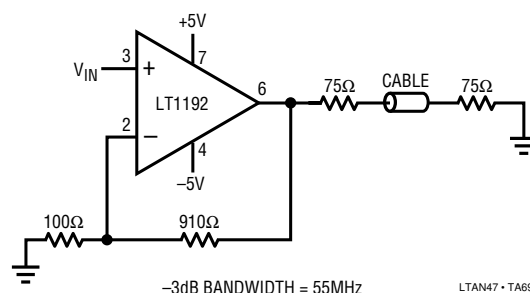
circuit, differential gain and phase errors measure 0.2% and 0.1°, respectively. A separate input permits DC level adjustment.

DC Stabilization – Summing Point Technique

Often it is desirable to obtain the precision offset of a DC amplifier with the bandwidth of a fast device. There are a variety of techniques for doing this. Which method is best is heavily application dependent, so several configurations are presented.

Figure 71 shows a composite made up of an LT1097 low drift device and an LT1191 high speed amplifier. The overall circuit is a unity gain inverter with the summing node located at the junction of the two 1k resistors. The LT1097 monitors this summing node, compares it to ground and drives the LT1191's positive input, completing a DC stabilizing loop around the LT1191. The 100kΩ-0.01μF time constant at the LT1097 limits its response to

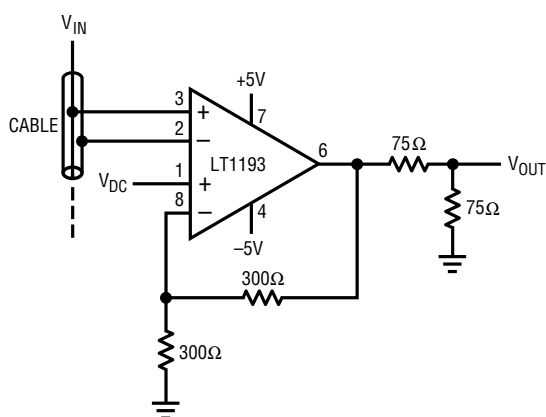
low frequency signals. The LT1191 handles high frequency inputs while the LT1097 stabilizes the DC operating point. The 4.7k-220Ω divider at the LT1191 prevents excessive input overdrive during start-up. This circuit combines the LT1097's 35μV offset and 1.5V/°C drift with the LT1191's 450V/μs slew rate and 90MHz bandwidth. Bias current, dominated by the LT1191, is about 500nA.



-3dB BANDWIDTH = 55MHz

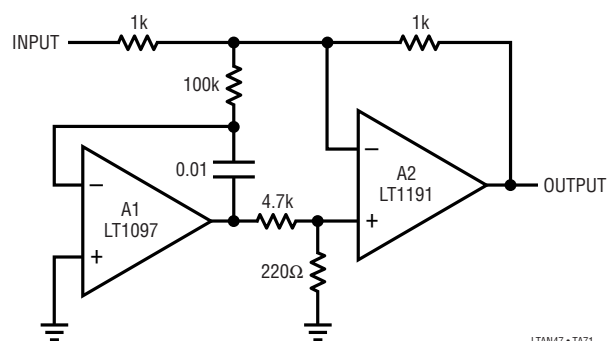
LTAN47 • TA69

Figure 69. Double Terminated Cable Driver



LTAN47 • TA70

Figure 70. Cable Sense Amplifier for Loop Through Connections with DC Adjust



LTAN47 • TA71

Figure 71. A1 DC Stabilizes A2 by Forcing the Summing Point to Zero

Application Note 47

DC Stabilization – Differentially Sensed Technique

Figure 72 is similar to Figure 71, except that the sensing is done differentially, preserving access to both fast amplifier inputs. The LT1097 measures the DC error at the LT1220's input terminals and biases its offset pins to force offset within $50\mu\text{V}$. The offset pin biasing at the LT1220 is arranged so the LT1097 will always be able to find the servo point. The $0.01\mu\text{F}$ capacitor rolls off the LT1097 at low frequency and the LT1220 handles high frequency signals. The combined characteristics of these amplifiers yield the following performance:

Offset Voltage $50\mu\text{V}$
Offset Drift $1\mu\text{V}/^\circ\text{C}$
Slew Rate $250\text{V}/\mu\text{s}$
Gain-Bandwidth 45MHz

DC Stabilization – Servo Controlled FET Input Stage

Figure 73 shows a wideband, highly stable gain-of-ten with high input impedance. Input capacitance is about 3pF . Q1 and Q2 constitute a simple, high speed FET input buffer. Q1 functions as a source follower, with the Q2 current source load setting the drain-source channel current. The LT1223 provides a 100MHz bandwidth gain of ten. Normally, this open loop configuration would be quite drifts because there is no DC feedback. The LT1097 contributes this function to stabilize the circuit. It does this by comparing the filtered circuit output to a similarly

filtered version of the input signal. The amplified difference between these signals is used to set Q2's bias, and hence Q1's channel current. This forces Q1's V_{GS} to whatever voltage is required to match the circuit's input and output potentials. The capacitor at A1 provides stable loop compensation. The RC network in A1's output prevents it from seeing high speed edges coupled through Q2's collector-base junction.

This circuit constitutes an extremely wideband (Q1 does not degrade A2's 100MHz performance), high input impedance amplifier. With an input capacitance of 3pF and

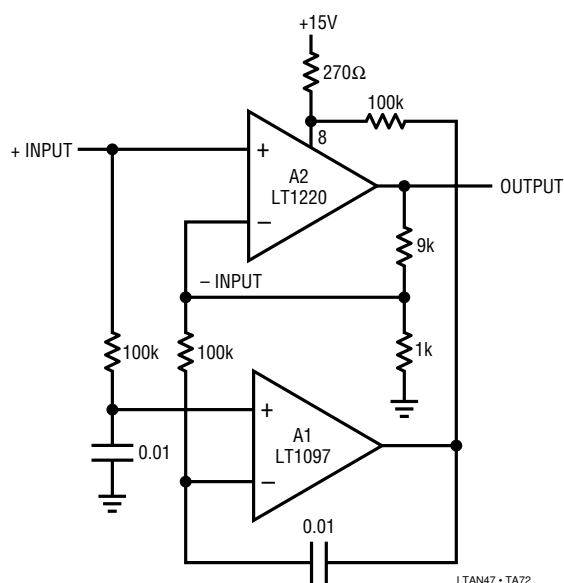


Figure 72. A1 DC Stabilizes A2 by Forcing the Offset Pins to Produce a 0V Difference at A2's Inputs

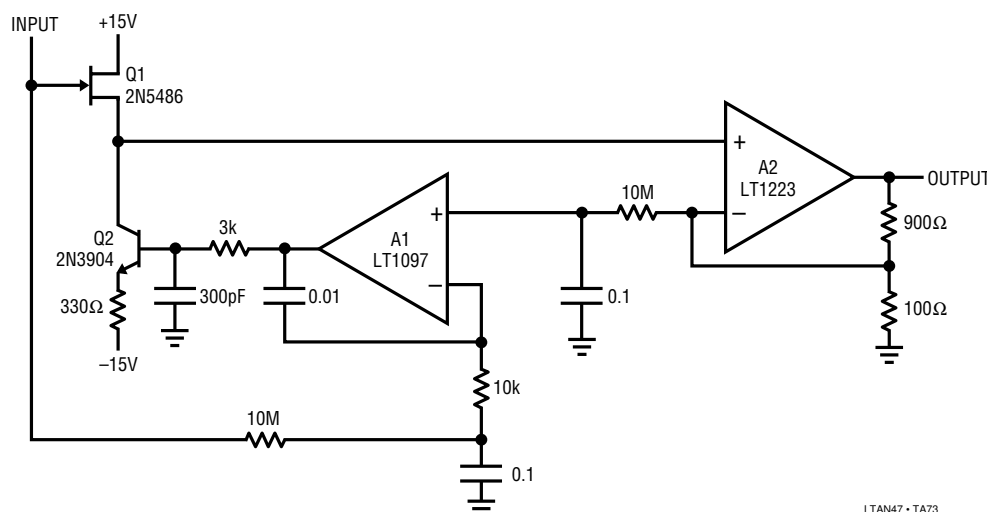


Figure 73. A1 DC Stabilizes the Circuit by Controlling Q1's Channel Current

bias current of 100pA, it is well suited for probing or as an ATE pin amplifier. As shown, gain is ten, but other gains are possible by varying the feedback ratio.

DC Stabilization – Full Differential Inputs with Parallel Paths

Figure 74 shows a way to get full differential inputs with DC stabilized operation. This circuit combines the output of two differential input amplifiers for overall DC corrected wideband operation. A1 and A2 both differentially sense the input at gains of ten. Wideband A1 feeds output amplifier A3 via a highpass network, while the slower A2 contributes DC and low frequency information to A3. A2 does not see high frequency inputs, because they are filtered by the 2k-200pF lowpass networks at its inputs. If the gain and bandwidth of the high and low frequency paths complement each other, A3's output should be an undistorted, amplified version (in this case $\times 10$) of the input. Figure 75 shows this to be the case. Trace A is one side of a differential input applied to the circuit. Trace B is A1's output taken at the 500 Ω potentiometer - 0.001 μ F junction. Trace C is A2's output. With the AC gain and DC gain match trims properly adjusted, the two paths' contributions match up and Trace D is singularly clean, with no residue. The adjustments are optimized by trimming the AC gain for the squarest corners and the DC gain match for a flat top. Bandwidth for this circuit exceeds 35MHz, slew rate is 450V/ μ s and DC offset about 200 μ V.

Note 11: For assistance in following circuit signal flow, the schematic of this device is included in the figure.

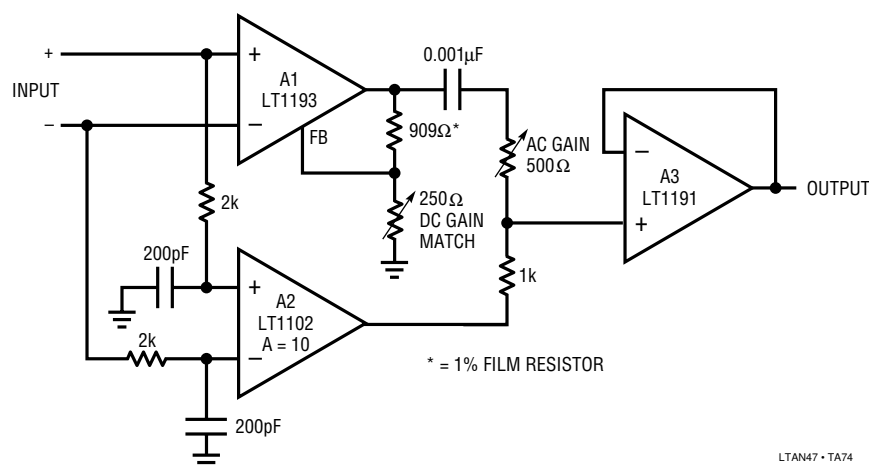


Figure 74. A Parallel Path DC Stabilized Differential Amplifier. High Frequency Signals Go through A1, while A2 Handles DC and Low Frequency. A3 Sums Both Paths

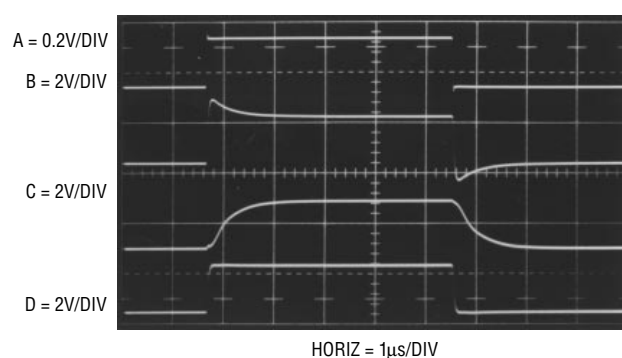


Figure 75. Waveforms for the Parallel Path Differential Amplifier. Trace A is the Input; B, C and D are the High Pass, Low Pass and Output Nodes, Respectively

DC Stabilization – Full Differential Inputs, Gain-of-1000 with Parallel Paths

Figure 76 is a very powerful extension of the previous circuit. Operation is similar, but gain is increased to 1000. Bandwidth is about 35MHz, rise time equals 7ns and delay is inside 7.5ns. Full power response is available to 10MHz, with input noise about 15 μ V broadband. This kind of speed, coupled with full differential inputs, the gain of 1000, DC stability, and low cost make the circuit broadly applicable in wideband instrumentation. As before, two differential amplifiers, A1 and A2, simultaneously sense the inputs. In this case A1 is the popular and economical 592-733 type, operating at a gain of 100.¹¹ A1's differential outputs feed output amplifier A3 via 1 μ F-1k Ω high pass networks which strip off A1's DC content. A2, a precision DC differential type, operates in similar fashion to the previous circuit, supplying DC and low frequency

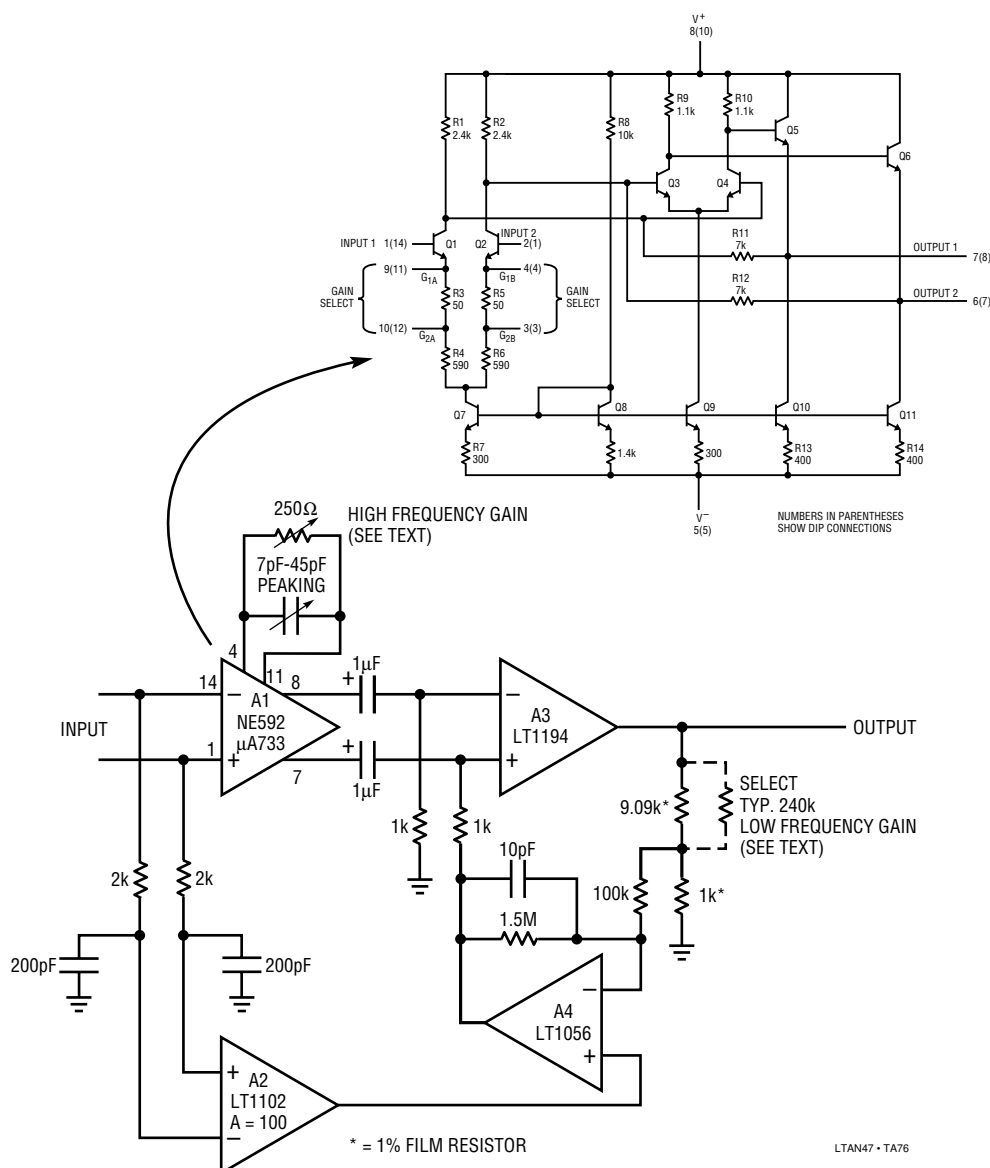


Figure 76. A Full Differential, Parallel Path Amplifier. Gain is 1000, with 38MHz Bandwidth. Delay is Inside 7.5ns and Rise Time Under 7ns

information to A3 at a trimmed gain of 100. In this case output amplifier A3 is a differential gain block with a nominal committed gain of 10. This change is necessitated by A1's differential output, which must be single-ended to obtain the circuit's output. As such A2 does not directly apply its low frequency information to A3 as it did before. Instead, A4 measures the difference between A2's output and a divided down portion of A3's output. A4's output, biasing A3's positive input via the 1kΩ resistor, closes a loop around the circuit's DC-low frequency path. The divider feeding A4's negative input is adjusted so that the circuit's DC gain is known and equal to its AC gain.

Figure 77 shows the circuit's response to a 60ns, 2.5mV amplitude pulse (Trace A). The X1000 output (Trace B) responds cleanly, with delay and rise time in the 5ns-7ns range. Some small amount of peaking is evident, although it may be trimmed with the peaking adjustment at A1. Figure 78 plots the circuit's gain vs frequency. Gain is flat within 1/2dB to 20MHz, with the -3dB point at 38MHz. Figure 77's edge peaking shows up here as a very slight gain increase starting around 1MHz and continuing out to about 15MHz. The peaking trim will eliminate this effect.

To use this circuit, put in a low frequency or DC signal of known amplitude and adjust the low frequency gain for a

X1000 output after the output has settled. Next, adjust the high frequency gain so that the signal's front and rear corners have amplitudes identical to the settled portion. Finally, trim the peaking adjustment for best settling of the output pulse's front and rear corners.

Figure 79 shows input (Trace A) and output (Trace B) waveforms with all adjustments properly set. Fidelity is excellent, with no aberrations or other artifacts of the parallel path operation evident. Figure 80 shows the effects of too much AC gain; excessive peaking on the edges with proper amplitude indicated only after the AC channel transitions through its highpass cut off. Similarly, excessive DC gain produces Figure 81's traces. The AC gain path

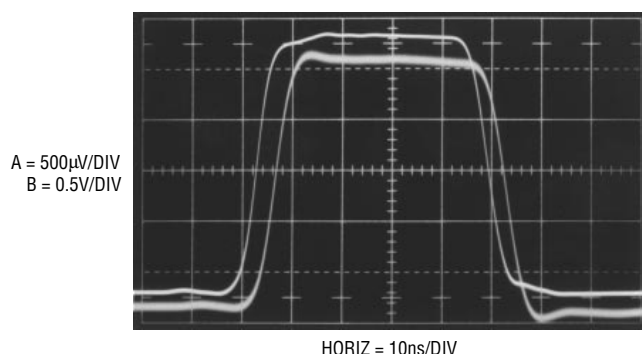


Figure 77. Pulse Response for the X1000 Differential Amplifier. Fidelity is Quite Good, with Only Slight Output Peaking (Trace B)

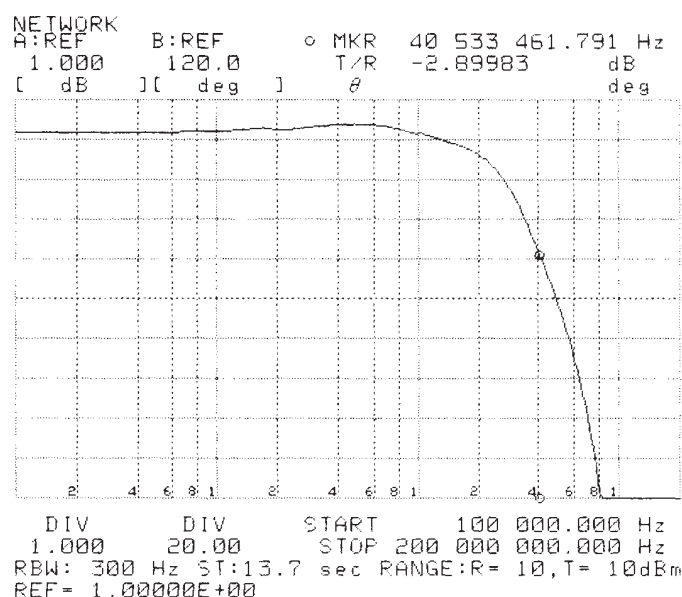


Figure 78. Gain vs Bandwidth for the X1000 Differential Amplifier. Peaking Noted in Figure 77 Shows up as 0.25dB Peak at 5MHz, Which Could be Trimmed Out

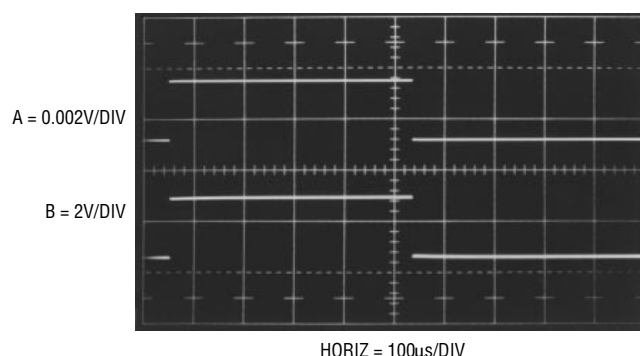


Figure 79. Response of X1000 Amplifier with Bandwidth Crossover Points Properly Adjusted. A = Input; B = Output

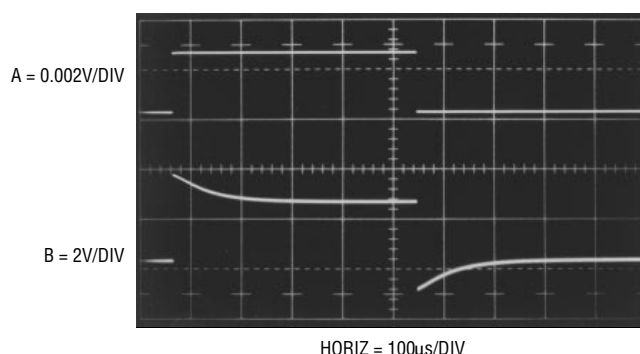


Figure 80. Response of X1000 Amplifier with Excessive AC Gain. A = Input; B = Output

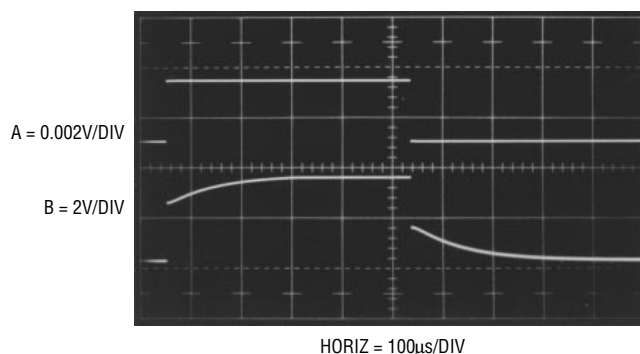


Figure 81. Response of X1000 Amplifier with Too Much DC Gain. A = Input; B = Output

provides proper initial response, but too much DC gain forces a long, tailing response to an incorrect amplitude.

High Speed Differential Line Receiver

High speed analog signals transmitted on a line often pick up substantial common-mode noise. Figure 82 shows a simple, fast differential line receiver using the LT1194

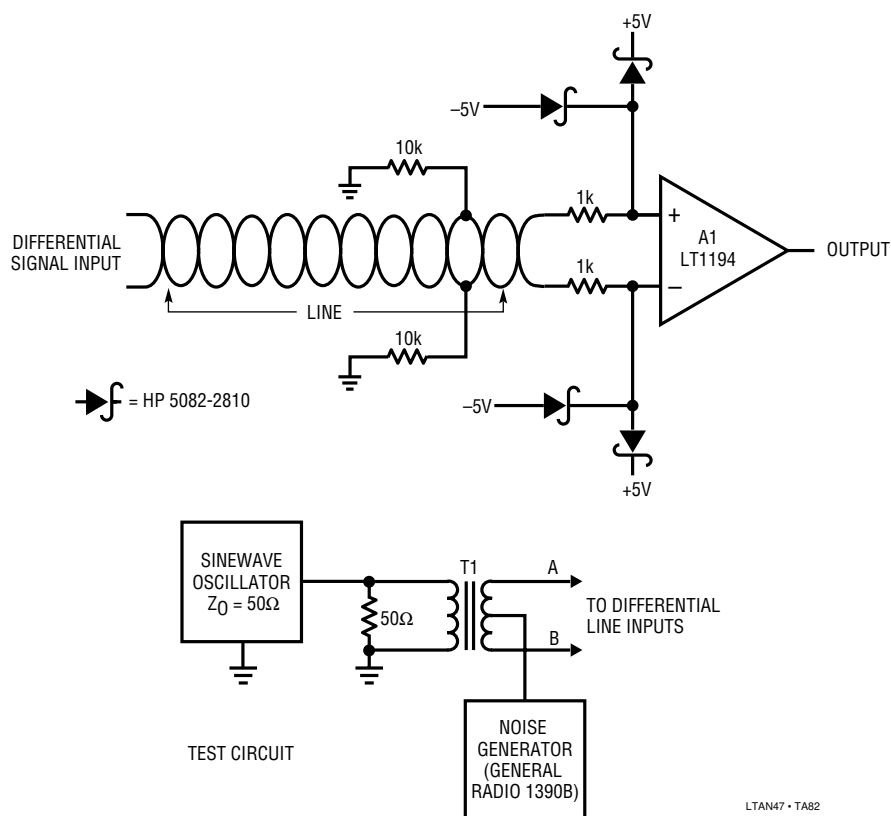


Figure 82. Simple, Full Differential Line Receiver

gain-of-ten differential amplifier. The differential line is fed to A1. The resistor-diode networks prevent overload and insure input bias for A1 under all conditions. A1's output represents the difference of the two line input times a gain of ten. In theory, all common-mode noise should be rejected. The test circuit shown in the figure confirms this. The sinewave oscillator drives T1 (Trace A, Figure 83), producing a differential line output at its secondary. T1's secondary is returned to ground through a broadband noise generator, flooding the line inputs with common-mode noise (traces B and C are A1's inputs). Trace D, A1's X10 version of the differential signal at its inputs, is clean with no visible noise or disturbances. This circuit will easily provide a clean output with DC-5MHz noise dominating signal by a 100:1 ratio.

Transformer Coupled Amplifier

Figure 84 shows another way to achieve high common-mode rejection. Additionally, this circuit has the advantage of true 3 port isolation. The input, gain stage, and output are all galvanically isolated from each other. As such, this

configuration is useful where large common-mode differences are encountered or where ground integrity is uncertain. A1 is set up in a simple gain of 11. T1 feeds its input, and the output is taken from T2. Figure 85 shows results for a 4MHz input, with all "•" designated transformer leads referred to ground. The input (Trace A, Figure 85) is applied to T1, whose output (Trace B) feeds A1. A1 takes gain, and its output (Trace C) feeds T2. T2's output

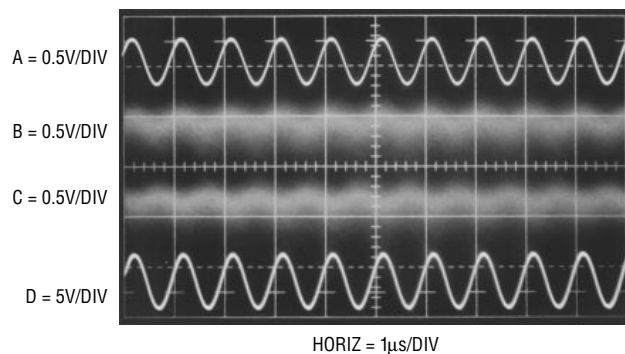
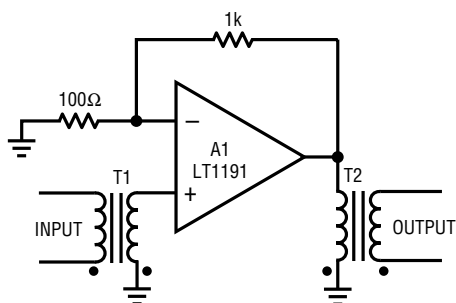


Figure 83. Differential Line Receiver Easily Pulls Out a Signal Buried in Common-Mode Noise. Output is Clean, Despite 100:1 Noise-to-Signal Ratio

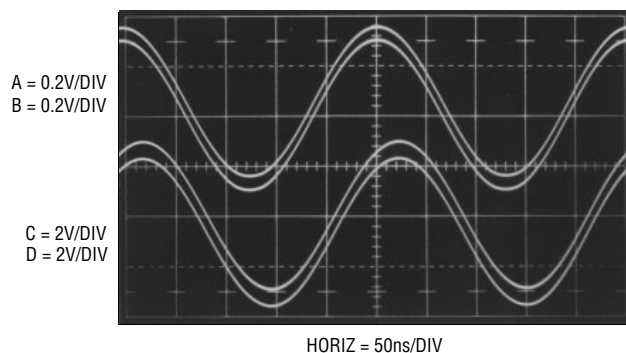
(Trace D) is the circuit's output. Phase shift is evident, although tolerable. T1 and T2 are very wideband devices, with low phase shift. Note the negligible phase difference between the A-B and C-D trace pairs. A1 contributes essentially the entire phase error. Using the transformers specified, the circuit's low frequency cut-off is about 10kHz.



T1, T2 = MINI CIRCUITS LAB # T1-6

I TAN47 • TA84

Figure 84. Transformer Coupled Amplifier. Note That A1 is Galvanically Isolated From Input and Output Nodes



I TAN47 • TA85

Figure 85. Transformer Coupled Amplifier Responds to an Input (Trace A) with A Slightly Phase Shifted Output (Trace D). Traces B and C are T1 Secondary and T2 Primary, Respectively

Differential Comparator Amplifier with Adjustable Offset

It is often desirable to examine or amplify one particular portion of a signal while rejecting all other portions. At high speed this can be difficult, because the amplifier may see fast, large common-mode swings. Recovery from such activity usually is dominated by saturation effects, making the amplifier's output questionable. The LT1193's differential amplifier's fast overload recovery permits this function, maintaining output fidelity to the input signal. Additionally, the input level amplitude at which amplification begins is settable, allowing any amplitude defined point to be selected. In Figure 86, A1, the LT1019 reference and associated components form an adjustable, bipolar voltage source which is coupled to differential amplifier A2's negative input. The input signal biases A2's positive input with A2's gain set by R1 and R2, in accordance with the equation given.

Input signals below A2's negative input levels maintain A2's output in saturation, and no signal is seen at the output. When the positive input rises above the negative input's bias point A2 becomes active, providing an amplified version of the instantaneous difference between its inputs. Figure 87 shows what happens when the output of a triangle wave generator (Trace A) is applied to the circuit. Setting the bias level just below the triangle peak permits high gain, detailed operation of the turnaround at the peak. Switching residue in the generator's output is clearly observable in Trace B. Appropriate variations in the voltage source setting would permit more of the triangle

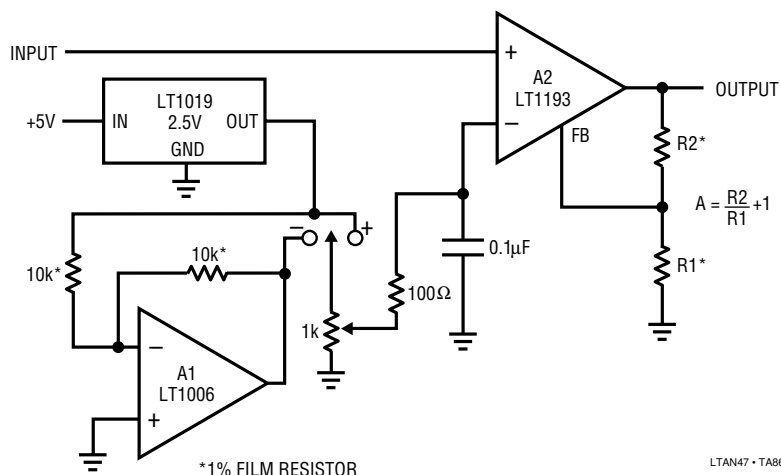


Figure 86. Fast Differential Comparator Amplifier with Settable Offset

Application Note 47

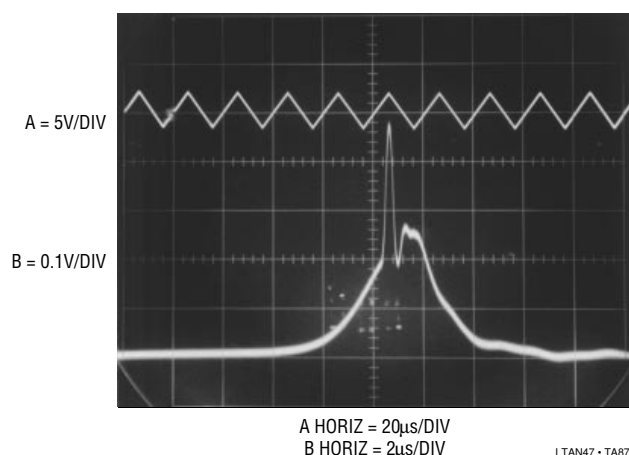


Figure 87. The Differential Comparator Amplifier Extracting Signal Detail From a Triangle Waveform's Peak. Triangle Generator's Switching Artifacts are Clearly Evident

slopes to be observed, with attendant loss of resolution due to oscilloscope overload limitations. Similarly, increasing A2's gain allows more amplitude detail while placing restrictions on how much of the waveform can be displayed. It is worth noting that this circuit performs the same function as differential plug-in units for oscilloscopes. This circuit's output is accurate and settled to 0.1% 100ns after it enters its linear region.

Differential Comparator Amplifier with Settable Automatic Limiting and Offset

Figure 88 extends the previous circuit's operation, allowing amplified observation of information between two settable, amplitude defined points. The amplitude setpoints are settable in both magnitude and sign. In this circuit the polarity of the offset applied to A2's negative input is

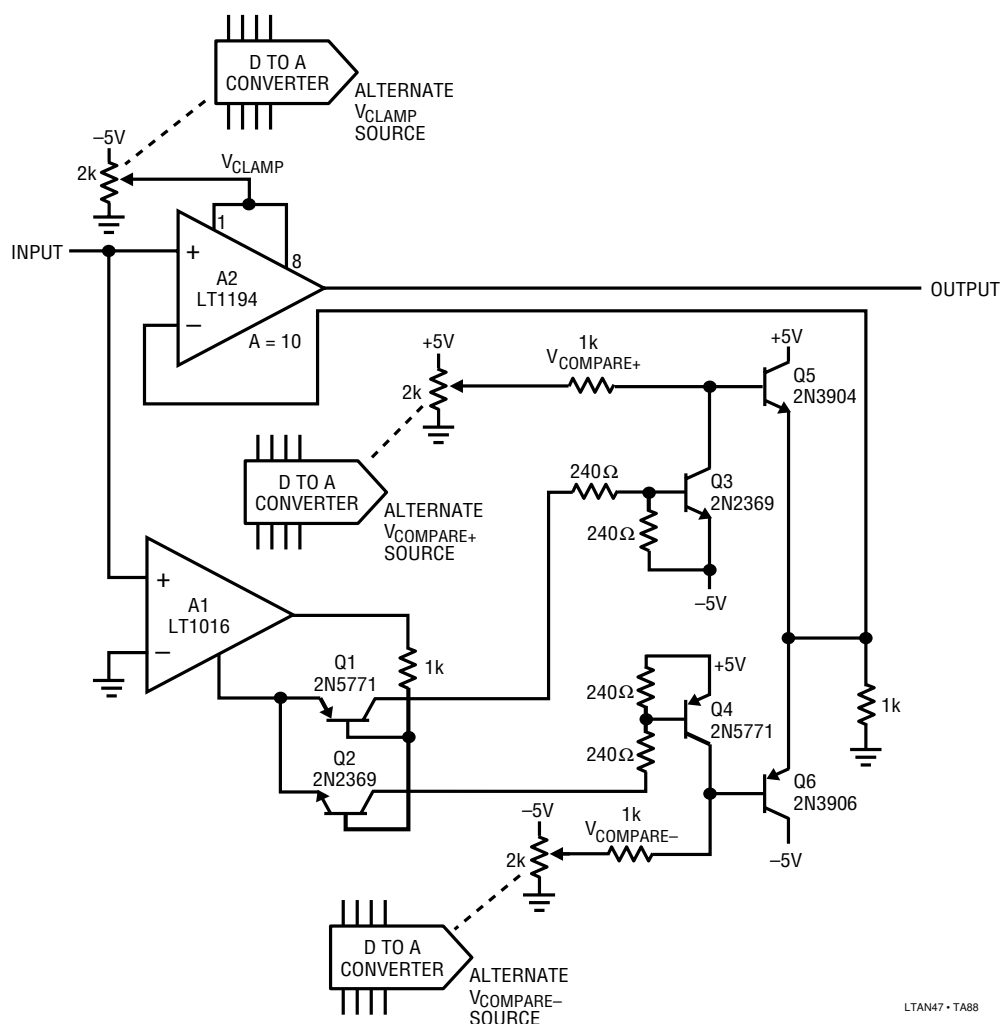


Figure 88. Differential Comparator Amplifier with Settable Automatic Limiting and Offset

determined by comparator A1's output state. A1 compares the circuit's input to ground, generating polarity information at its outputs. Level shifters Q1-Q3 and Q2-Q4 bias followers Q5 and Q6. Positive circuit inputs result in Q5 supplying the " $V_{\text{COMPARE+}}$ " potential to A2, while negative inputs route " $V_{\text{COMPARE-}}$ " to A2. This eliminates the previous circuit's manual polarity switch, permitting automatic selection of the differencing polarity and amplitude. Additionally, this circuit takes advantage of A2's input clamp feature. This feature (See LT1194 Data Sheet) limits the dynamic range of the input, clamping the amplifier's input operating range. Signals inside the clamp limit are processed normally, while signals outside the limit are precluded from influencing the amplifier. This combination of circuit controls allows very tightly defined windows on a waveform to be selected for accurate amplification without overload restrictions.

Figure 89 shows the circuit output for a sine input (Trace A) from the same function generator used to test the previous circuit. The V^+ and V^- compare voltages are set just below the sinewave peaks, with " V_{clamp} " programmed to restrict amplification to the peak's excursion. Trace B, the circuit's output, simultaneously shows amplitude detail of *both* sine peaks. The observed distortion is directly traceable to this generator's imperfect internal triangle waveform (see Figure 87), as well as its sine shaper characteristics.

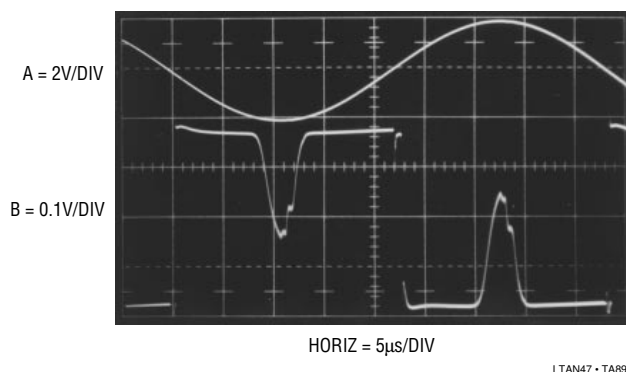


Figure 89. The Automatic Differential Comparator Amplifier Finds Triangle Wave and Switching Residuals (Trace B) in Trace A's Peaks

Photodiode Amplifier

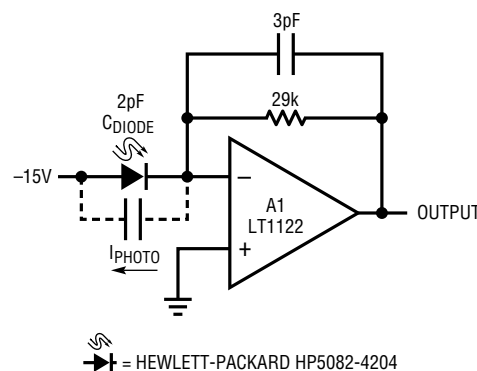
Amplification of fast photodiode signals over a wide range of intensity is a common requirement. Figure 90's fast FET amplifier serves well, giving wideband operation with 5

decades of photocurrent. The photodiode is set up in the conventional manner. Photocurrent is fed directly to A1's summing point, causing A1's output to move to the level required to maintain virtual ground at the negative input. The -15V diode bias aids diode response. The table in the figure details circuit operating characteristics with the diode specified.

Some care in frequency compensating this configuration is required. The diode has about 2pF of parasitic capacitance, forming a significant lag at A1's summing point. If no feedback capacitor is used, high speed dynamics are poor. Figure 91 shows circuit response to a photo input (Trace A) with the indicated 3pF feedback capacitor removed. A1's output overshoots and saturates before finally ringing down to final value. In contrast, replacing the 3pF capacitor provides Figure 92's results. The same input pulse (Trace A, Figure 92) produces a cleanly damped output (Trace B). The capacitor imposes a 50% speed penalty (note faster horizontal scale for Figure 92). This is unavoidable because suppressing the parasitic ringing's relatively low frequency mandates significant roll-off.

Fast Photo Integrator

A related circuit to the photodiode amplifier is Figure 93's photo integrator. Here, the output represents the integral of the diode's photocurrent over some period of time. This



RESPONSE DATA

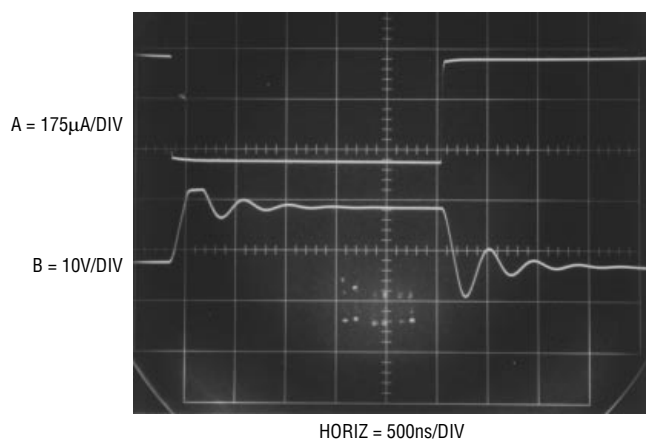
LIGHT (900nm)	DIODE CURRENT	CIRCUIT OUTPUT
1mW	350μA	10.0V
100μW	35μA	1V
10μW	3.5μA	0.1V
1μW	350nA	0.01V
100nW	35nA	0.001V

LTAN47 • TA90

Figure 90. A Simple Photodiode Amplifier

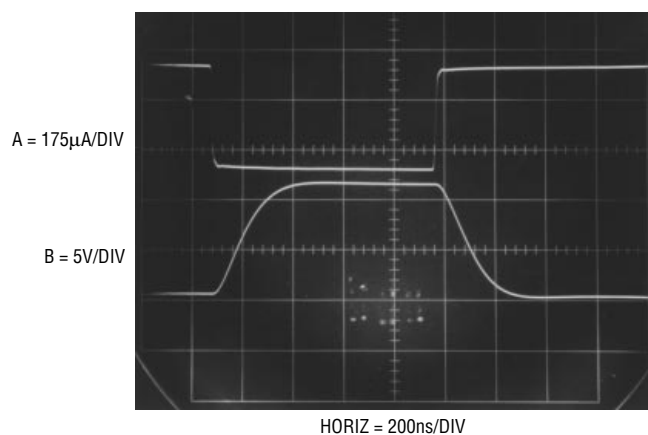
Application Note 47

circuit is particularly applicable in situations where the total energy in a light pulse (or pulses) must be measured. The circuit is a very fast integrator, with S1 used as a reset switch. S2, switched simultaneously with S1, compensates S1's charge injection error. With the control input (Trace A, Figure 94) low and no photocurrent, S1 is closed and A1 looks like a grounded follower. Under these conditions A1's output (Trace C) sits at 0V. When the control input goes high, A1 becomes an integrator as soon as S1 opens. Due to switch delay, this occurs about 150ns after the control input goes high. When S1 opens it delivers some parasitic charge to A1's summing point. S2 provides a compensatory charge based pulse at A1's positive terminal to cancel the effects of S1's charge error. This action shows up as a fast, small amplitude event in A1's output which settles rapidly back to 0V.



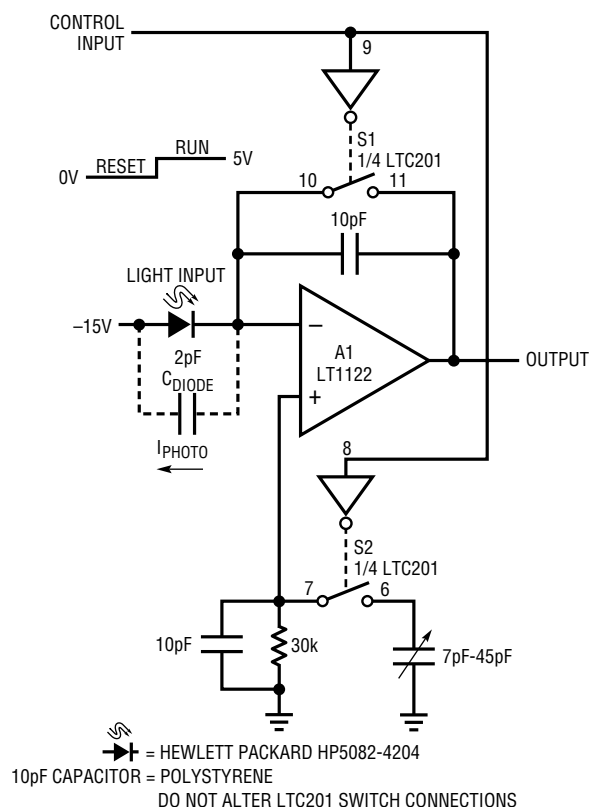
LTAN47 • TA91

Figure 91. Response of Figure 90 Without a Feedback Capacitor



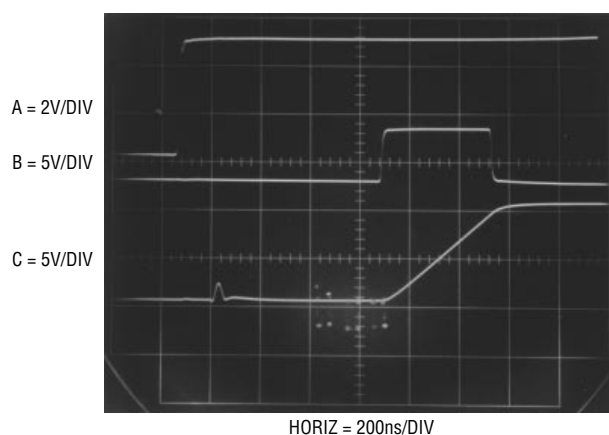
LTAN47 • TA92

Figure 92. Figure 90 Responding with a Feedback Capacitor



LTAN47 • TA93

Figure 93. A Very Fast Photo Integrator. S2 Compensates Reset Switch S1's Small Charge Injection



LTAN47 • TA94

Figure 94. The Photo Integrator Acquires (Trace C) an Input Light Pulse (Trace B) with the Control Line (Trace A) in the Run Mode. Charge Cancellation Action is Evident at Trace C's 400ns Point

At this point in time the integrator is ready to receive and record a photo pulse. When light falls on the photodiode (Trace B triggers a light pulse seen by the photodiode) A1 responds by integrating. In this case A1's output integrates rapidly until the light pulse ceases. A1's voltage after the light event is over is related to the total energy seen by the diode during the event. A monitoring A-D converter can acquire A1's output. In typical operation the control line returns low, resetting A1 until the next event is to be integrated.

With only 10pF of integration capacitor, the circuit has an output droop rate of about 0.2V/ μ s. This can be increased, although integration speed will suffer accordingly. Integration times of nanoseconds to milliseconds and photocurrents ranging from nanoamperes to hundreds of microamperes are accommodated by the circuit as shown. Thus, light intensities spanning microwatts to milliwatts over wide ranges of duration are practical inputs. The primary accuracy restrictions are A1's 75pA bias current, its 12V output swing and the effectiveness of the charge cancellation network. Typically, full-scale accuracy of several percent is achievable if the charge cancellation network is trimmed. To do this, assure that the diode sees no light while repetitively pulsing the control line. Adjust the trimmer capacitor for 0V output at A1 immediately after the disturbance associated with the S1-S2 switching settles.

Fiber Optic Receiver

A simple high speed fiber optic receiver appears in Figure 95. A1, a photocurrent-to-voltage converter similar to Figure 90, feeds comparator A2. A2 compares A1's output

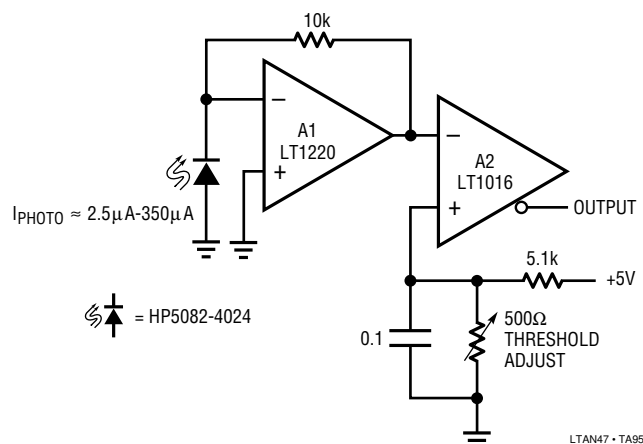


Figure 95. A Simple Fiber Optic Receiver

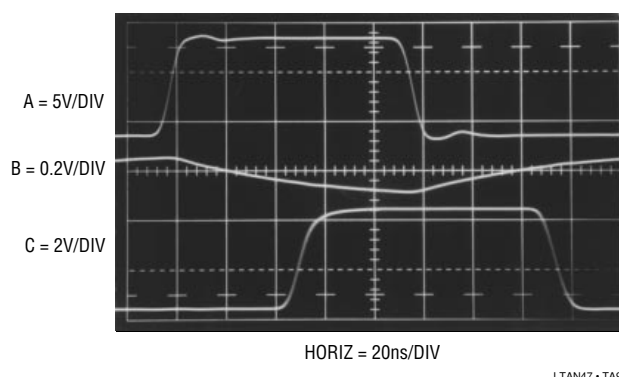


Figure 96. Waveforms for the Simple Fiber Optic Receiver. A1 (Trace B) Lags the Input (Trace A), but Output (Trace C) is Clean

to a DC level established by the threshold adjust setting, producing a logic compatible output. Figure 96 shows typical waveforms. Trace A is a pulse associated with a photo input. Trace B is A1's response and Trace C is A2's output. The phase shift between the photo input and A2's output is due to A1's delay in reaching the threshold level. Reducing the threshold level will help, but moves operation closer to the noise floor. Additionally, the fixed threshold level cannot account for response changes in the emitter and detector diodes and fiber optic line over time and temperature.

40MHz Fiber Optic Receiver with Adaptive Trigger

Receiving high speed fiber optic data with wide input amplitude variations is not easy. The high speed data and uncertain intensity of the light level can cause erroneous results unless the receiver is carefully designed. Figure 97 addresses the previous circuit's limitations, offering significant performance advantages. This receiver will reliably condition fiber optic inputs of up to 40MHz with input amplitude varying by >40dB. Its digital output features an adaptive threshold trigger which accommodates varying signal intensities due to component aging and other causes. An analog output is also available to monitor the detector output. The optical signal is detected by the PIN photodiode and amplified by A1. A second stage, A2, gives further amplification. The output of this stage biases a 2-way peak detector (Q1-Q4). The maximum peak is stored in Q2's emitter capacitor, while the minimum excursion is retained in Q4's emitter capacitor. The DC value of A2's output signal's mid-point appears at the junction of the 500pF capacitor and the 22M Ω units. This point will always sit midway between the signal's excursions, re-

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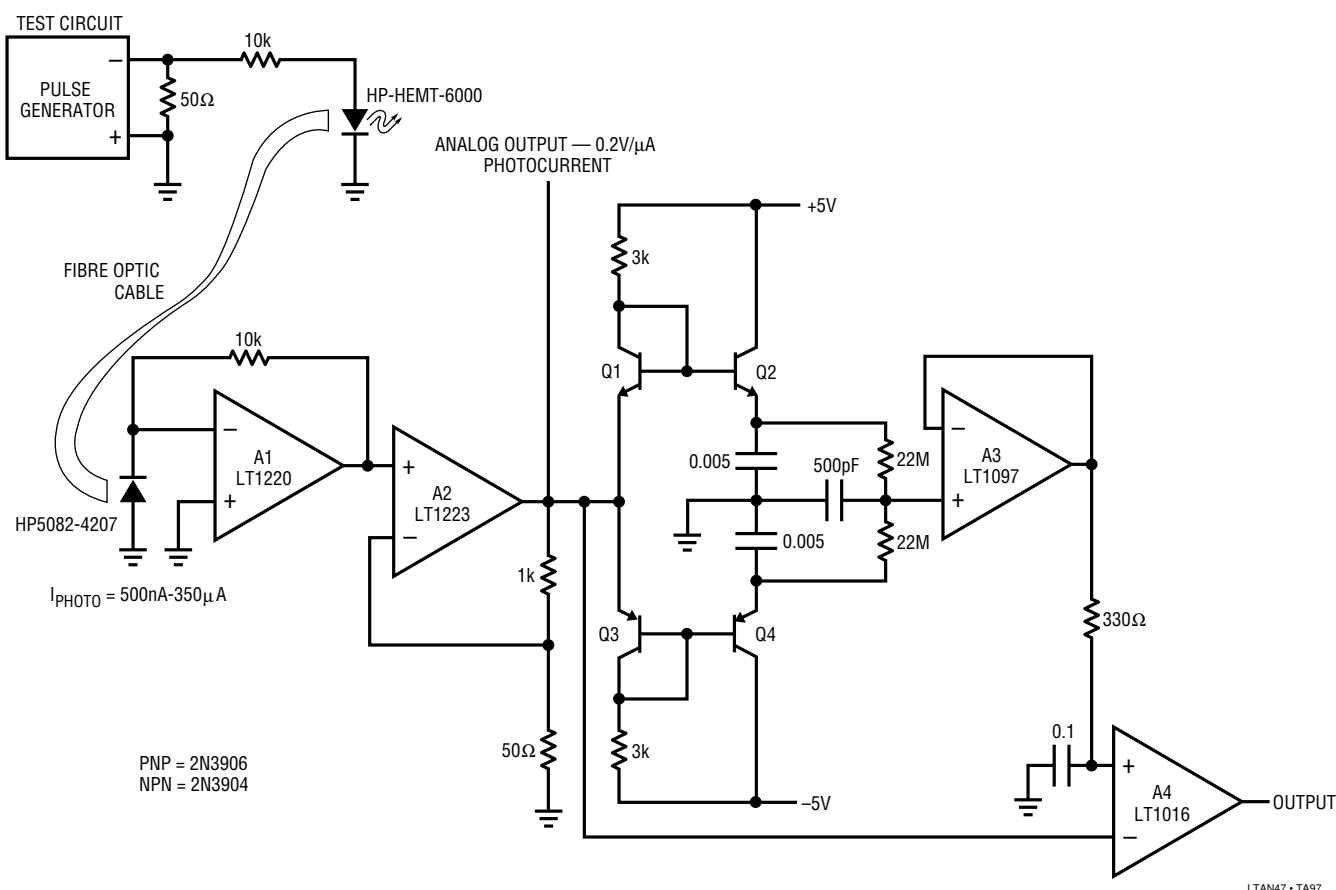


Figure 97. Adaptively Triggered 40MHz Fiber Optic Receiver is Immune to Shifts in Operating Point

regardless of absolute amplitude. This signal-adaptive voltage is buffered by the low bias LT1097 to set the trigger voltage at the LT1016's positive input. The LT1016's negative input is biased directly from A2's output. Figure 98 shows the results using the test circuit indicated in Figure 97. The pulse generator's output is Trace A, while A2's output (analog output monitor) appears in Trace B. The LT1016 output is Trace C. The waveforms were recorded with a 5μA photocurrent at about 20MHz. Note that A4's output transitions correspond with the midpoint of A2's output (plus A4's 10ns propagation delay) in accordance with the adaptive trigger's operation.

50MHz High Accuracy Analog Multiplier

Although highly accurate, very wideband analog multipliers are available, their output takes a differential form. These differential outputs, which have substantial common mode content, are frequently inconvenient to work with. RF transformers can be used to single end the outputs, but DC and low frequency information is lost.

Figure 99 uses the LT1193 differential amplifier to accomplish the differential-to-single ended transition. The AD834 is set up in the recommended configuration (see Analog Devices AD834 Data Sheet, Reference 26). The LT1193 takes the differential signal from the AD834's 50Ω terminated output and provides a single ended output. The gain of two yields a ±1V output at full-scale.

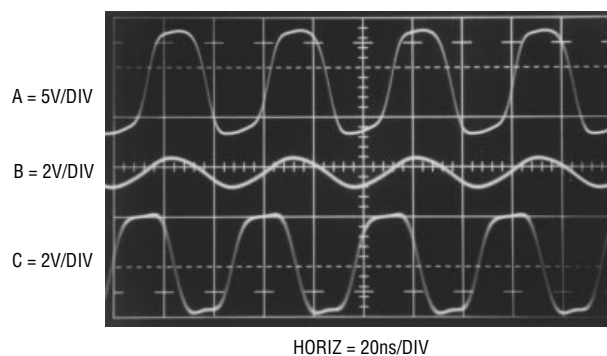


Figure 98. Adaptively Triggered Fiber Optic Receiver's Waveforms at 20MHz with 5μA Diode Current

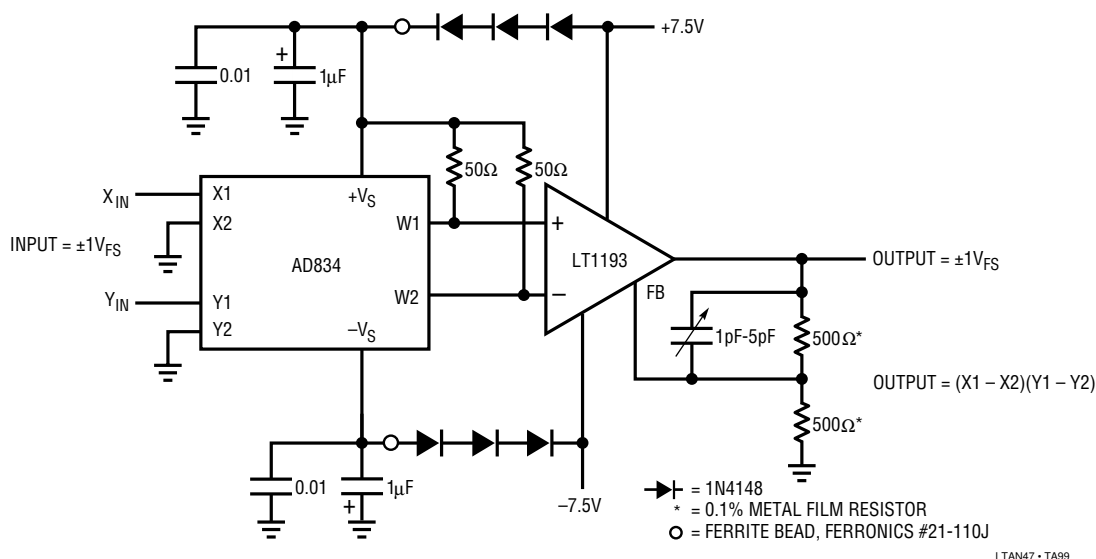


Figure 99. Analog Multiplier with 2% Accuracy Over DC to 50MHz Has a Single Ended Output

The AD834 outputs come out riding on a common-mode level very close to the device's positive supply. This common-mode level falls outside the LT1193's input common-mode range. The diodes in the 7.5V supply rails drop the supply at the AD834, biasing its outputs within the LT1193's input range. This scheme avoids the attenuation and matching problems presented by placing a level shift between the multiplier and amplifier. The ferrite beads combine with the diode's impedance to ensure adequate bypassing for the multiplier, a very wideband device.

Performance for this circuit is quite impressive. Error remains within 2% over DC-50MHz, with feedthrough below -50dB. Trimming the circuit involves adjusting the variable capacitor at the amplifier for minimal output square wave peaking. Figure 100 shows performance when a 20MHz sine input is multiplied by Trace A's waveform. The output (Trace B) is a singularly clean instantaneous representation of the X•Y input products, with strict fidelity to their components.

Power Booster Stage

Occasionally, it is necessary to supply larger output currents than an amplifier is capable of delivering. The power gain stage, sometimes called a booster, is usually placed within the monolithic amplifier's feedback loop, preserving the IC's low drift and stable gain characteristics.

Because the output stage resides in the amplifier's feedback path, loop stability is a concern. This is particularly the case with high speed amplifiers. The output stage's gain and AC characteristics must be considered if good dynamic performance is to be achieved. Overall circuit phase shift, frequency response and dynamic load handling capabilities are issues that cannot be ignored when designing a power gain stage for a monolithic amplifier. The output stage's added gain and phase shift can cause poor AC response or outright oscillation. Judicious application of frequency compensation methods is needed for good results (see Appendix C, "The Oscillation Problem – Frequency Compensation Without Tears", for discussion and details on compensation methods).

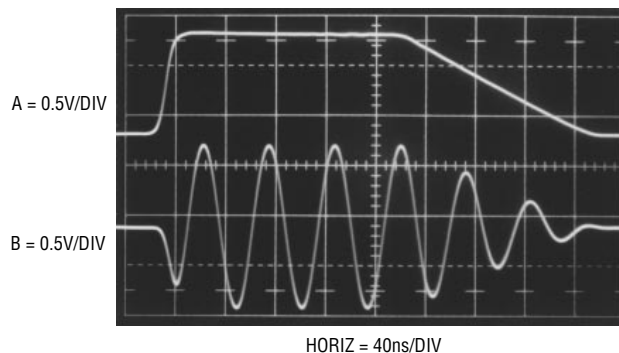


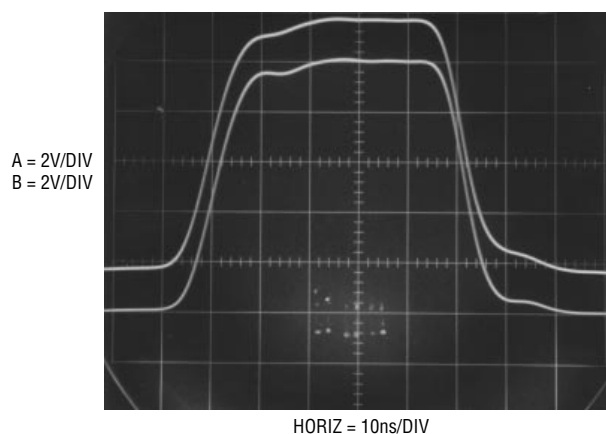
Figure 100. The Multiplier Produces a Modulated Sine Output (Trace B) in Accordance with Trace A's Envelope

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Figure 101 shows a 200mA power booster used with an LT1220 amplifier. Complementary emitter followers Q1-Q5 provide current gain for positive signals, with Q2 and Q6 handling negative excursions. Q3 and Q4 are V_{BE} based current limits, coming on and robbing drive from the appropriate output transistor when current exceeds about 300mA. The diodes prevent Q1 and Q2 from seeing reverse V_{BE} during current limit. The 100 Ω resistor and ferrite beads prevent the low impedance amplifier output from causing oscillation in Q1 and Q2 (see Appendix C).

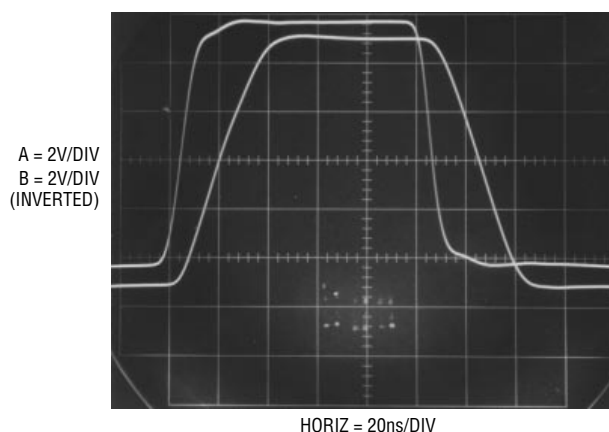
To be effective, the booster must be exceptionally fast. A slow design will obviate the AC performance of the amplifier controlling it, or in the worst case, cause oscillation (again, see Appendix C). Figure 102 shows booster performance with the LT1220 removed from the circuit. The input pulse (Trace A) is applied to the booster input, with the output (Trace B) taken at the indicated spot. Evaluation of the photograph shows that booster rise and fall times are limited by the input pulse generator. Additionally, delay is in the 1ns range. This kind of speed makes the circuit a good candidate for acceptable AC performance within a fast amplifier's loop.

Figure 103 shows pulse response with the LT1220 installed in the circuit with a 50 Ω load. The booster's high speed contributes negligible delay and overall response is clean and predictable. The local 3pF roll-off at the LT1220 optimizes response, but is not absolutely necessary in this circuit. The input (Trace A) produces a nicely shaped LT1220 slew-limited output (Trace B).



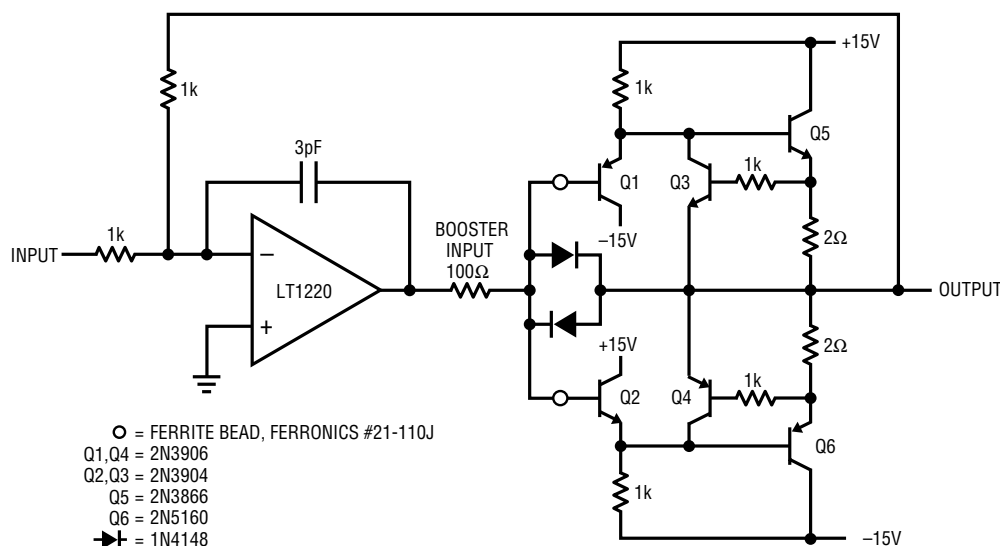
LTAN47 • TA102

Figure 102. Response of Figure 101's Booster Stage



LTAN47 • TA103

Figure 103. The Booster's Response When Inside an Amplifier's Loop



LTAN47 • TA101

Figure 101. A 200mA Output Wideband Booster Stage

High Power Booster Stage

In theory, higher power booster stages should be achievable by utilizing bigger devices. This is partly the case, but lack of availability of wideband PNP power transistors is an issue. Figure 104 shows a way around this problem.

The circuit is essentially a 1A output version of Figure 101, with several differences. In the positive signal path output transistor Q4 is an RF power type, driven by Darlington connected Q3. The diode in Q1's emitter compensates the additional V_{BE} introduced by Q3, preventing crossover distortion.

The negative signal path substitutes the Q5-Q6 connection to simulate a fast PNP power transistor. Although this configuration acts like a fast PNP follower, it has voltage gain and tends to oscillate. The local 2pF feedback capacitor suppresses these parasitic oscillations and the composite transistor is stable.

This circuit also includes a feedback capacitor trim to optimize AC response. This difference from the previous circuit is necessitated by this circuit's slightly slower characteristics and much heavier loading. Current limit operation and other characteristics are similar to the lower power circuit.

Figure 105 shows waveforms for a 10V negative input step (Trace A) with a 10Ω load. The amplifier responds (Trace B), driving the booster to the voltage required to close the loop. For this positive step, the amplifier provides about 1.5V overdrive to overcome Q3 and Q4's V_{BE} drops. The booster output, lagging by a few nanoseconds (Trace C), drives the load cleanly, with only minor peaking. This peaking may be minimized with the feedback capacitance trimmer.

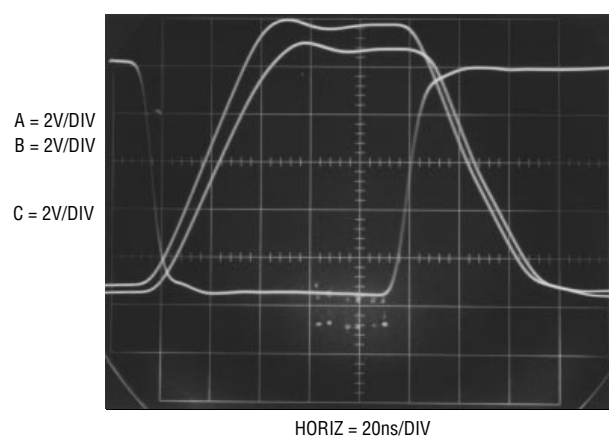


Figure 105. The Boosted Op Amp Drives a 1A Load to 10V in 50ns

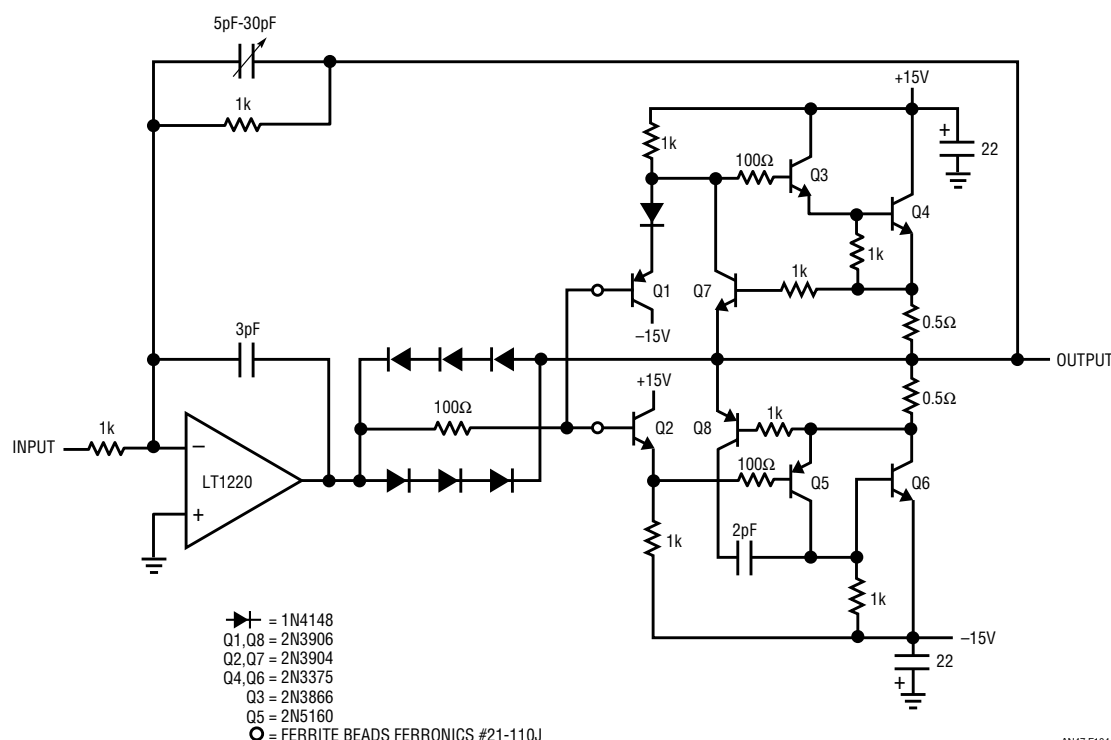


Figure 104. Fast, 1A Booster Stage

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Ceramic Bandpass Filters

Figure 106 is a highly selective bandpass filter using a resonant ceramic element and a single amplifier. The ceramic element nominally looks like a high impedance off its resonant frequency, in this case 400kHz. For off resonance inputs, A1 acts like a grounded follower, producing no output. At resonance, the ceramic element has a low impedance and A1 responds as an inverter with gain. The 100Ω resistor isolates the ceramic element's capacitance from A1's summing point. This capacitance is quite substantial and limits the circuit's out of band rejection capability. Figure 107 shows this. This plot shows very steep rejection, with A1's output down almost 20dB at 300kHz and 40dB at 425kHz. The device's stray parasitic capacitance causes the gentle rise in output at higher frequencies and also sets the -20dB floor at 300kHz.

Figure 108 partially corrects this problem with a nulling technique. This circuit is similar to the previous one, except that a portion of the input is fed to A1's positive input. The RC network at this input is scaled to look like the ceramic resonator's off null impedance. As such, A1's inputs see similar signals for out of band components,

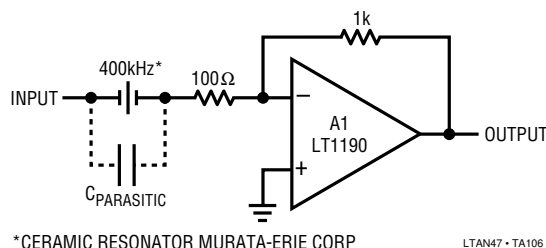


Figure 106. A Piezo-Ceramic Based Filter

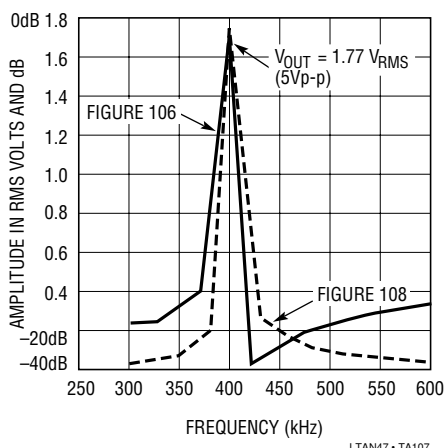


Figure 107. Response of Both Piezo-Ceramic Filters. Differential Network's Activity is Evident in Figure 108's Performance

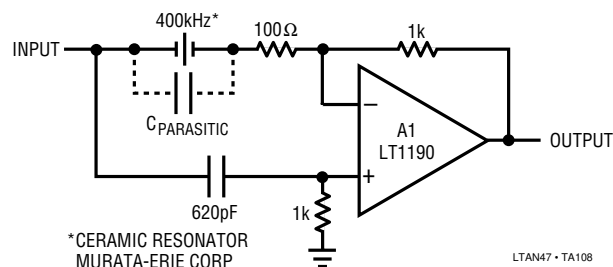


Figure 108. Differential Network Nulls Parasitic Capacitance of Ceramic Element

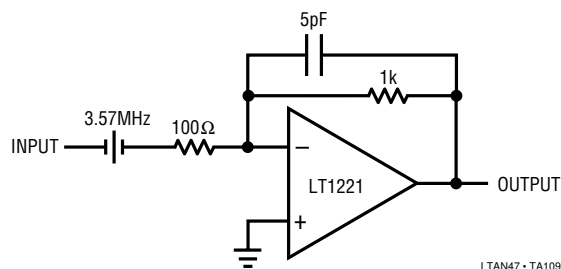


Figure 109. Crystal Filter

resulting in attenuation via A1's common-mode rejection. At resonance, the added RC network appears as a much higher impedance than the ceramic element and filter response is similar to Figure 106's circuit. Figure 107 shows that this circuit has much better out of band rejection than Figure 106. The high frequency roll-off is smooth, and over 20dB deeper than Figure 106 at 475kHz. The low frequency side of resonance has similar characteristics at 375kHz and below.

Crystal Filter

Quartz crystals can also be used to make even higher selectivity filters at higher frequencies. Figure 109 replaces Figure 106's ceramic element with a 3.57MHz quartz crystal. Figure 110 shows almost 30dB attenuation only a few kHz on either side of resonance! The differential nulling technique used with the ceramic elements is less effective with quartz crystals. Crystals have significantly lower parasitic terms, making the cancellation less effective.

APPLICATIONS SECTION II - OSCILLATORS

Sine Wave Output Quartz Stabilized Oscillator

Figure 111 places a crystal within the amplifier's feedback path, creating an oscillator. With the crystal removed, the

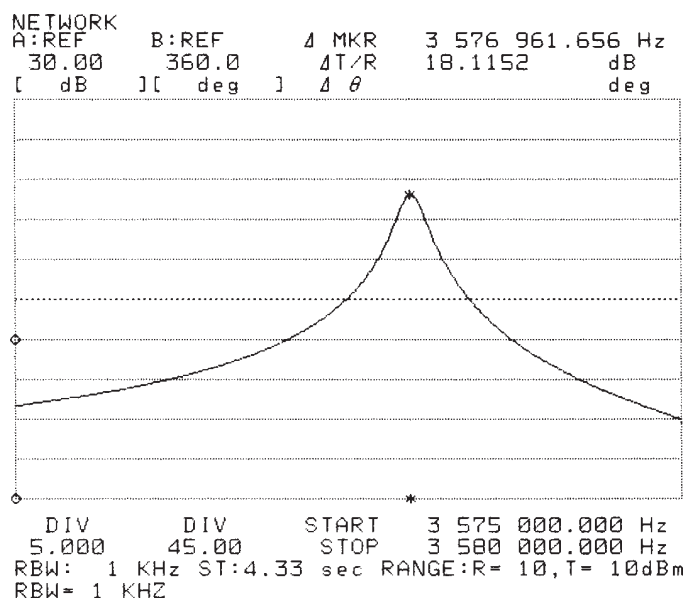


Figure 110. The Crystal Filter's Response

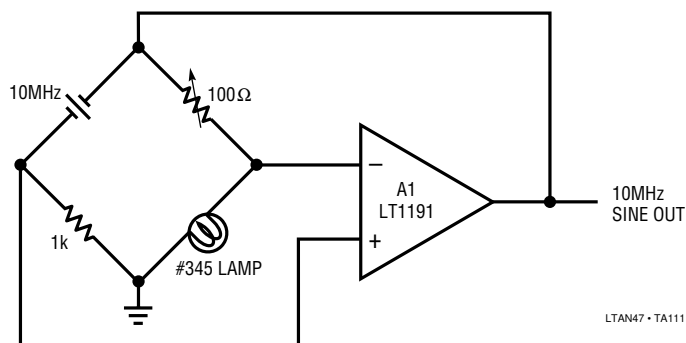


Figure 111. 10MHz Quartz Stabilized Sine Wave Oscillator

circuit is a familiar non-inverting amplifier with a grounded input. Gain is set by the impedance ratio of the elements associated with A1's negative input. Inserting the crystal closes a positive feedback path at the crystal's resonant frequency and oscillations commence.

In any oscillator it is necessary to control the gain as well as the phase shift at the frequency of interest. If gain is too low, oscillation will not occur. Conversely, too much gain produces saturation limiting. Here, gain control comes from the positive temperature coefficient of the lamp at A1's negative input. When power is applied, the lamp is at a low resistance value, gain is high and oscillation amplitude builds. As amplitude builds, lamp current increases, heating occurs, and the lamp's resistance goes up. This causes a reduction in amplifier gain and the circuit finds a

stable operating point. This circuit's sine wave output has all the stability advantages associated with quartz crystals. Although shown at 10MHz, it works well with a wide variety of crystal types over a 100kHz-20MHz range. The use of the lamp to control amplifier gain is a classic technique, first described by Meacham in 1938.¹² Electronic gain control, while more complex, offers more precise control of amplitude.

Sine Wave Output Quartz Stabilized Oscillator with Electronic Gain Control

Figure 112's quartz stabilized oscillator replaces the lamp with an electronic amplitude stabilization loop. A2 compares the A1 oscillator's positive output peaks with a DC reference. The diode in the DC reference path temperature compensates the rectifier diode. A2 biases Q1, controlling its channel resistance. This influences loop gain, which is reflected in oscillator output amplitude. Loop closure around A1 occurs, stabilizing oscillator amplitude. The 1μF capacitor compensates the gain control loop.

The DC reference network is set up to provide optimum temperature compensation for the rectifier diode, which sees a 2Vp-p 20MHz waveform out of A1. A1's small amplitude swing minimizes distortion introduced by channel resistance modulation in Q1. To use this circuit, adjust the 50Ω trimmer until 2Vp-p oscillations appear at A1's output.

Figure 113 is a spectrum analysis of the oscillator's output. The fundamental sits at 20MHz, with the second harmonic 47dB down at 40MHz. A third harmonic, 50dB down, occurs at 60MHz. Resolution bandwidth for the spectrum analysis is 1kHz.

DC Tuned 1MHz-10MHz Wien Bridge Oscillator

In Figure 114 the quartz crystal is replaced with a Wien network at A2's positive input. A1 controls Q1 to amplitude stabilize A2's oscillations in identical fashion to the previous figure. Although the Wien network is not nearly as stable as a quartz crystal, it has the advantage of a variable frequency output. Normally, this is facilitated by varying either R, C, or both. Usually, manually adjustable elements such as dual potentiometers and two section variable

Note 12: See Reference 20, as well as References 19 and 21 for supplemental information.

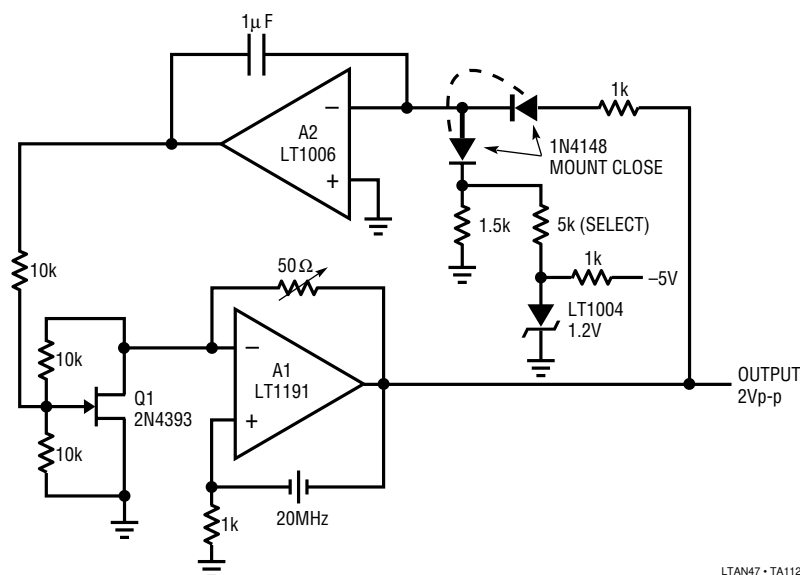


Figure 112. 20MHz Quartz Stabilized Sine Wave Oscillator with Electronic AGC

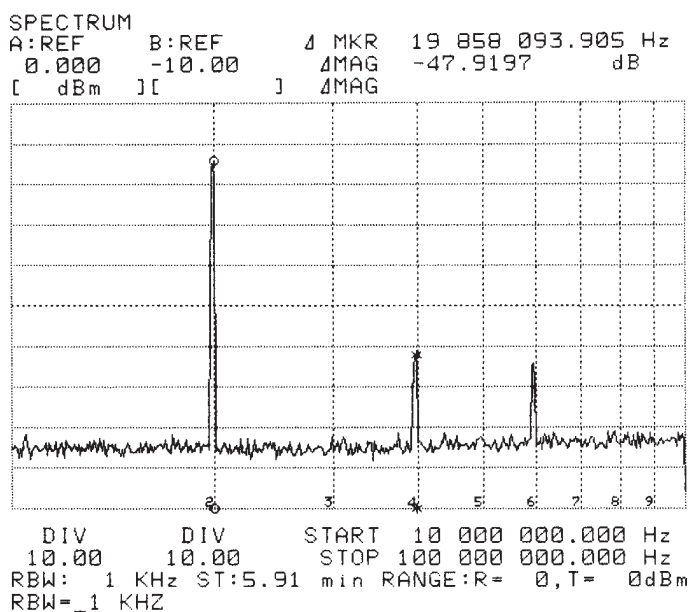


Figure 113. Spectrum Analysis of the 20MHz Quartz Oscillator. Harmonic Content is at Least 47dB Down

capacitors are used. Here, the Wien network resistors are fixed at 360Ω, while the capacitive elements are realized with varactor diodes. The varactor diodes voltage-variable-capacitance characteristic allows DC tuning of the oscillator. DC inputs of 0V-10V to the varactors result in a 1MHz to 10MHz shift in oscillation frequency. The 0.1μF capacitor blocks the DC bias from A2's positive input while permitting the Wien network to function normally. A2's 2Vp-p output minimizes the varactor's junction effects, aiding distortion.

This ±5V powered circuit requires voltage step-up to develop adequate varactor drive. A3 and the LT1172 switching regulator form a simple voltage step-up regulator. A3 controls the LT1172 to produce the output voltage required to close a loop at A3's negative input. L1's high voltage inductive flyback events, rectified by the diode and zener connected Q2, are stored in the 22μF output capacitor. The 7.5k-2.5k divider provides a sample of the output's value to A3's negative input, closing the loop. The 0.1μF capacitor stabilizes this feedback action. Q2's zener drop allows the circuit to produce controlled outputs all the way down to 0V. This arrangement permits a 0V-2.5V input at A3 to produce a corresponding 0V-10V varactor bias. Figure 115, a spectral plot of the circuit running at 7.6MHz, shows the second harmonic down 35dB, with the third harmonic down almost 60dB. Resolution bandwidth is 3kHz.

Complete AM Radio Station

A complete microphone-to-antenna AM radio station appears in Figure 116.¹³ The carrier is generated by A1, set up as a quartz stabilized oscillator similar to the one described in Figure 111. A1's output feeds A2, functioning as a modulated RF power output stage. A2's input signal range is restricted by the bias applied to offset pins 1 and

Note 13: The construction and operation of this apparatus may require Federal Communications Commission review and/or licensing. See Appendix G for FCC licensing and application information.

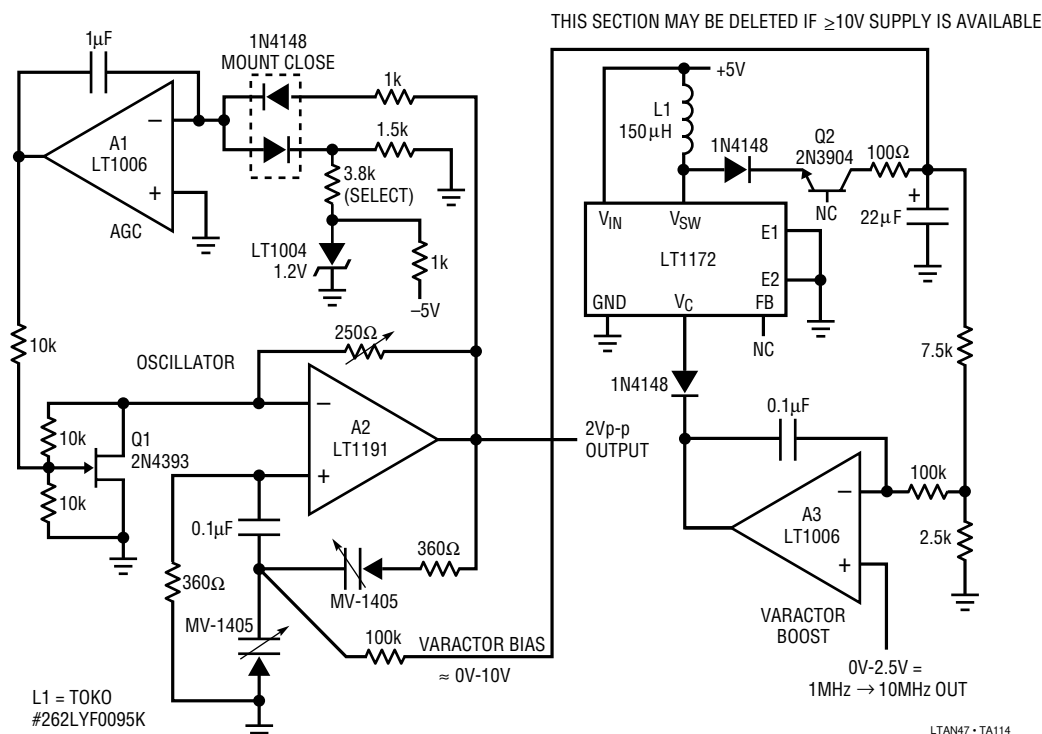


Figure 114. Varactor Tuned 1MHz-10MHz Wien Bridge Oscillator

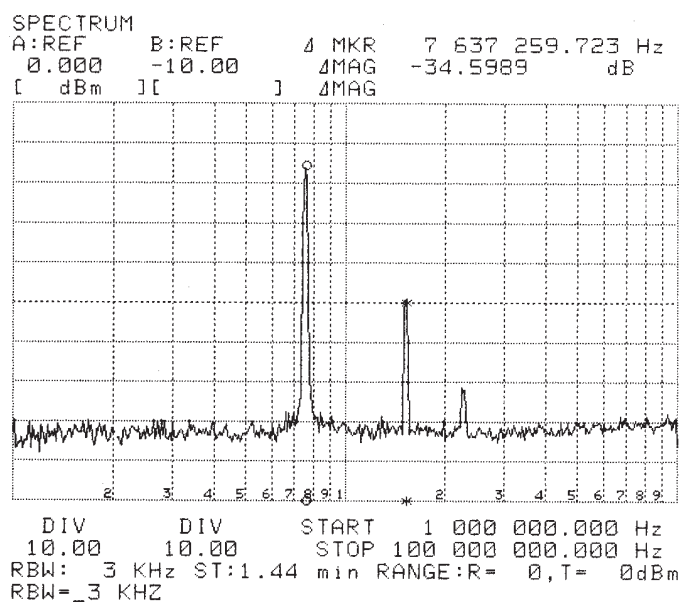


Figure 115. Spectrum Analysis for the Varactor Tuned Wien Bridge. Harmonics are at Least 34dB Down From Fundamental

8 (see LT1194 data sheet for details). A3, a microphone amplifier, supplies bias to these pins, resulting in an amplitude modulated RF carrier at A2's output. The DC term summed with the microphone biases A3's output to the appropriate level for good quality modulation characteristics. Calibration of this circuit involves trimming the

100 Ω potentiometer in the oscillator for a stable 1Vp-p 1MHz A1 output.¹⁴

Figure 117 shows typical AM carrier output at the antenna. In this case the modulation is supplied by Mr. Chuck Berry, singing "Johnny Be Goode".

APPLICATIONS SECTION III - DATA CONVERSION

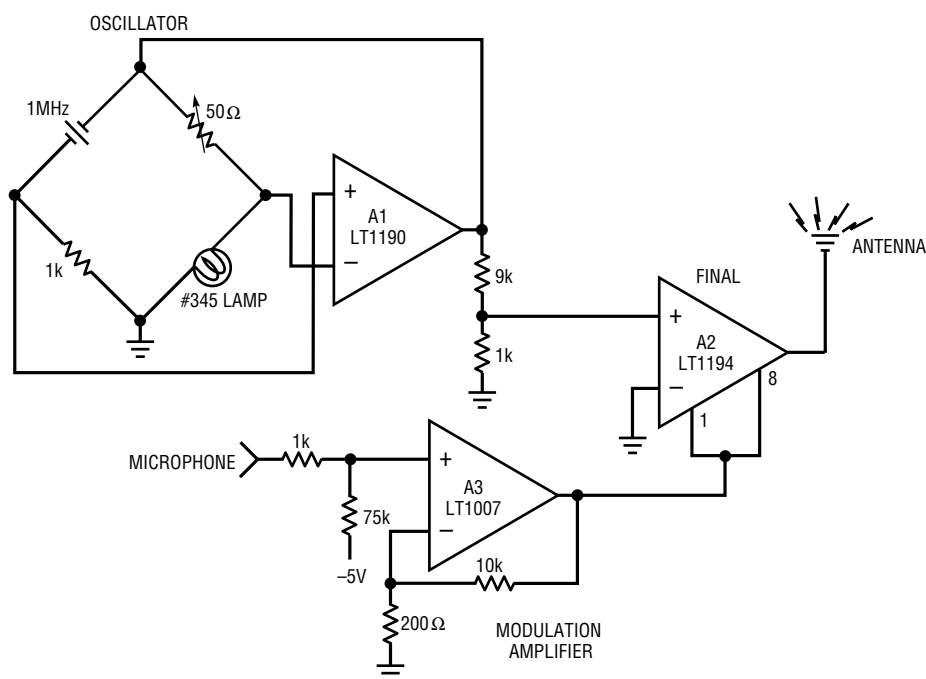
1Hz-1MHz Voltage-Controlled Sine Wave Oscillator

The oscillators presented to this point have limited frequency tuning range. Although Figure 118 is not a true oscillator, it produces a synthesized sine wave output over a wide dynamic range. Many applications such as audio, shaker table driving and automatic test equipment require voltage-controlled oscillators (VCO) with a sine wave output. This circuit meets this need, spanning a 1Hz-1MHz range (120dB or 6 decades) for a 0V to 10V input. It maintains 0.25% frequency linearity and 0.40% distortion specifications.¹⁵ To understand the circuit, assume Q5 is

Note 14: Operating frequency subject to FCC approval and assignment. See Footnote 13 and Appendix G.

Note 15: Seasoned readers of LTC literature, a hardened corps, may recognize this and other circuits in this publication as updated versions of previous LTC applications. The partial repetition is justified based on improved specifications and/or simplification of the original circuit.

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Figure 116. A Complete AM Radio Station. Don't Forget Your Advertisers and FCC License (See Appendix G)

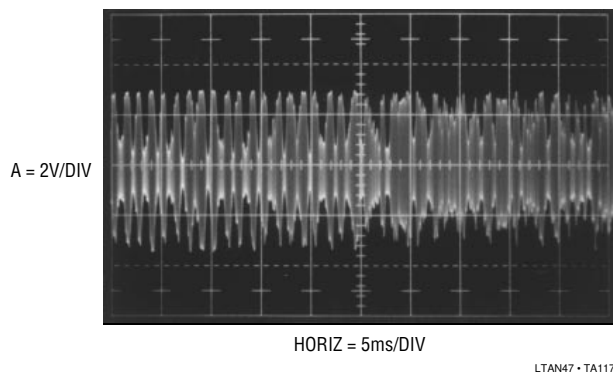


Figure 117. Chuck Berry Lays a Little Modulation on the 1MHz Carrier

on and its collector (Trace A, Figure 119) is at -15V , cutting off Q1. The positive input voltage is inverted by A3, which biases the summing node of integrator A1 through the $3.6\text{k}\Omega$ resistor and the self-biased FETs. A current, $-I$, is pulled from the summing point. A2, a precision op amp, DC stabilizes A1. A1's output (Trace B, Figure 119) integrates positive until C1's input (Trace C) crosses 0V . When this happens, C1's inverting output goes negative, the Q4-Q5 level shifter turns off, and Q5's collector goes to $+15\text{V}$. This allows Q1 to come on. The resistors in Q1's path are scaled to produce a current, $+2I$, exactly twice the absolute magnitude of the current, $-I$, being removed from the

summing node. As a result, the net current into the junction becomes $+I$ and A1 integrates negatively at the same rate as its positive excursion.

When A1 integrates far enough in the negative direction, C1's "+" input crosses zero and its outputs reverse. This switches the Q4-Q5 level shifter's state. Q1 goes off and the entire cycle repeats. The result is a triangle waveform at A1's output. The frequency of this triangle is dependent on the circuit's input voltage and varies from 1Hz to 1MHz with a 0V - 10V input. The LT1009 diode bridge and the series-parallel diodes provide a stable bipolar reference which always opposes the sign of A1's output ramp. The Schottky diodes bound C1's "+" input, assuring it clean recovery from overdrive.

The AD639 trigonometric function generator, biased via A4, converts A1's triangle output into a sine wave (Trace D).

The AD639 must be supplied with a triangle wave which does not vary in amplitude or output distortion will result. At higher frequencies, delays in the A1 integrator switching loop result in late turn on and turn off of Q1. If these delays are not minimized, triangle amplitude will increase with frequency, causing distortion level to also increase with frequency. The total delay generated by the LT1016,

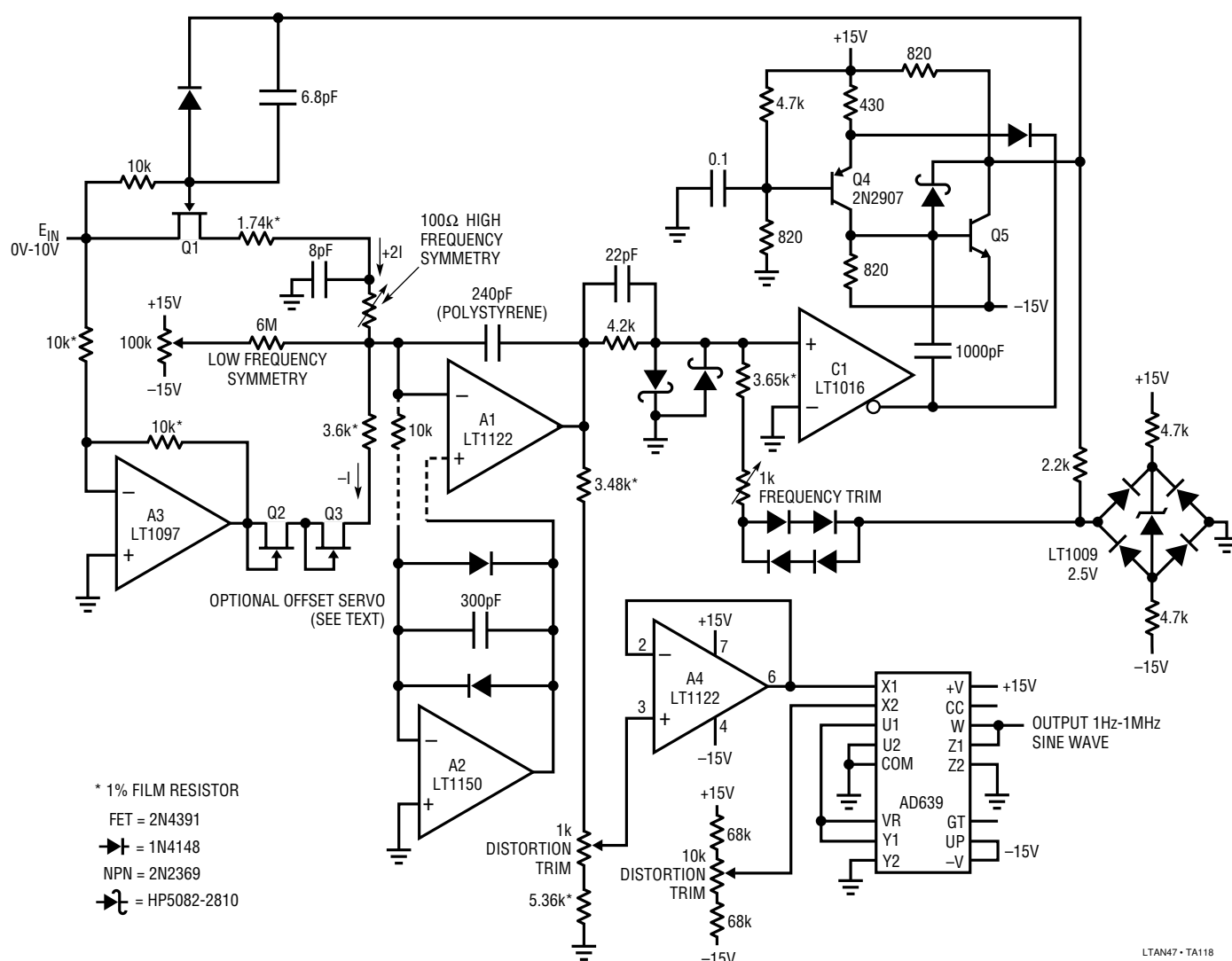


Figure 118. 1Hz-1MHz Sine Wave Output VCO Has 0.25% Linearity and 0.4% Distortion

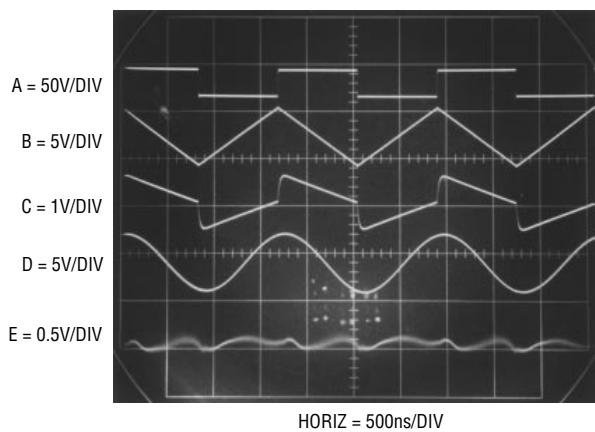


Figure 119. Sine Wave VCO Waveforms

the Q4-Q5 level shifter, and Q1 is 14ns. This small delay, combined with the 22pF feedforward network at the LT1016's input, keeps distortion to just 0.40% over the entire 1MHz range. At 100kHz, distortion is typically inside 0.2%. The effects of gate-source charge transfer, which happens whenever Q1 switches, are minimized by the 8pF unit in Q1's source line. Without this capacitor, a sharp spike would occur at the triangle peaks, increasing distortion. The Q2-Q3 FETs compensate the temperature-dependent on-resistance of Q1, keeping the $+2I/-I$ relationship constant with temperature.

This circuit features extremely fast response to input changes, something most sine wave circuits cannot do.

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Figure 120 shows what happens when the input switches between two levels (Trace A). A1's triangle output (Trace B) shifts frequency immediately, with no glitching or poor dynamics. The sine output (Trace C), reflecting this action, is similarly clean. To adjust this circuit, put in 10.00V and trim the 100Ω pot for a symmetrical triangle output at A1.

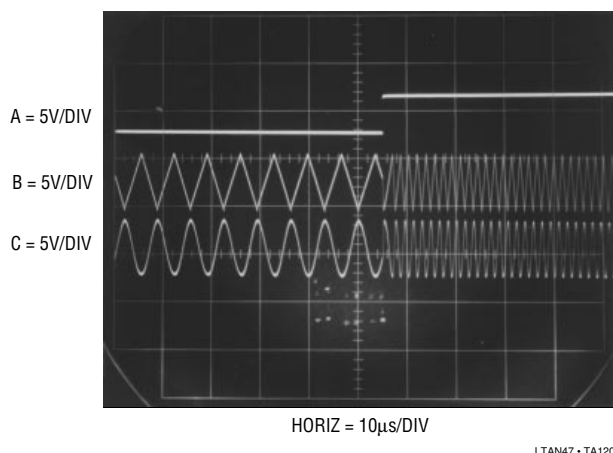


Figure 120. Sine Wave Output VCO Step Response is Quick and Clean

Next, put in 100 μ V and trim the 100k pot for triangle symmetry. Then, put in 10.00V again and trim the 1k frequency trim adjustment for a 1MHz output frequency. Finally, adjust the distortion trim potentiometers for minimum distortion as measured on a distortion analyzer (Trace E, Figure 119). Slight readjustment of the other potentiometers may be required to get lowest possible distortion. If operation below 100Hz is not required, the A2 based DC stabilization stage may be deleted. If this is done, A1's positive input should be grounded.

1Hz-10MHz V → F Converter

The LT1016 and the LT1122 high speed FET amplifier combine to form a high speed $V \rightarrow F$ converter in Figure 121. A variety of circuit techniques are used to achieve a 1Hz to 10MHz output. Overrange to 12MHz ($V_{IN} = 12$) is provided. This circuit has a wider dynamic range (140dB or 7 decades) than any commercially available unit. The 10MHz full-scale frequency is 10 times faster than currently available monolithic $V \rightarrow Fs$. The theory of operation is based on the identity $Q = CV$.

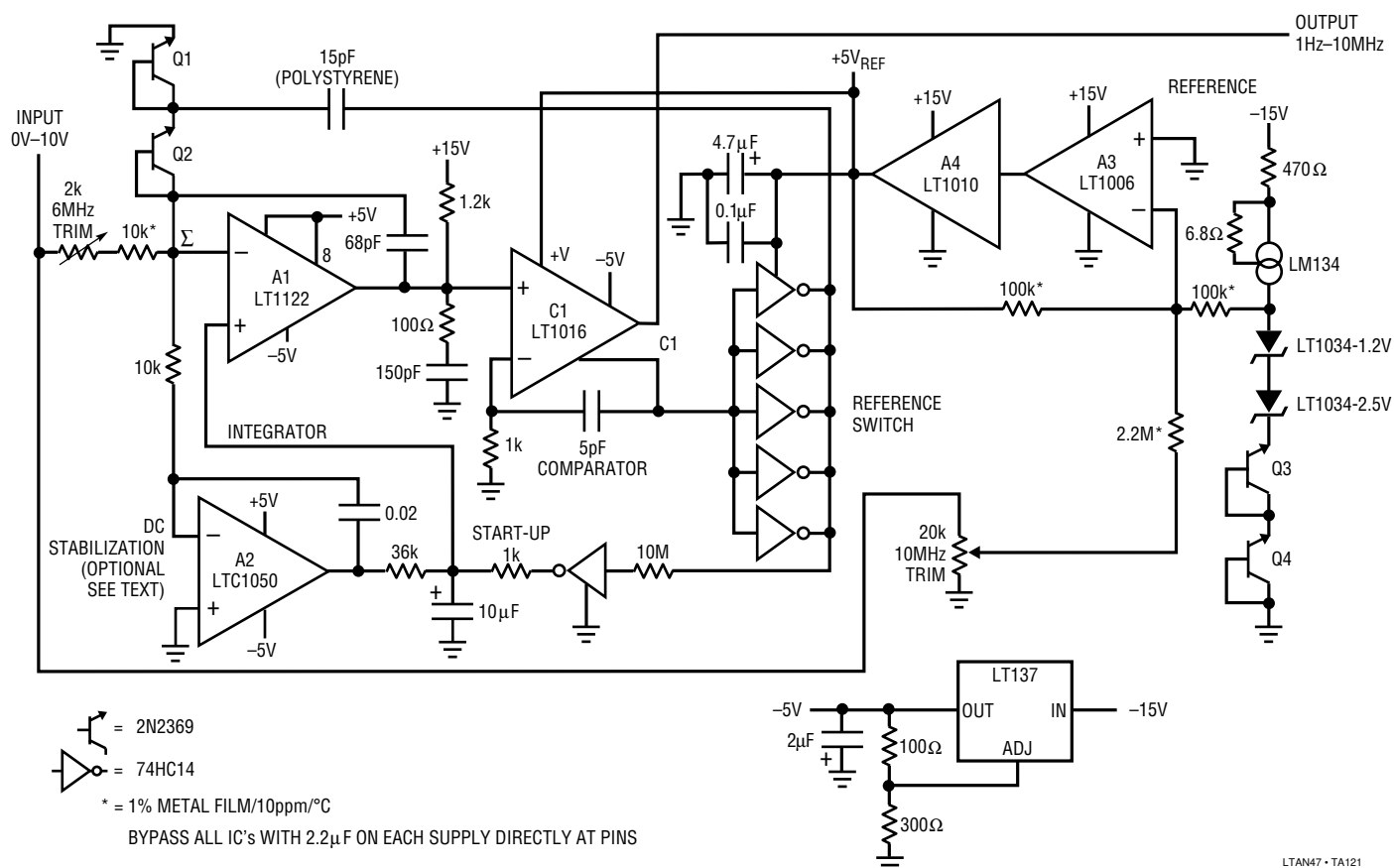


Figure 121. 1Hz-10MHz V-to-F Converter. Linearity is 0.03% with 50ppm/°C Drift

Each time the circuit produces an output pulse, it feeds back a fixed quantity of charge (Q) to a summing node (Σ). The circuit's input furnishes a comparison current at the summing node. The difference signal at the node is integrated in a monitoring amplifier's feedback capacitor. The amplifier controls the circuit's output pulse generator, completing a feedback loop around the integrating amplifier. To maintain the summing node at zero, the pulse generator runs at a frequency which permits enough charge pumping to offset the input signal. Thus, the output frequency will be linearly related to the input voltage. A1 is the integrating amplifier.

$0.05\mu\text{V}/^\circ\text{C}$ offset drift performance is obtained by stabilizing A1 with A2, a chopper stabilized op amp. A2 measures the DC value of the negative input, compares it to ground, and forces the positive input to maintain offset balance in A1. Note that A2 is configured as an integrator and cannot see high frequency signals. It functions only at DC and low frequency.

A1 is arranged as an integrator with a 68pF feedback capacitor. When a positive voltage is applied to the input, A1's output integrates in a negative direction (Trace A, Figure 122). During this period, C1's inverting output is low. The paralleled HCMOS inverters form a reference voltage switch. The reference voltage is established by the LM134 current source driven LT1034's and the Q3-Q4 combination. Additionally, a small input voltage related term is summed into the reference, improving overall circuit linearity. A3-A4 provides low drift buffering, presenting a low impedance reference to the paralleled inverter's supply pin. The HCMOS outputs give low resistance, essentially errorless switching. The reference switch's output charges the 15pF capacitor via Q1's path.

When A1's output crosses zero, C1's inverting output goes high and the reference switch (Trace B) goes to ground. This causes the 15pF unit to dispense charge into the summing node via Q2's V_{BE} . The amount of charge dispensed is a direct function of the voltage the 15pF unit was charged to ($Q = CV$). Q1 and Q2 are temperature compensated by Q3 and Q4 in the reference string. The current through the 15pF unit (Trace C) reflects the charge pumping action. The removal of current from A1's summing junction (Trace D) causes the junction to be driven very quickly negative. The initial negative-going 15ns

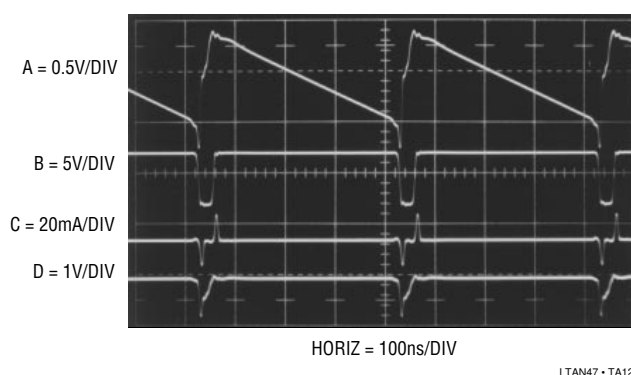


Figure 122. 10MHz V-to-F's Operating Waveforms. LT1122 Integrator is Completely Reset in 60ns

transient at A1's output is due to amplifier delay. The input signal feeds directly through the feedback capacitor and appears at the output. When the amplifier finally responds, its output (Trace A) slew limits as it attempts to regain control of the summing node. The class A $1.2\text{k}\Omega$ pull-up and the RC damper at A1's output minimizes erroneous output movement, enhancing this slew recovery. The amount of time the reference switch remains at ground depends on how long it takes A1 to recover and the 5pF- 1000Ω hysteresis network at C1. This 60ns interval is long enough for the 15pF unit to fully discharge. After this, C1 changes state, the reference switch swings positive, the capacitor is recharged and the entire cycle repeats. The frequency at which this oscillation occurs is directly related to the voltage-input-derived current into the summing junction. Any input current will require a corresponding oscillation frequency to hold the summing point at an average value of 0V.

Maintaining this relationship at megahertz frequencies places severe restrictions on circuit timing. The key to achieving 10MHz full-scale operating frequency is the ability to transmit information around the loop as quickly as possible. The discharge-reset sequence is particularly critical and is detailed in Figure 123. Trace A is the A1 integrator output. Its ramp output crosses 0V at the first left vertical graticule division. A few nanoseconds later, C1's inverting output begins to rise (Trace B), switching the reference switch to ground (Trace C). The reference switch begins to head towards ground about 16ns after A1's output crosses 0V. 2ns later, the summing point (Trace D) begins to go negative as current is pulled from it through the 15pF capacitor. At 25ns, C1's inverting

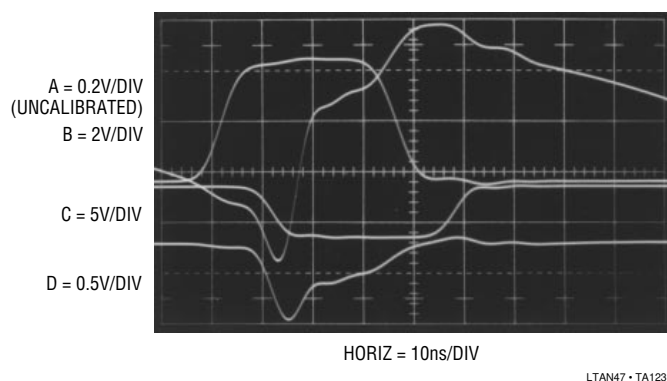


Figure 123. Detail of 60ns Reset Sequence (Whoosh!)

output is fully up, the reference switch is at ground, and the summing point has been pulled to its negative extreme. Now, A1 begins to take control. Its output (Trace A) slews rapidly in the positive direction, restoring the summing point. At 60ns, A1 is in control of the summing node and the integration ramp begins again.

Start-up and overdrive conditions could force A1's output to go to the negative rail and stay there. The AC-coupled nature of the charge dispensing loop can preclude normal operation and the circuit may latch. The remaining HCMOS inverter provides a watchdog function for this condition. If A1's output rails negative the reference switch tries to stay at ground. The remaining inverter goes high, lifting A1's positive input. This causes A1's output to slew positive, initiating normal circuit action. The 1k-10 μ F combination and the 10M-inverter input capacitance limit start-up loop bandwidth, preventing unwanted outputs.

The LM134 current source driving the reference string has a built in 0.33%/°C thermal coefficient, causing slight voltage modulation in the Q3-Q4 pair over temperature. This small change ($\approx +120$ ppm/°C) opposes the -120 ppm/°C drift in the 15pF polystyrene capacitor, aiding overall circuit tempco.

To trim this circuit, apply exactly 6V at the input and adjust the 2k Ω potentiometer for 6.000MHz output. Next, put in exactly 10V and trim the 20k unit for 10.000MHz output. Repeat these adjustments until both points are fixed. A2's low drift eliminates a zero adjustment. If operation below 600Hz is not required, A2 and its associated components may be deleted.

Linearity of this circuit is 0.03% with full-scale drift of 50ppm/°C. Zero point error, controlled by A2, is 0.05Hz/°C.

8-Bit, 100ns Sample-Hold

Figure 124 shows a simple, very fast sample-hold circuit. This circuit will acquire a ± 5 V input to 8-bit accuracy in 100ns. Hold step is inside 1/4 LSB with hold settling inside 25ns. Aperture time is 4ns and droop rate about 1/2 LSB/ μ s.

The input is fed to a Schottky switching bridge via inverting buffer A1. The Schottky bridge, similar to types used in sampling oscilloscopes¹⁶, gives 1ns switching and eliminates the charge pump-through that a FET switch would contribute. The switching bridge's output feeds output amplifier A2. A2, configured as an integrator, is the actual hold amplifier. Its output is fed back to the switching bridge's input, forming a summing point with A1's output resistor. This feedback loop places the bridge within a loop, enhancing accuracy.

The bridge is switched by driving the sample-hold input line. Q1 and Q2 drive L1's primary. L1's secondaries provide complementary drive to the bridge with almost no time skewing.

Figure 125 shows the circuit acquiring a full scale step. Trace A is the input command while Trace B is A2's output. The aberration visible in A2's output when switching into hold (hold step) is due to minute residual AC imbalances in the bridge. Figure 126 studies this effect in high resolution detail, with the hold step trim deliberately disconnected. After A2's output nominally settles at final value, the circuit is switched into hold. The bridge imbalance allows a small parasitic charge to be displaced into A2's summing point, causing A2 to step 10mV higher (in this case). If the trim is connected and properly adjusted, it supplies a small compensatory charge during switching. Figure 127 shows the effect of this on the output. The settled hold output is the same as the acquired value. To trim this circuit, ground the input while pulsing the sample-hold control line. Next, adjust the trim for minimal amplitude step between the sample and hold states.

In contrast to low frequency sample-hold circuits this design cannot pass signal if left in the sample mode. The transformer's inherent AC coupling precludes such operation. Similarly, extended sample mode duration (e.g., >500ns) will cause transformer saturation, resulting in erroneous outputs and excessive Q1-Q2 dissipation. If

Note 16: See References 7, 8 and 28.

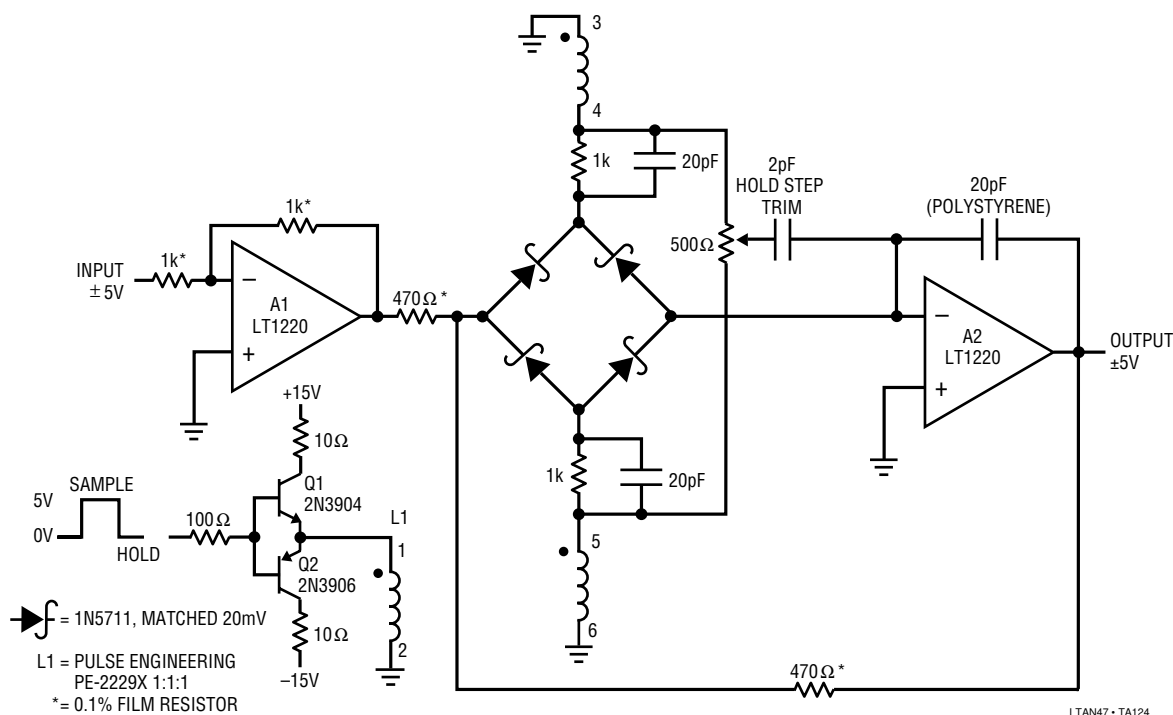


Figure 124. 8-Bit, 100ns Sample-Hold

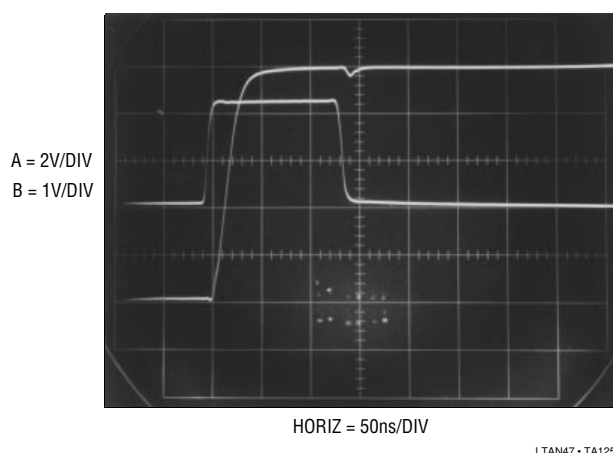


Figure 125. Fast Sample-and-Hold Acquiring a Full-Scale Input

extended logic high durations are possible at the control input, it should be AC coupled.

15ns Current Summing Comparator

Figure 128 shows a way to build a high speed current comparator with resolution in the 12-bit range. Current comparison, the fastest way to compare D→A outputs and analog values, is commonly used in high speed A→D converters and instrumentation. A1 is set up as a Schottky bounded amplifier. The bound diodes prevent A1 from

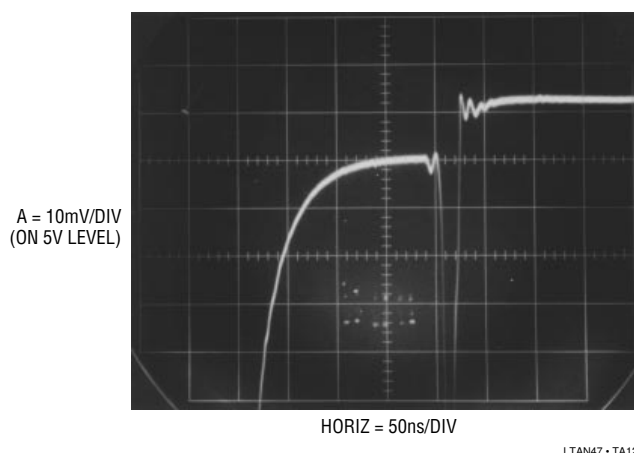


Figure 126. Hold Step with Mis-Adjusted Compensation

saturating due to excessive summing point overdrive, aiding response time. The 3pF capacitor, a typical value, compensates DAC output capacitance and is selected for best amplifier damping. The 10k feedback resistor, also typical, is chosen for best gain-bandwidth performance. Voltage gains of 4 to 10 are common. Figure 129 shows performance. Trace A, a test input, causes A1's output (Trace B) to slew through zero (screen center horizontal line). When A1 crosses zero, C1's input biases negative and it responds (C1's output is Trace C) 10ns later with a TTL

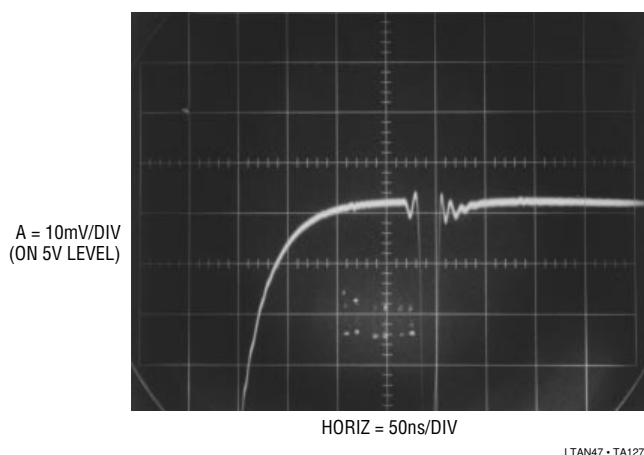


Figure 127. Hold Step with Properly Adjusted Compensation

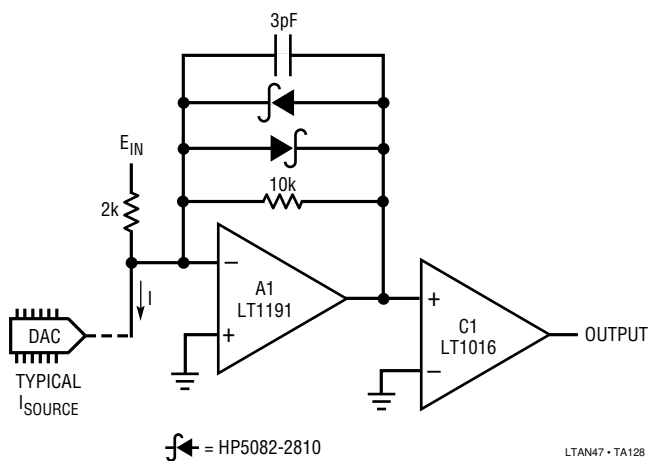


Figure 128. Fast Summing Comparator

output. Total elapsed time from the test input arriving at a TTL high until the comparator output achieves a TTL high is inside 15ns.

50MHz Adaptive Threshold Trigger Circuit

Figure 130 is an extremely versatile trigger circuit. Designing a fast, stable trigger is not easy and often entails a considerable amount of discrete circuitry. This circuit reliably triggers from DC-50MHz over a 2mV-300mV input range with no level adjustment required.

A1, a gain of ten preamplifier, feeds an adaptive trigger configuration identical to the one described in Figure 97's fiber optic receiver. The adaptive trigger maintains the A3 output comparator's trip point at 1/2 input signal amplitude, regardless of its magnitude. This insures reliable automatic triggering over a wide input amplitude range, even for very low level inputs. As an option, the network

shown in dashed lines permits changing the trip threshold. This allows any point on the input waveform edge to be selected as the actual trigger point.¹⁷

Figure 131 shows performance for a 40MHz input sine wave (Trace A). A1's output (Trace B) takes gain and the A3 comparator gives a clean logic output (Trace C). At the highest frequencies, any bandwidth limiting in A1 is irrelevant; the adaptive trigger threshold will simply vary ratiometrically to maintain circuit output.

Fast Time-to-Height (Pulsewidth-to-Voltage) Converter

The circuit of Figure 132 allows very short pulsewidths (in this case 250ns full-scale) to be determined to a typical accuracy of 1%. Digital methods of achieving similar results dictate clock speeds of 1GHz, which is cumbersome. In addition, processor based approaches using averaging techniques require repetitive pulses which this circuit does not. Circuits of this type are frequently required in automatic test equipment and nuclear and high energy physics work where determination of short pulsewidths is a common requirement.

The circuit functions by charging a capacitor during the period of a pulsewidth. When the pulse ends, charging ceases and the voltage across the capacitor is proportional to the width of the pulse.

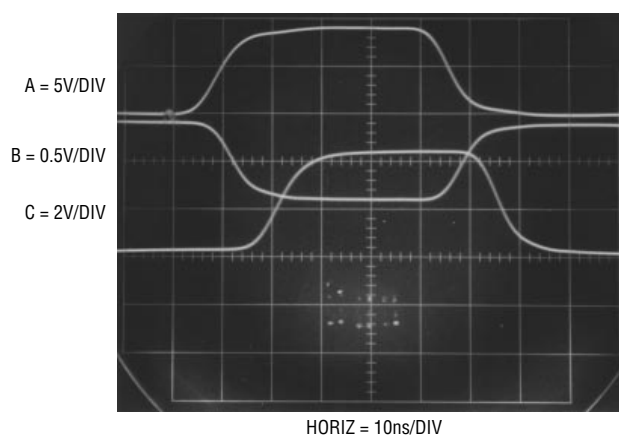


Figure 129. Fast Summing Comparator's Waveforms. Total Delay is 15ns

Note 17: This technique is borrowed from oscilloscope trigger circuitry. See Reference 29.

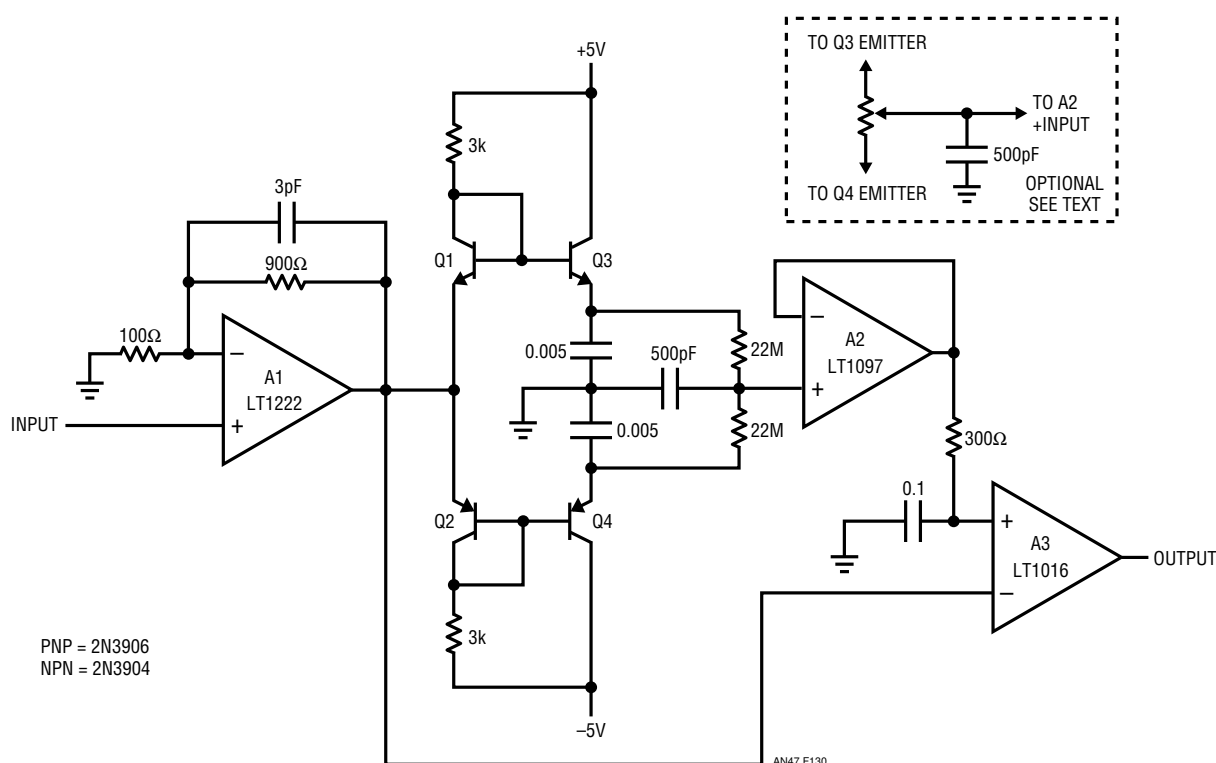


Figure 130. 50MHz Trigger with Adaptive Threshold

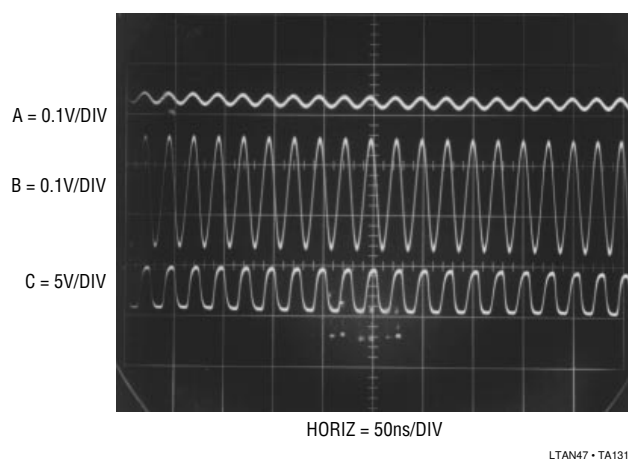


Figure 131. The Trigger Responds to a 40MHz Input. Input Amplitude Variations from 2mV-300mV Have No Effect

The input pulse to be measured (Trace A, Figure 133) simultaneously biases the 74C221 dual one shot and Q3. Q3, aided by Baker¹⁸ clamping, capacitive feedforward and optimized DC base biasing, turns off in a few nanoseconds. Current source Q2's emitter forward biases and Q2

supplies constant current to the 100pF integrating capacitor. Q1 supplies temperature compensation for Q2, with the 2.5V LT1009 referencing the current source. Q2's collector (e.g., the 100pF capacitor) charges in ramp fashion (Trace B). A1 supplies a buffered output (Trace C). When the input pulse ends, Q3 rapidly turns on, reverse biasing Q2's emitter and turning off the current source. A1's voltage is directly proportional to the input pulse width. A monitoring A → D converter can acquire this data.

After a time set by the 74C221's RC programmed delay, a pulse appears at its Q2 output (Trace D). This pulse turns on Q4, discharging the 100pF capacitor to zero and readying the circuit for the next input pulse.

This circuit's accuracy and resolution are crucially dependent on minimizing delay in switching the Q1-Q2 current source. Figure 134 provides amplitude and time expanded versions of critical circuit waveforms. Trace A is the input pulse and Trace B is A1's input, showing the beginning of the ramp's ascent. Trace C, A1's output, shows about 13ns delay from A1's input. Traces D and E, A1's input and

Note 18: See Reference 45.

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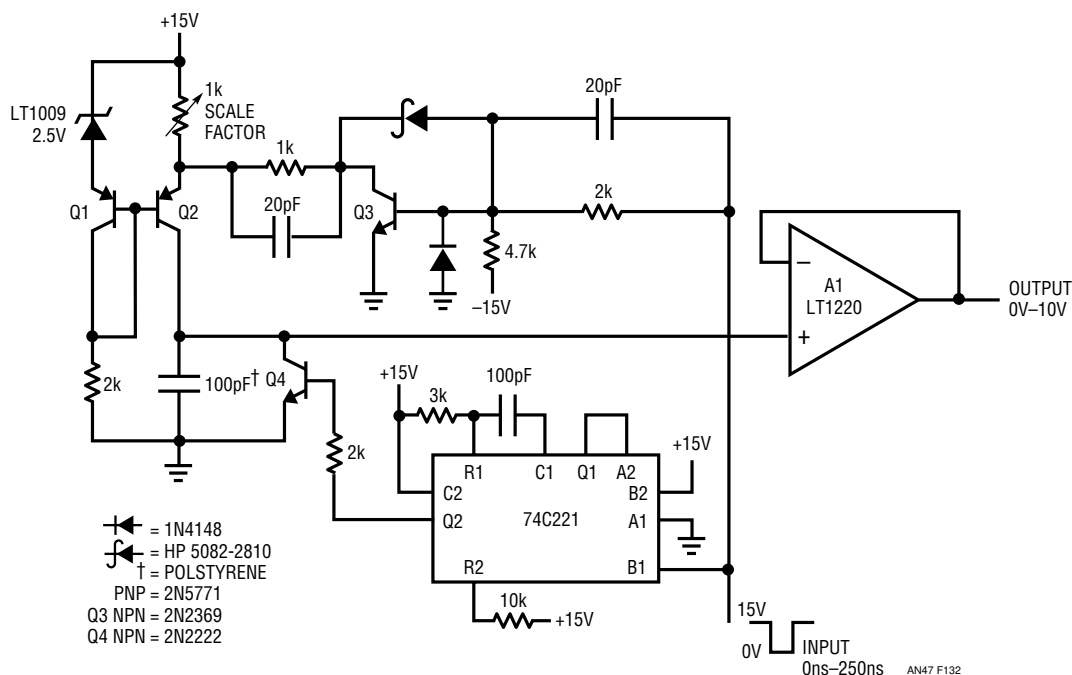


Figure 132. Fast Time-to-Height Converter

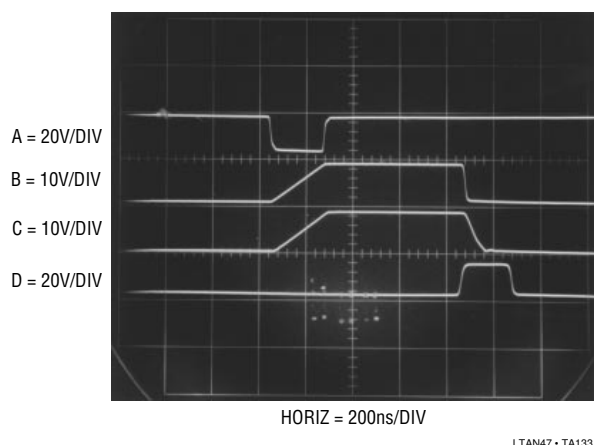


Figure 133. Time-to-Height Converter Acquires a 250ns Pulse

output respectively, record similar A1 delays for ramp turn-off. The photo reflects the extremely fast current source switching; the vast majority of delay is due to A1's delay. A1's delay is far less critical than current source switching delays; A1 will always settle to the correct value well before the one shot resets the circuit. In practice, a monitoring A → D converter should not be triggered until about 50ns after the circuit's input pulse has ceased. This gives A1 plenty of time to catch up to the 100pF capacitor's settled value.

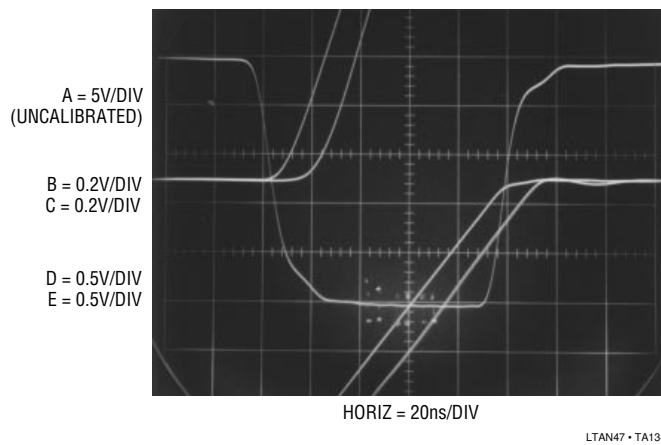


Figure 134. Detail of Time-to-Height Converter's Ramp Switching

As mentioned, current source switching speed is essential for good results. Figure 135 details current source turn off. Trace A is the circuit's input pulse rising edge and Trace B shows the top of the ramp. Turn off occurs in a few nanoseconds. Similar speed is characteristic of the input's falling edge (current source turn on). Additionally, it is noteworthy that circuit accuracy and resolution limits are set by the *difference* in current source turn on and off delays. As such, the *effective* overall delay is extremely small.

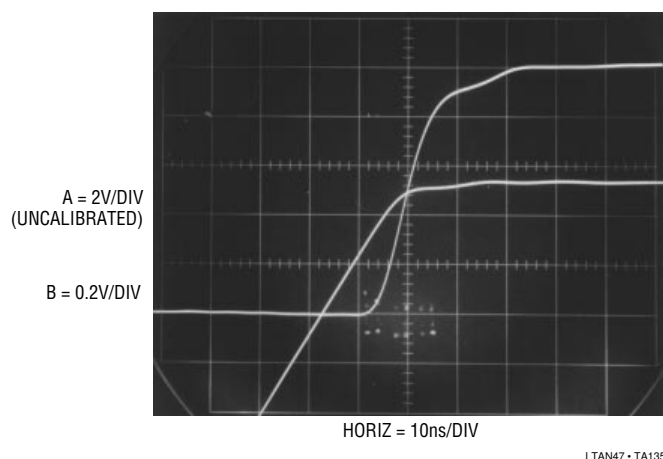


Figure 135. Current Source Turn-Off Detail for the Time-to-Height Converter

To calibrate this circuit, put in a 250ns width pulse and trim the 1k Ω potentiometer for 10V output. The circuit will convert pulse widths from 20ns to 250ns to a typical accuracy of 1%. The 20ns minimum measurable width is due to inability to fully discharge the 100pF capacitor. If this is objectionable, Q4 can be replaced with a lower saturation device or A1's output can be offset.

True RMS Wideband Voltmeter

Most AC RMS measurements use logarithmic techniques to compute the waveform's RMS value. This method limits bandwidth to below 1MHz and crest factor performance to about 10:1. Practically speaking, a waveform's RMS value is defined as its heating value in the load. Specialized instruments employ thermally based assemblies that compute the RMS value of the input. The thermal method provides substantially improved bandwidth and crest factor capability compared to logarithmically based converters.

Thermal RMS-DC converters are direct acting, thermo-electronic analog computers. The thermal technique is explicit, relying on first principles. The simple operation permits wideband performance unattainable with implicit, indirect methods based on logarithmic computing.

Figure 136 shows a classic scheme for implementing a thermally based RMS-DC converter. Here, the DC amplifier forces a second, identical, heater-sensor pair to the same thermal conditions as the input driven pair. This differentially sensed, feedback enforced loop makes ambient temperature shifts a common-mode term, eliminating their effect. Also, although the voltage and thermal

interaction is non-linear, the input-output voltage relationship is linear with unity gain. The ability of this arrangement to reject ambient temperature shifts depends on the heater-sensor pairs being isothermal. This is achievable by thermally insulating them with a time constant well below that of ambient shifts. If the time constants to the heater-sensor pairs are matched, ambient temperature terms will affect the pairs equally in phase and amplitude. The DC amplifier will reject this common-mode term. Note that, although the pairs are isothermal, they are insulated from each other. Any thermal interaction between the pairs reduces the system's thermally based gain terms. This would cause unfavorable signal-to-noise performance, limiting dynamic operating range. Figure 136's output is linear because the matched thermal pair's non-linear voltage-temperature relationships cancel each other.

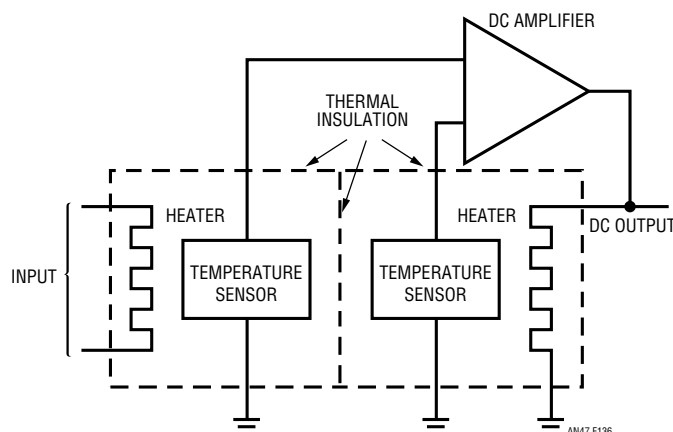


Figure 136. Conceptual Thermal RMS-DC Converter

The advantages of this approach have made its use popular in thermally based RMS-DC measurements. Typically, the assembly is composed of matched heater resistors, sensors and thermal insulation. These assemblies are relatively large and expensive to produce. Figure 137's economical wideband thermally based voltmeter is based on a monolithic thermal converter. The LT1223 provides gain, and drives the LT1088 RMS-DC thermal converter.¹⁹ The LT1088's temperature sensing diodes are biased from the supply. A1, set up as a differential servo amplifier with a gain of 9000, extracts the diode's difference signal and biases Q1. Q1 drives one of the LT1088's heaters, completing a loop. The 3300pF capacitor gives a stable roll-off.

Note 19: Complete details on this device and a discussion on thermal conversion considerations are found in Reference 40.

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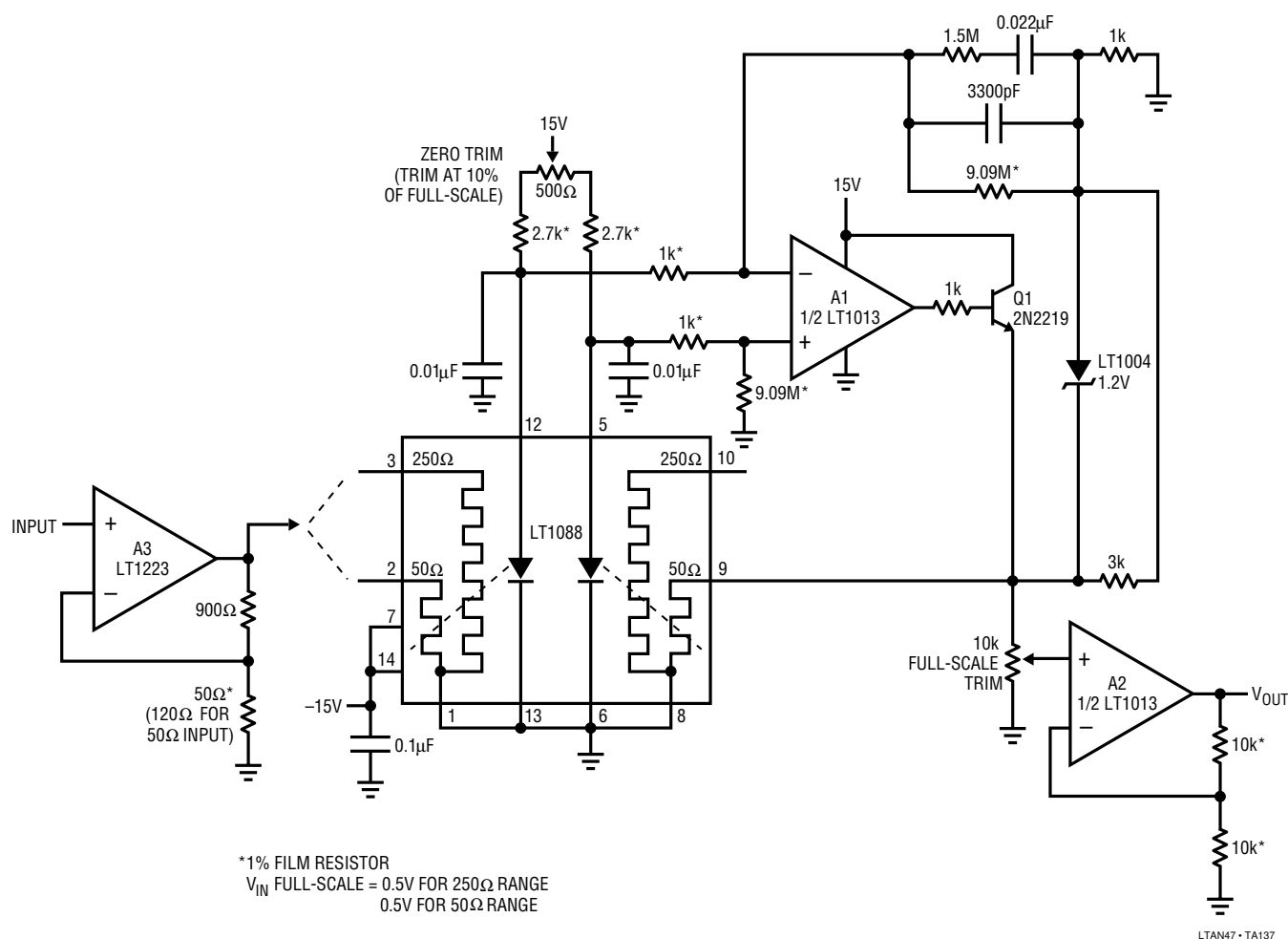


Figure 137. Wideband True RMS Voltmeter

The 1.5M-0.022 μ F combination improves settling by reducing gain during output slew. The LT1088's square-law thermal gain means overall loop gain is lower for small inputs. Normally, this would result in slow settling for values below about 10%-20% of scale. The LT1004 1k-3k network is a simple breakpoint, boosting amplifier gain in this region to improve settling. A2, a gain trimmable output stage, serves to compensate for gain variations in the two sides of the LT1088. To trim the circuit, put in about a 10% scale DC signal (e.g., 0.05V). Adjust the zero trim so that $V_{OUT} = V_{IN}$. Next, apply a full-scale DC input and set the full-scale trim to that value at the output. Repeat the trims until both are fixed well within 1% of full-scale. An alternate trim scheme involves applying no input, grounding Q1's base and setting the zero trim until A1's output is active. Then, unground Q1's base, apply a full-scale input and trim the full-scale adjustment for that value at the output.

Figure 138 is a plot of error vs input frequency. The LT1088 is specified at 2% to 100MHz (50Ω heater) or 1% to 20MHz (250Ω heater). As such, most of the error shown is due to bandwidth restrictions in A3, but performance is still impressive. The plots include data taken at various input levels into both heaters. A 500mV input into 250Ω dips to 1% at 8MHz and 2.5% at 14MHz before peaking badly beyond 17MHz. This input level forces a 9.5 V_{RMS} output at A3, introducing large signal bandwidth limitations. The 400mV input to the 250Ω heater shows essentially flat results to 20MHz, the LT1088's 250Ω heater specification limit.

The 50Ω heater provides significantly wider bandwidth, although A3's 50mA output limits maximum input to about 100mV_{RMS} (1.76V_{RMS} at the LT1088).

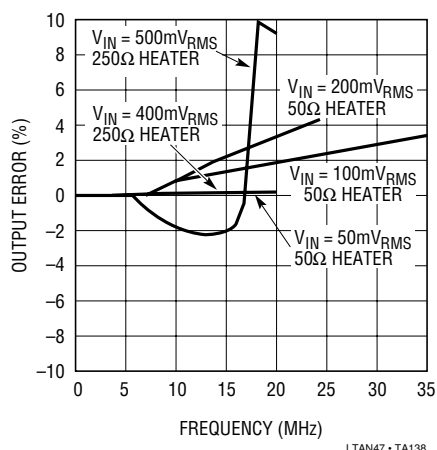


Figure 138. Accuracy Plot for the RMS Voltmeter

APPLICATIONS SECTION IV - MISCELLANEOUS CIRCUITS

RF Leveling Loop

Figure 137's wideband AC conversion can be applied in other areas. A common RF requirement is to stabilize the amplitude of a waveform against variations in input, time and temperature. Instruments and transmitters frequently require this function, which is not easy if waveform purity must be maintained. Figure 139A shows a 25MHz RF leveling loop. The RF input is applied to the AD539 wideband multiplier. The multiplier's output drives A1. A1's output is converted to DC by the LT1088 based RMS-DC converter (see previous circuit). A servo amplifier compares this output with a settable DC reference and biases the multiplier's control channel, completing a loop. The 0.33μF capacitor provides frequency compensation by rolling off gain at a frequency well below the response of the LT1088 servo. The loop maintains the output's 25MHz RMS amplitude at the DC reference's value. Changes in load, input, power supply and other variables are rejected.

Figure 139B, a similar circuit, offers significantly lower cost although performance is not quite as good. The RF input is applied to LT1228 A1, an operational transconductance amplifier. A1's output feeds LT1228 A2, a current feedback amplifier. A2's output, the circuit's output, is sampled by the A3 based gain control configuration. This arrangement, similar to the gain control loops described in Figures 112 and 114, closes a gain control loop back at A1. The 4pF capacitor compensates rectifier diode

capacitance, enhancing output flatness vs frequency. A1's I_{SET} input current controls its gain, allowing overall output level control. This approach to RF leveling is simple and inexpensive, although output drift, distortion and regulation are somewhat higher than in the previous circuit.

Voltage Controlled Current Source

Figure 140 shows a voltage controlled current source with load and control voltage referred to ground. This simple, powerful circuit produces output current in accordance with the sign and magnitude of the control voltage. The circuit's scale factor is set by resistor R. A1, biased by V_{IN} , drives current through R (in this case 10Ω) and the load. A2, sensing differentially across R, closes a loop back to A1. The load current is constant because A1's loop forces a fixed voltage across R. The 2k-100pF combination sets roll off and the configuration is stable. Figure 141 shows dynamic response. Trace A is the voltage control input while Trace B is the output current. Response is quick and clean, with delay of 5ns and no slew residue or aberration.

High Power Voltage Controlled Current Source

Figure 142 is identical to the basic current source, except that it adds a 1A booster stage (adapted from Figure 104) for increased output power. Including the booster inside A1's feedback loop eliminates its DC errors. Note that the booster's current limiting features have been removed, because of this circuit's inherent current limiting nature of operation. Figure 143 shows this circuit's response to be as clean as the lower power version, although delay is about 20ns slower. It is worth mentioning that the loop stability considerations involved in placing A2 and the booster in A1's feedback path are significant. This circuit receives treatment in Appendix C, "The Oscillation Problem - Frequency Compensation Without Tears".

18ns Circuit Breaker

Figure 144 shows a simple circuit which will turn off current in a load 18ns after it exceeds a preset value. This circuit has been used to protect integrated circuits during developmental probing and is also useful for protecting expensive loads during trimming and calibration. The circuit's versatility is enhanced because one side of the load is grounded. Under normal conditions, Q1's emitter (Trace A, Figure 145, is Q1's current, and Trace C is its

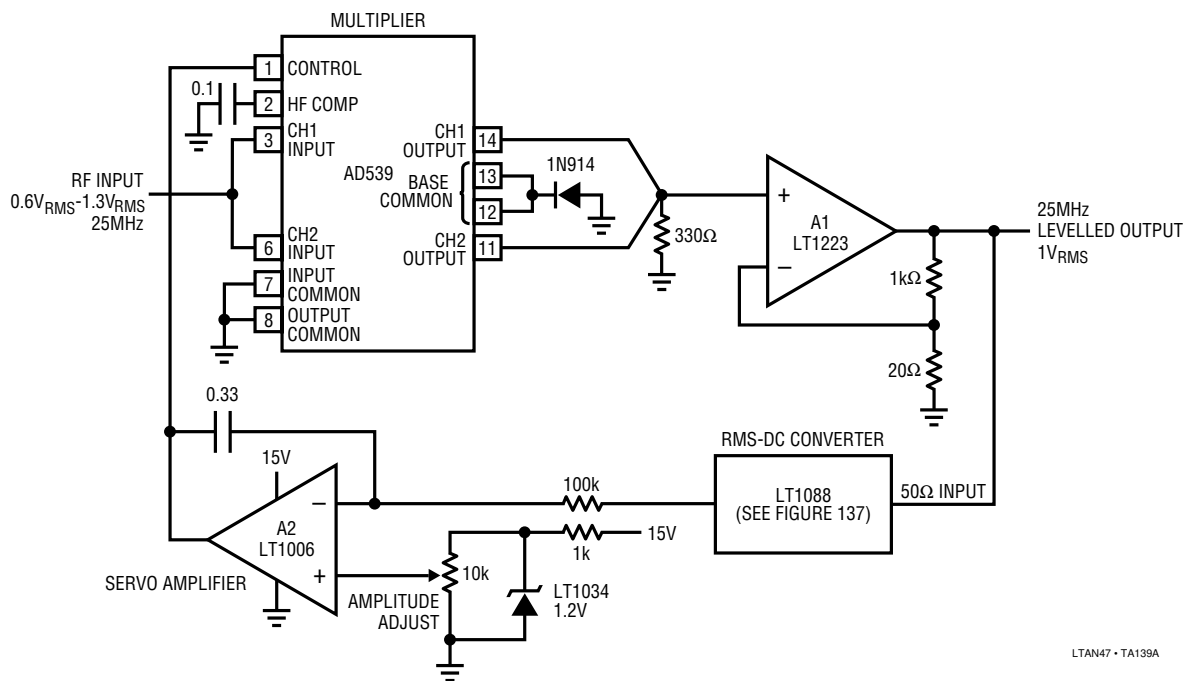


Figure 139A. RF Leveling Loop

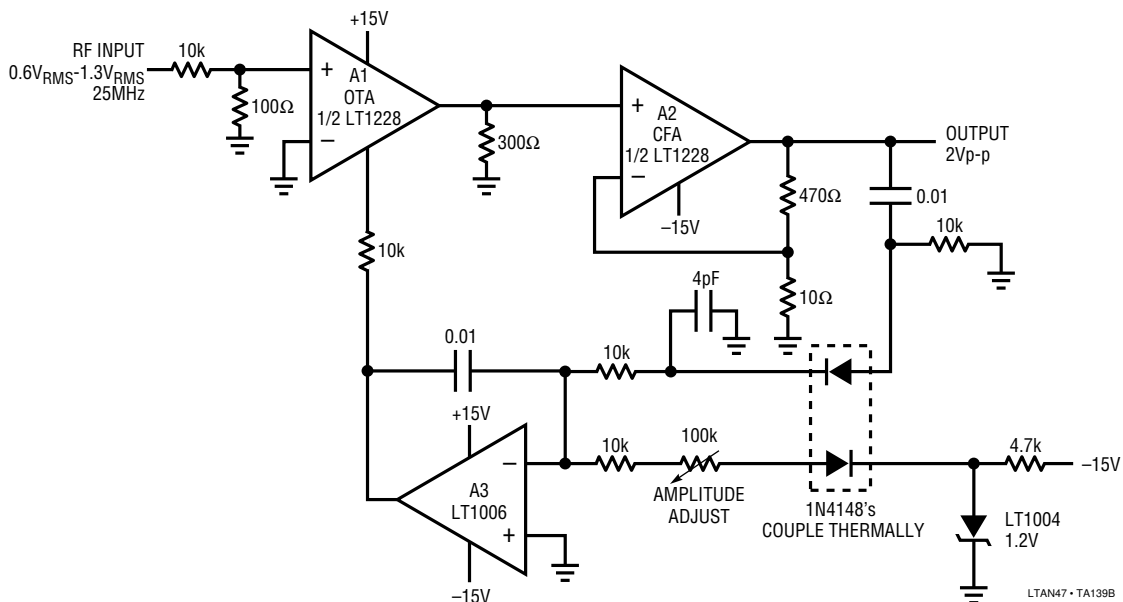


Figure 139B. Simple RF Leveling Loop

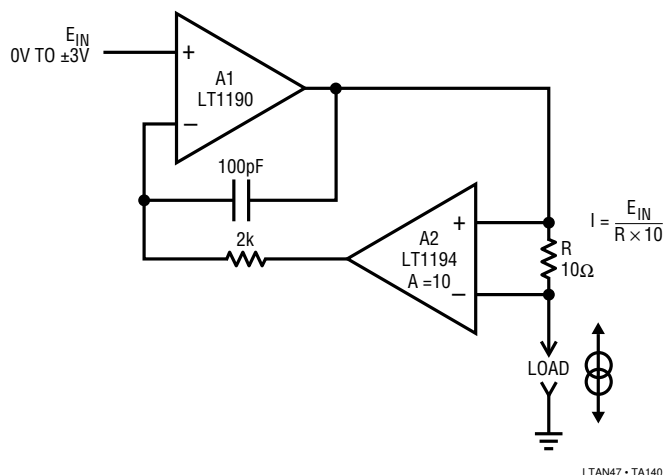


Figure 140. Fast, Precise, Voltage Controlled Current Source with Grounded Load

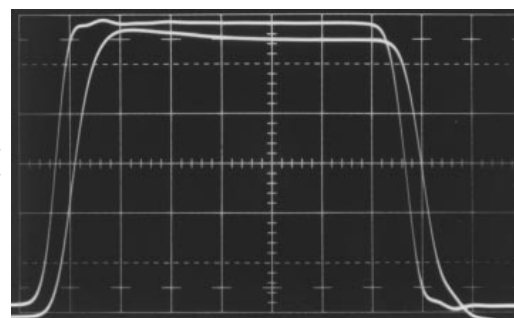


Figure 141. Dynamic Response of the Current Source. Delay is 4ns, with Clean Settling

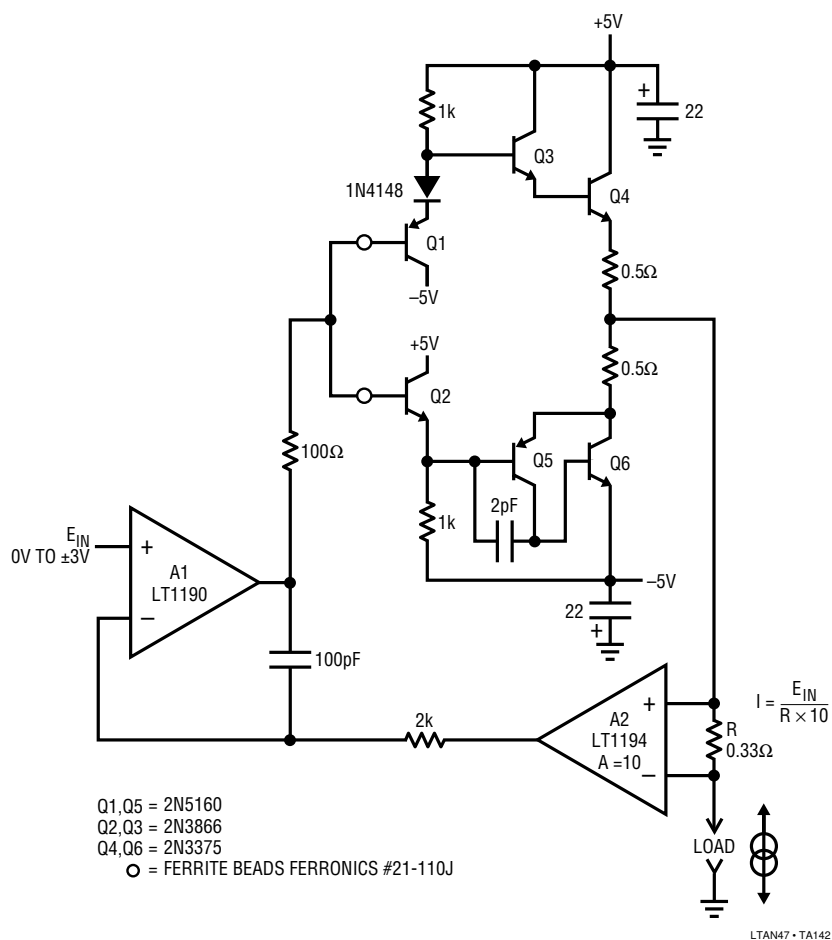


Figure 142. High Power, Wideband Voltage Controlled Current Source

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voltage) is biased on, supplying power to the load via the 10Ω current shunt. Differential amplifier A1's output resides below comparator A2's voltage programmed trip point and Q2 is off. When an overload occurs, Q1's emitter current begins to increase (Trace A, just prior to the third vertical division). A1's output (Trace B) begins to rise as it tracks the increase in the 10Ω shunt's voltage. The $9k-1k$ dividers keep A1 inputs inside their common-mode range. Simultaneously, Q1's emitter voltage (Trace C) begins to drop as it beta limits. When A1's version of the load current exceeds A2's trip point, A2 (Trace D) goes high, turning on Q2. Q2's turn on steals Q1's base drive, turning off the load

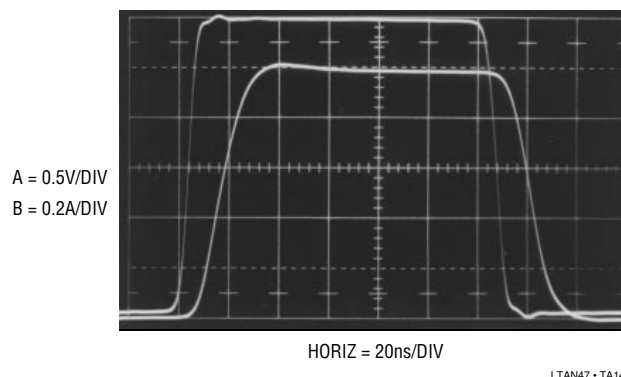


Figure 143. 1A Pulse Response of the High Power Current Source

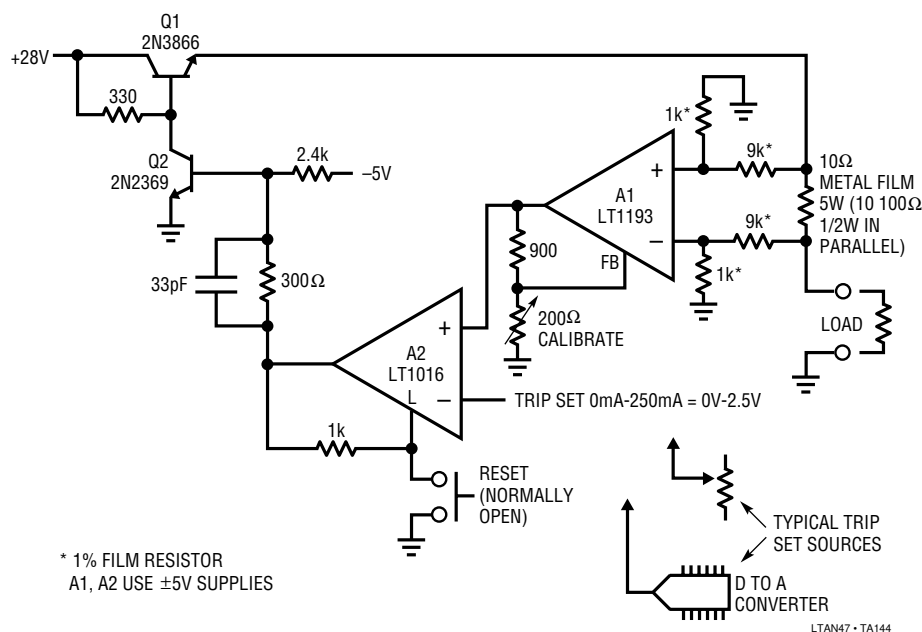


Figure 144. 18ns Circuit Breaker with Voltage Programmable Trip Point

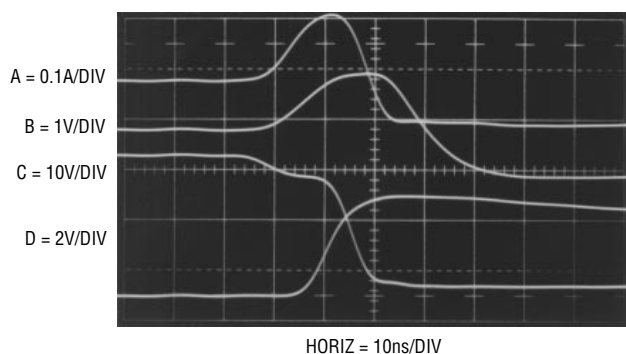


Figure 145. Operating Waveforms for the 18ns Circuit Breaker. Circuit Output (Trace C) is Shut Down 18ns After Output Current (Trace A) Begins to Rise

current. Local positive feedback at A2's latch pin causes it to latch in this off state. When the load fault has been cleared, the pushbutton can be used to reset the circuit. The delay from the onset of excessive load current to complete shutdown is inside 18ns. The 4ns delay of Trace A's current probe should be factored in when interpreting waveforms. To calibrate this circuit, ground Q2's base and install a 250mA load. Adjust the 200Ω trim for a 2.5V output at A1. Next, remove the load, unground Q2's base and press the reset button. Finally, put in the desired trip set voltage and the circuit is ready for use.